Modicon M580
Safety Manual

09/2017

QGH46982.00

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All pertinent state, regional, and local safety regulations must be observed when installing and using this product. For reasons of safety and to help ensure compliance with documented system data, only the manufacturer should perform repairs to components.

When devices are used for applications with technical safety requirements, the relevant instructions must be followed.

Failure to use Schneider Electric software or approved software with our hardware products may result in injury, harm, or improper operating results.

Failure to observe this information can result in injury or equipment damage.

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Safety Information

Important Information

NOTICE

Read these instructions carefully, and look at the equipment to become familiar with the device before trying to install, operate, service, or maintain it. The following special messages may appear throughout this documentation or on the equipment to warn of potential hazards or to call attention to information that clarifies or simplifies a procedure.



The addition of this symbol to a “Danger” or “Warning” safety label indicates that an electrical hazard exists which will result in personal injury if the instructions are not followed.



This is the safety alert symbol. It is used to alert you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.

⚠ DANGER

DANGER indicates a hazardous situation which, if not avoided, **will result in** death or serious injury.

⚠ WARNING

WARNING indicates a hazardous situation which, if not avoided, **could result in** death or serious injury.

⚠ CAUTION

CAUTION indicates a hazardous situation which, if not avoided, **could result** in minor or moderate injury.

NOTICE

NOTICE is used to address practices not related to physical injury.

PLEASE NOTE

Electrical equipment should be installed, operated, serviced, and maintained only by qualified personnel. No responsibility is assumed by Schneider Electric for any consequences arising out of the use of this material.

A qualified person is one who has skills and knowledge related to the construction and operation of electrical equipment and its installation, and has received safety training to recognize and avoid the hazards involved.

BEFORE YOU BEGIN

Do not use this product on machinery lacking effective point-of-operation guarding. Lack of effective point-of-operation guarding on a machine can result in serious injury to the operator of that machine.

⚠ WARNING

UNGUARDED EQUIPMENT

- Do not use this software and related automation equipment on equipment which does not have point-of-operation protection.
- Do not reach into machinery during operation.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

This automation equipment and related software is used to control a variety of industrial processes. The type or model of automation equipment suitable for each application will vary depending on factors such as the control function required, degree of protection required, production methods, unusual conditions, government regulations, etc. In some applications, more than one processor may be required, as when backup redundancy is needed.

Only you, the user, machine builder or system integrator can be aware of all the conditions and factors present during setup, operation, and maintenance of the machine and, therefore, can determine the automation equipment and the related safeties and interlocks which can be properly used. When selecting automation and control equipment and related software for a particular application, you should refer to the applicable local and national standards and regulations. The National Safety Council's Accident Prevention Manual (nationally recognized in the United States of America) also provides much useful information.

In some applications, such as packaging machinery, additional operator protection such as point-of-operation guarding must be provided. This is necessary if the operator's hands and other parts of the body are free to enter the pinch points or other hazardous areas and serious injury can occur. Software products alone cannot protect an operator from injury. For this reason the software cannot be substituted for or take the place of point-of-operation protection.

Ensure that appropriate safeties and mechanical/electrical interlocks related to point-of-operation protection have been installed and are operational before placing the equipment into service. All interlocks and safeties related to point-of-operation protection must be coordinated with the related automation equipment and software programming.

NOTE: Coordination of safeties and mechanical/electrical interlocks for point-of-operation protection is outside the scope of the Function Block Library, System User Guide, or other implementation referenced in this documentation.

START-UP AND TEST

Before using electrical control and automation equipment for regular operation after installation, the system should be given a start-up test by qualified personnel to verify correct operation of the equipment. It is important that arrangements for such a check be made and that enough time is allowed to perform complete and satisfactory testing.

⚠ WARNING

EQUIPMENT OPERATION HAZARD

- Verify that all installation and set up procedures have been completed.
- Before operational tests are performed, remove all blocks or other temporary holding means used for shipment from all component devices.
- Remove tools, meters, and debris from equipment.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Follow all start-up tests recommended in the equipment documentation. Store all equipment documentation for future references.

Software testing must be done in both simulated and real environments.

Verify that the completed system is free from all short circuits and temporary grounds that are not installed according to local regulations (according to the National Electrical Code in the U.S.A, for instance). If high-potential voltage testing is necessary, follow recommendations in equipment documentation to prevent accidental equipment damage.

Before energizing equipment:

- Remove tools, meters, and debris from equipment.
- Close the equipment enclosure door.
- Remove all temporary grounds from incoming power lines.
- Perform all start-up tests recommended by the manufacturer.

OPERATION AND ADJUSTMENTS

The following precautions are from the NEMA Standards Publication ICS 7.1-1995 (English version prevails):

- Regardless of the care exercised in the design and manufacture of equipment or in the selection and ratings of components, there are hazards that can be encountered if such equipment is improperly operated.
- It is sometimes possible to misadjust the equipment and thus produce unsatisfactory or unsafe operation. Always use the manufacturer's instructions as a guide for functional adjustments. Personnel who have access to these adjustments should be familiar with the equipment manufacturer's instructions and the machinery used with the electrical equipment.
- Only those operational adjustments actually required by the operator should be accessible to the operator. Access to other controls should be restricted to prevent unauthorized changes in operating characteristics.

About the Book

At a Glance

Document Scope

This Safety Manual describes the modules of the M580 safety system with special regard as to how they meet the Safety requirements of the IEC 61508 standard. It provides detailed information on how to install, run, and maintain the system correctly in order to protect human beings as well as to prevent damage to environment, equipment, and production.

This documentation is intended for qualified personnel familiar with Functional Safety and Unity Pro XLS. Commissioning and operating the M580 Safety System may only be performed by persons who are authorized to commission and operate systems in accordance with established Functional Safety standards.

NOTE:

- The English language version of this manual is the original version.
- In case of a change request or quality issue relating to the M580 Safety offer, please contact your local Customer Care Center for Technical support. You can find more information in the *Support / Contact us* section of your Schneider Electric website at: *http://www.schneider-electric.com/b2b/en/support/*

Validity Note

This document is valid for Unity Pro XLS 13.0 or later.

For product compliance and environmental information (RoHS, REACH, PEP, EOLI, etc.), go to *www.schneider-electric.com/green-premium*.

The technical characteristics of the devices described in this document also appear online. To access this information online:

Step	Action
1	Go to the Schneider Electric home page *www.schneider-electric.com*.
2	In the **Search** box type the reference of a product or the name of a product range. ● Do not include blank spaces in the reference or product range. ● To get information on grouping similar modules, use asterisks (*).
3	If you entered a reference, go to the **Product Datasheets** search results and click on the reference that interests you. If you entered the name of a product range, go to the **Product Ranges** search results and click on the product range that interests you.
4	If more than one reference appears in the **Products** search results, click on the reference that interests you.

Step	Action
5	Depending on the size of your screen, you may need to scroll down to see the data sheet.
6	To save or print a data sheet as a .pdf file, click **Download XXX product datasheet**.

The characteristics that are presented in this manual should be the same as those characteristics that appear online. In line with our policy of constant improvement, we may revise content over time to improve clarity and accuracy. If you see a difference between the manual and online information, use the online information as your reference.

Related Documents

Title of documentation	Reference number
M580 Safety System Planning Guide	QGH60283 (English), QGH60284 (French), QGH60285 (German), QGH60286 (Spanish), QGH60287 (Italian), QGH60288 (Chinese)
Unity Pro Safety Block Library	QGH60275 (English), QGH60278 (French), QGH60279 (German), QGH60280 (Italian), QGH60281 (Spanish), QGH60282 (Chinese)
Modicon Controllers Platform Cyber Security, Reference Manual	EIO0000001999 (English), EIO0000002001 (French), EIO0000002000 (German), EIO0000002002 (Italian), EIO0000002003 (Spanish), EIO0000002004 (Chinese)
Modicon M580, Hardware, Reference Manual	EIO0000001578 (English), EIO0000001579 (French), EIO0000001580 (German), EIO0000001582 (Italian), EIO0000001581 (Spanish), EIO0000001583 (Chinese)
Modicon M580 Standalone System Planning Guide for Frequently Used Architectures	HRB62666 (English), HRB65318 (French), HRB65319 (German), HRB65320 (Italian), HRB65321 (Spanish), HRB65322 (Chinese)
Modicon M580 System Planning Guide for Complex Topologies	NHA58892 (English), NHA58893 (French), NHA58894 (German), NHA58895 (Italian), NHA58896 (Spanish), NHA58897 (Chinese)
Unity Loader, A SoCollaborative, User Manual	33003805 (English), 33003806 (French), 33003807 (German), 33003809 (Italian), 33003808 (Spanish), 33003810 (Chinese)
Unity Pro, Operating Modes	33003101 (English), 33003102 (French), 33003103 (German), 33003104 (Spanish), 33003696 (Italian), 33003697 (Chinese)

Title of documentation	Reference number
Unity Pro, System Bits and Words, Reference Manual	EIO0000002135 (English), EIO0000002136 (French), EIO0000002137 (German), EIO0000002138 (Italian), EIO0000002139 (Spanish), EIO0000002140 (Chinese)

You can download these technical publications and other technical information from our website at *http://www.schneider-electric.com/en/download*.

Chapter 1
M580 Safety Function

M580 Safety Function

Introducing the Schneider Electric M580 Safety Function

Using Unity Pro XLS, you can program, configure and maintain a safety application. When designing and programming your safety application, apply safety functions only to components of a safety loop.

NOTE: Include only safety modules, their configuration settings, and their data in a safety loop.

After commissioning, while your M580 safety system is operating in safety mode, the safety system periodically reads safety inputs, processes the application program safety logic, performs diagnostics, and applies the logic results to safety outputs.

If CPU or I/O diagnostics detect an error, the safety system places the affected part of the system into a safe state. Depending on the nature of the detected error, the scope of the response may place a single I/O channel, an I/O module, or the entire system into the safe state.

The safe state is always the de-energized state. For example:

- If the BMXSAI0410 analog input module or the BMXSDI1602 digital input module detects a dangerous internal condition, it sets the value of their inputs to the CPU to “0” (the de-energized state), which remain in that state until the underlying condition has been resolved.
- If the BMXSDO0802 digital output module or BMXSRA0405 digital relay output module detect a dangerous internal condition, it sets its outputs to the de-energized state, which remain in that state until the underlying condition has been resolved and the module is restarted.
- If the BMXSDO0802 digital output module or BMXSRA0405 digital relay output module detects a communication error on a black channel link to the CPU, the output module sets its outputs to their fallback state.
  NOTE: You can use Unity Pro XLS to configure the fallback state (energized, de-energized, or maintain last value) in the event black channel communication between the CPU and output module is lost.
- If the BMEP584040S CPU detects a communication error on a black channel link to a safety input module, it sets the state of the affected inputs to “0” (the de-energized state) until the black channel again becomes operational and the CPU can again read actual input values.

Safety Loop

A safety loop is the collection of equipment and logic that executes a safety process. A safety project can include multiple safety loops. For each safety loop, you need verify that:

- The process safety time *(see page 139)* is greater than the system reaction time *(see page 139)*.
- The sum of the PFD or PFH values *(see page 133)* for all components in the safety loop does not exceed the maximum permitted value for the intended:
  - safety integrity level (1, 2, 3, or 4)
  - mode of operation (low demand or high demand)
  - proof test interval

Include only safety equipment in a safety loop. Although you can include non-interfering modules *(see page 23)* in your safety project, use them only for non-safe (MAST, FAST, AUX0, or AUX1) tasks.

⚠ WARNING

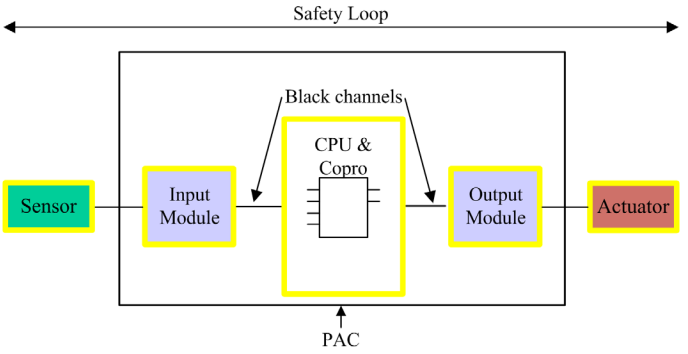
LOSS OF THE ABILITY TO PERFORM SAFETY FUNCTIONS

- Use only safety modules to perform safety functions.
- Do not use inputs or outputs of non-interfering modules for safety-related functions.
- Do not use variables from the Global area for safety-related functions.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Refer to the topic *Data Separation in an M580 Safety Project (see page 150)* for a description of global area variables.

Safety Loop:



Safety equipment includes the following Schneider Electric M580 safety modules:

- BMEP584040S CPU & BMEP58CPROS3 Copro:
  The CPU & Copro together perform the tasks of reading safety inputs, processing safety logic, performing diagnostics, and applying results to outputs. All of these tasks are part of the safety loop. Ports used for black channel communications are also part of the safety loop. However, other CPU components – for example the USB port, SD memory card, and non-volatile static random access memory (nvSRAM) area – are not part of the safety loop.
  NOTE: On both a cold and a warm system start, the CPU & Copro do not load data stored in nvSRAM into the safety task. (nvSRAM data is used only in the non-safe MAST, FAST, and AUX tasks). Instead, the CPU & Copro initially apply default configuration settings from the SD memory card, then apply values received directly from inputs during operation.
- Safety I/O (BMXSAI0410, BMXSDI1602, BMXSDO0802, and BMXSRA0405):
  The functions of sending input signals, receiving output signals, and performing diagnostics are part of the safety loop.
- BMXCPS4002S power supply:
  The safety power supply provides over-voltage detection, and this is part of the safety loop. Because the power supply reliability (i.e. its dangerous failure rate) is more than 100 times better than the threshold for the SIL3 standard, the safety power supply is not included in safety integrity level calculations for the safety loop.

The safety loop also includes the following non-safety equipment:

- Sensors, actuators and the cabling that connects them to safety I/O modules.The safety I/O perform wiring diagnostics for sensors and actuators to help manage the safety loop.
  NOTE: When you design your safety application, you need to identify sensor and actuator characteristics (in particular PFD/PFH values).

Chapter 2
Certification Standards

Certifications

M580 Safety PAC Certification Standards

The M580 safety PAC is certified by TÜV Rheinland Group for use in applications up to and including safety integrity level 3 (SIL3) *(see page 286)*.

Programmable controller specifications

- IEC 61131-2: Programmable controllers- Part 2: Equipment requirements and tests.
- IEC/EN 61010-2-201, UL 61010-2-201, CSA -C22.2 No. 61010-2-201: Safety requirements for electrical equipment - Part 2-201: Particular requirements for control equipment.

Environmental specifications

Refer to "M580 Safety Standards and Certifications" for environment tests levels.

Ex areas specifications

For USA and Canada: Hazardous location class I, division 2, groups A, B, C and D

- CSA 22.2 No213, ANSI/ISA12.12.01 and FM3611

For other countries: CE ATEX (directive 2014/34/EU) or IECEx in defined atmosphere Zone 2 (gas) and/or Zone 22 (dust)

- IEC/EN 60079-0 ; IEC/EN 60079-7; IEC/EN 60079-15

Power utility automation systems specifications

- IEC/EN 61000-6-5: Electromagnetic compatibility - Part 6-5: Generic standards - Immunity for power station and substation environments.
- IEC/EN 61850-3: Communication networks and systems for power utility automation - Part 3: General requirements

Functional safety specifications

- IEC/EN 61000-6-7: Electromagnetic compatibility - Part 6-7: Generic standards - Immunity requirements for equipment intended to perform functions in a safety-related system (functional safety) in industrial locations.
- IEC 61326-3-1: Electrical equipment for measurement, control and laboratory use - Part 3-1: Immunity requirements for safety-related systems and for equipment intended to perform safety-related functions - General industrial application.
- IEC 61508-1: Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 1: General requirements.
- IEC 61508-2: Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 2: Requirements for electrical/electronic/programmable electronic safety related systems.
- IEC 61511-1: Functional safety - Safety instrumented systems for the process industry sector - Part 1: Framework, definitions, system, hardware and software requirements.
- IEC 61511-2: Functional safety - Safety instrumented systems for the process industry sector - Part 2: Guidelines for the application of IEC 61511-1.
- IEC 61511-3: Functional safety - Safety instrumented systems for the process industry sector - Part 3: Guidance for the determination of the required safety integrity levels.

Safety machinery specifications

- IEC/EN 62061: Safety of machinery - Functional safety of safety-related electrical, electronic and programmable electronic control systems.
- o ISO EN 13849-1: Safety of machinery - Safety-related parts of control systems - Part 1: General principles for design.

Functional safety in systems specifications

- EN 54-2: Fire detection and fire alarm systems Part 2: Control and indicating equipment.
- EN 50156-1: Electrical equipment for furnaces and ancillary equipment - Part 1: Requirements for application design and installation.
- EN 50130-4: Alarm systems - Part 4: Electromagnetic compatibility - Product family standard: Immunity requirements for components of fire, intruder, hold up, CCTV, access control and social alarm systems.
- EN 298: Automatic burner control systems for burners and appliances burning gaseous or liquid fuels.
- NFPA 85: Boiler and Combustion Systems Hazards Code.
- NFPA 86: Standard for Ovens and Furnaces.
- NFPA 72: National Fire Alarm and Signaling Code.

Notes:

For the complete list of the standards (with their revisions and dates) which are certified by TÜV, please refer to the TÜV certificate in the web site at: http://www.tuvasi.com.

Chapter 3
M580 Safety System Supported Modules

Introduction

An M580 safety project can include both safety modules and non-safety modules. You can use:

- Safety modules in the SAFE task.
- Non-safety modules only for the non-safe tasks (MAST, FAST, AUX0, and AUX1).
  NOTE: Only non-safety modules that do not interfere with the safety function can be added to a safety project.

Use only the Unity Pro XLS programming software of Schneider Electric for programming, commissioning, and operating your M580 safety application. Unity Pro XLS provides all the functionality of Unity Pro XL and is the exclusive version for programming the M580 safety PAC.

This chapter lists the safety and non-safety modules supported by the M580 safety system.

What Is in This Chapter?

This chapter contains the following topics:

Topic	Page
M580 Safety System Certified Modules	22
Non-Interfering Modules	23

M580 Safety System Certified Modules

Certified Modules

The M580 safety PAC is a safety-related system certified by TÜV Rheinland Group, according to:

- SIL3/IEC 61508/IEC 61511
- SIL CL3/IEC 62061
- PLe, Cat. 4 / ISO 13849-1

It is based on the M580 family of programmable automation controllers (PACs). The following Schneider Electric M580 safety modules are certified:

- BMEP584040S CPU
- BMEP58CPROS3 co-processor
- BMXSAI0410 analog input module
- BMXSDI1602 digital input module
- BMXSDO0802 digital output module
- BMXSRA0405 digital relay output module
- BMXCPS4002S power supply

NOTE: In addition to the safety modules listed above, you can also include non-interfering, non-safety modules *(see page 23)* in your safety project.

You can find the most recent information on the certified product versions on the TÜV Rheinland Group website *http://www.tuvasi.com/* under *Information* and further *List of Type Approved Programmable Electronic Systems*.

Non-Interfering Modules

Introduction

An M580 safety project can include both safety modules and non-safety modules. You can use non-safety modules only for non-safe tasks (MAST, FAST, AUX0, and AUX1). Only non-safety modules that do not interfere with the safety function can be added to a safety project.

Definition of a Non-Interfering Module

⚠ CAUTION

INCORRECT USE OF SAFETY-RELATED DATA

Confirm that neither input data nor output data from non-interfering modules are used for controlling safety-related outputs. Non-safety modules can process only non-safety data.

Failure to follow these instructions can result in injury or equipment damage.

A non-interfering module is a module which cannot interfere with the safety function. There are two types of non-interfering, non-safety modules:

- **Type 1**: A type 1 module can be installed in the same rack as safety modules.
- **Type 2**: A type 2 non-interfering module cannot be installed in the same rack as safety modules.
  NOTE: Suitable installations for a type 2 non-interfering module would be:
- As distributed equipment in an Ethernet DIO network where no safety modules are installed, or
- In an Ethernet RIO drop where no safety modules are installed.

Type 1 Non-Interfering Modules for SIL3 Applications

The following non-safety modules can qualify as type-1 non-interfering modules in an M580 safety system.

NOTE: The list of type-1 non-interfering non-safety modules may change from time to time. For the current list, visit the TÜV Rheinland website at https://fs-products.tuvasi.com.

Module type	Module Reference
Backplane 4 slots	BME XBP 0400
Backplane 8 slots	BME XBP 0800
Backplane 12 slots	BME XBP 1200
Backplane 4 slots	BMX XBP 0400
Backplane 6 slots	BMX XBP 0600
Backplane 8 slots	BMX XBP 0800
Backplane 12 slots	BMX XBP 1200
Backplane 6 slots with dual slots for redundant power supplies	BME XBP 0602

Module type	Module Reference
Backplane 10 slots with dual slots for redundant power supplies	BME XBP 1002
Communication: Performance X80 Ethernet Drop Adapter 1 CH	BMX CRA 312 10
Communication: Performance X80 Ethernet Drop Adapter 1 CH	BME CRA 312 10
Communication: Ethernet module with standard web services	BME NOC 0301
Communication: Ethernet module with IP Forwarding	BME NOC 0321
Communication: Ethernet module with FactoryCast web services	BME NOC 0311
Communication: Rack extender module	BMX XBE 1000
Communication: AS-Interface	BMX EIA 0100
Communication: Global Data	BMX NGD 0100
Communication: Bus module 2 RS485/232 Port	BMX NOM 0200
Communication: Ethernet 1 Port 10/100 RJ45- RTU	BMX NOR 0200
Communication: Fiber Converter MM/LC 2CH 100Mb	BMX NRP 0200
Communication: Fiber Converter SM/LC 2CH 100Mb	BMX NRP 0201
Counting: SSI module 3 CH	BMX EAE 0300
Counting: High speed counter 2 CH	BMX EHC 0200
Counting: High speed counter 8 CH	BMX EHC 0800
Motion: Pulse Train Output 2 independent CH	BMX MSP 0200
Analog: Ana 8 In Current Isolated HART	BME AHI 0812
Analog: Ana 4 Out Current Isolated HART	BME AH0 0412
Analog: Ana 4 U/I In Isolated High Speed	BMX AMI 0410
Analog: Ana 4 U/I In Non Isolated High Speed	BMX AMI 0800
Analog: Ana 8 U/I In Isolated High Speed	BMX AMI 0810
Analog: Ana 4 In U/I 4 Out U/I	BMX AMM 0600
Analog: Ana 2 U/I Out Isolated	BMX AMO 0210
Analog: Ana 4 U/I Out Isolated	BMX AMO 0410
Analog: Ana 8 Out Current No Isolated	BMX AMO 0802
Analog: Ana 4 TC/RTD Isolated In	BMX ART 0414.2
Analog: Ana 8 TC/RTD Isolated In	BMX ART 0814.2
Discrete: Dig 8 In 220 Vac	BMX DAI 0805
Discrete: Dig 8 In 100 to 120 Vac Isolated	BMX DAI 0814
Discrete: Dig 16 In 24Vac/24Vdc Source	BMX DAI 1602
Discrete: Dig 16 In 48Vac	BMX DAI 1603
Discrete: Dig 16 In 100 to 120 Vac	BMX DAI 1604
Discrete: Dig 16 O Triacs	BMX DAO 1605

Module type	Module Reference
Discrete: Dig 16 In 24Vdc Sink	BMX DDI 1602
Discrete: Dig 16 In 48Vdc Sink	BMX DDI 1603
Discrete: Dig 16 In 125Vdc Sink	BMX DDI 1604
Discrete: Dig 32 In 24Vdc Sink	BMX DDI 3202K
Discrete: Dig 64 In 24Vdc Sink	BMX DDI 6402K
Discrete: Dig 8 In 24Vdc 8Q Source Tr	BMX DDM 16022
Discrete: Dig 8 In 24Vdc 8Q Relays	BMX DDM 16025
Discrete: Dig 16 In 24Vdc 16Q Source Tr	BMX DDM 3202K
Discrete: Dig 16Q Trans Source 0.5A	BMX DDO 1602
Discrete: Dig 16 O Trans Sink	BMX DDO 1612
Discrete: Dig 32Q Trans Source 0.1A	BMX DDO 3202K
Discrete: Dig 64Q Trans Source 0.1A	BMX DDO 6402K
Discrete: Dig 8Q 125Vdc	BMX DRA 0804T
Discrete: Dig 8Q Isolated Relays	BMX DRA 0805
Discrete: Dig 16Q Relays	BMX DRA 1605
Discrete: Dig 16In 24/125Vdc TSTAMP	BMX ERT 1604
Mx80 Network Option Switch	BME NOS 0300
Turbomachinery Frequency Input 2 CH	BMX ETM 0200

Type 2 Non-Interfering Modules for SIL2/3 Applications

The following in-rack non-safety modules can be considered to be type-2 non-interfering modules in an M580 safety system.

NOTE: The list of type-2 non-interfering non-safety modules may change from time to time. For the current list, visit the TÜV Rheinland website at https://fs-products.tuvasi.com.

Module type	Module Reference
Communication: Standard X80 Ethernet Drop Adapter 1 CH	BMX CRA 312 00
Standard AC power supply	BMX CPS 2000
Standard Isolated DC power supply	BMX CPS 2010
High Power Isolated 24 to 48 VDC power supply	BMX CPS 3020
Standard Redundant 125VDC power supply	BMX CPS 3522
Standard Redundant 24048VDC power supply	BMX CPS 4022
Standard Redundant AC power supply	BMX CPS 4002
High Power AC power supply	BMX CPS 3500
High Power DC power supply	BMX CPS 3540T

Module type	Module Reference
Weight module	PME SWT 0100
CANopen X80 Master	BME CXM 0100
Partner diagnostic module	PMX CDA 0400
WiFi bridge communication	PMX NOW 0300

NOTE: All authorized equipment of an M580 system that are linked to safety modules via Ethernet are considered as non-interfering. As a consequence, all modules from Quantum and STB Advantys ranges (not pluggable in the same rack as M580 safety modules) are Type 2 non-interfering modules.

Chapter 4
Cyber Security for the M580 Safety System

Cyber Security for the M580 Safety System

Cyber Security Reference

The purpose of a cyber security policy is to reduce, to the greatest extent possible, the vulnerability of your safety system to cyber attacks. For information on developing a cyber security policy for your M580 safety system, refer to the *Modicon Controllers Platform Cyber Security Reference Manual* (Reference Number EIO0000001999 (EN).

Chapter 5
Application Lifecycle

Application Lifecycle

Introduction

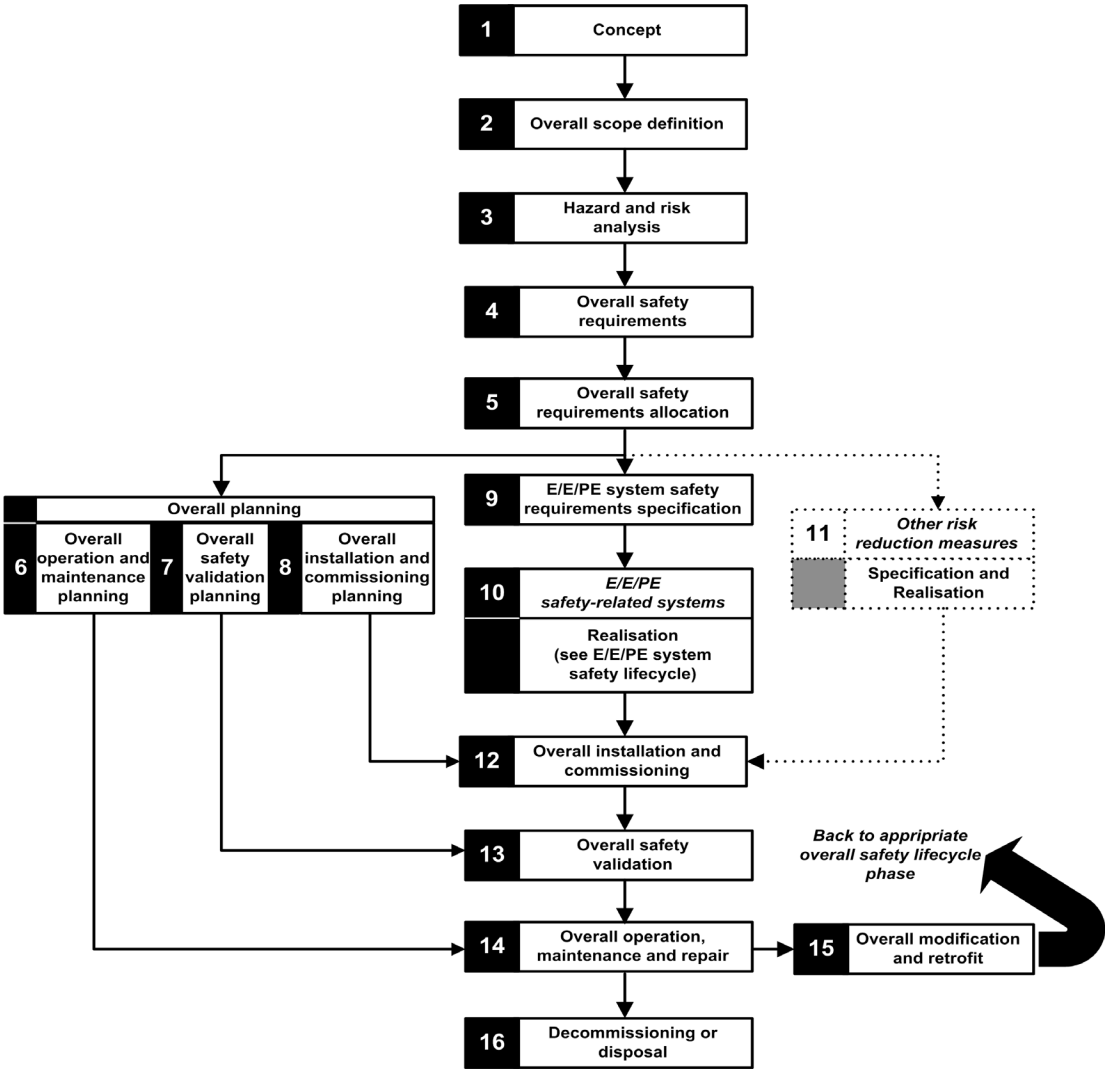
When designing a safe application, you will need to follow the recommendation of one of the safety standard which apply to your application domain. Most of the application standards derive from or are linked to the generic standard IEC 61508 including, for example, the process industry standard (IEC 61511), the machine industry standards (IEC 62061 and ISO 13489), the nuclear industry standard (IEC 61513), ans so forth.

IEC 61508 defines an application life cycle with a sequence of steps. Each step has a defined role, needs mandatory input documents, and produces output documents. The decision to use a safety integrated system (SIS) is made at the end of the Safety Requirements Allocation step (step 5).

This topic defines the necessary checks, related to the usage of a M580 safety system, that you need to perform at the following steps:

9. E/E/PE System safety requirements specification
10. E/E/PE Safety related systems realisation
12. Overall installation and commissioning
13. Overall safety validation
14. Overall operation, maintenance and repair
15. Overall modification and retrofit

The following diagram presents the overall safety lifecycle:



Step 9: E/E/PE System safety requirements specification

This step takes place when the risk analysis is completed and has provided, among other things, the following information:

- Definition of the safety integrated functions
- Their required performances (time, risk reduction, SIL…)
- Their failure modes

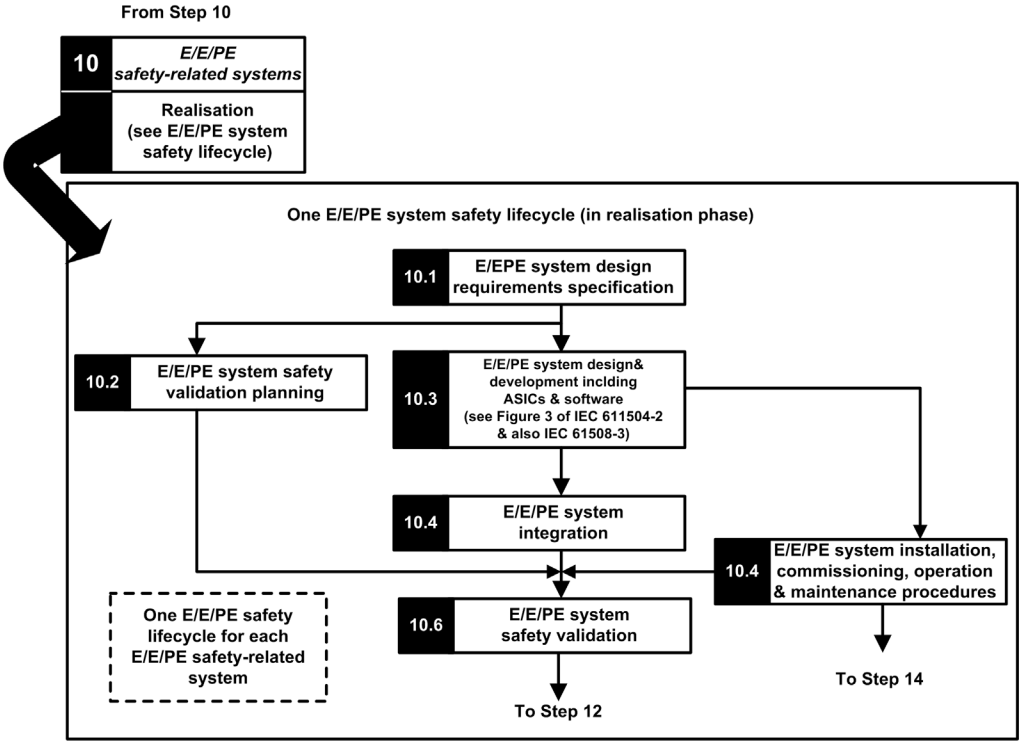
It should produce the safety requirement specifications which will include, at least, the following information necessary to design a safe application using any type of safety PAC:

- Safe state of the safety integrated functions
- SIS operating mode analysis (including the behavior in run, stop, power on sequence, maintenance, repair…)
- Test interval of the SIF
- MTTR of the SIS
- Choice of energized or de-energized SIF
- Performance of the logic solver (reaction time, precision …)
- Performance requirements
  - Fault tolerance
  - Integrity
  - Maximum spurious trip rate
  - Maximum dangerous fault rate
- Environmental specification (EMC, mechanical, chemical, climate…)

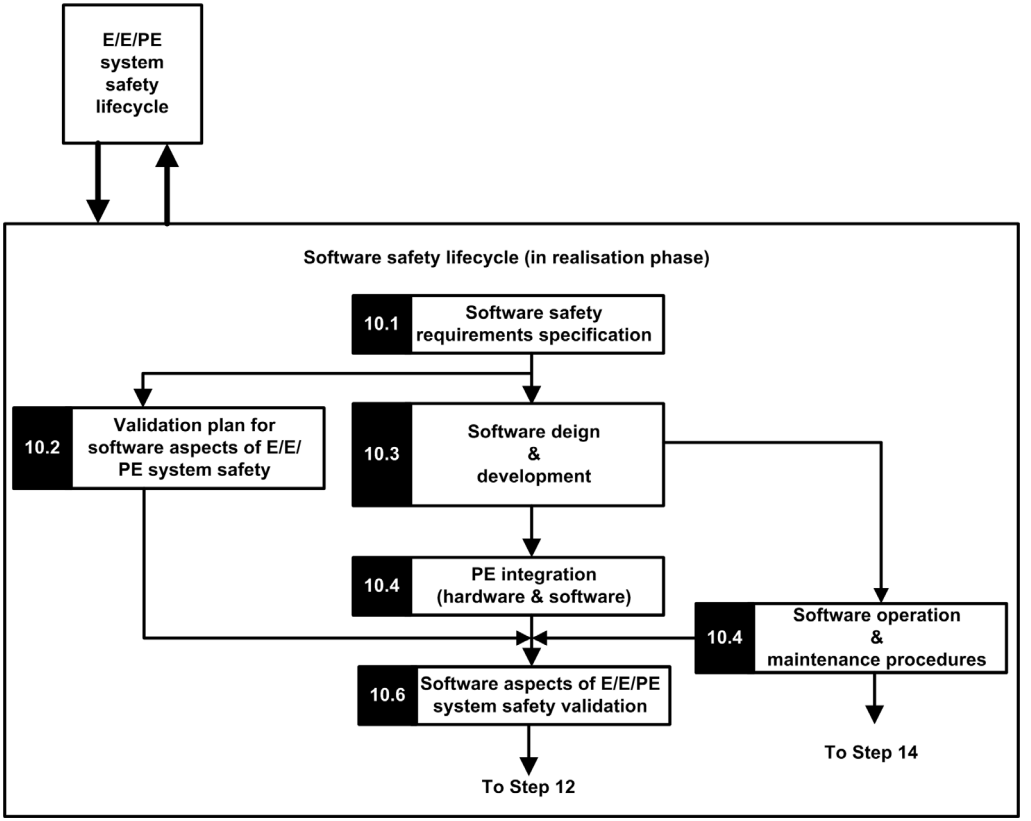
Step 10: E/E/PE Safety related systems realisation

The IEC 61508 divide this step into 2 sub life cycles, one for the system realisation, one for the software realisation.

System realisation:



Software realisation:



The goal of the first sub steps (10.1) is to convert the SIS safety requirements into specification of the hardware design, hardware tests, software design, software tests and integration tests. It should provide at least the following information necessary to design a safe application using safety M580:

- Hardware architecture taking care of:
  - The respect of M580 rules about mixing non safety and safety modules: all the safety modules (safe IO modules and safe CPU/COPRO) are placed in racks where the main rack and its extension are powered by safe power supply and contains only safe modules or non-interfering modules of type 1.
  - Electric consumption per rack.
  - Derating rules.
- Power supply architecture:
  - Only SELV/PELV power supply.

- Software architecture:
  - Including the usage of M580 global variables; a global variable should not prevent a safety action to be triggered unless a “safe application protocol” is used.
- Hardware integration (cabling, cabinet, and so forth):”
  - Fuse protection.
  - Accessories for wire diagnostic.
- Human machine interfaces:
  - Including the usage of M580 global variables; a global variable should not prevent a safety action to be triggered unless a “safe application protocol” is used.
- Electric/numerical interfaces:
  - Safe state.
  - Sensor and actuator.
- Algorithm
- Performances (including task period, watchdog and timeout definition, prediction of a good behavior using the formula:

$$\Sigma_{all\ tasks} \frac{Exe_{task}}{Period_{task}} < 80\%$$

- Behavior in case of:
  - Unlock configuration
  - Maintenance mode
  - Maintenance input
  - Invalid channel
  - Wiring failure
- Management of the UID of the safe IO modules (define when a UID should be changed).
- NTP server:
  - Choice of PAC as NTP server or external FTP server (depending on the usage of I/O time stamping in the process application, safe pier to pier communication, and so forth.)
  - Server redundancy
  - Server loss

The next sub steps refine the specifications into technical detailed specification, perform the design itself execute all the test plans and provides the reports.

Step 12: Overall installation and commissioning

The goal of this step is to define the requirements for installation, task planning, tooling, commissioning procedure and then build the system and verify its correctness.

In an M580 safety system, the commissioning procedure should include the following points:

- Verify Unity pro integrity, verify Unity pro version.
- Correctness of the CPU and Coprocessor firmware versions by supervising the system words %SW14 (Firmware version of PLC processor) and %SW142 (Firmware version of coprocessor).
- Correctness of each module addresses (position in rack, CRA switches).

- Correctness of the cabling:
  - Point to point verification: from internal variable to IO module and to actuator/sensor.
  - Fuses.
  - Equipment for wiring diagnostic.
- At the end of the procedure, all the safety modules are in “lock” mode (it is recommended that the safe application itself checks this condition).
- Correctness of each module configuration (including the timeouts):”
  - Read the configuration using the Unity pro screen and compare to specification.
- All the safety applications have been rebuilt using the “rebuild all project” option, then downloaded to each PLC, and their SAId saved as well as the application archive.
- The task period and task watchdog are correct.
- Module references and version.
- Usage of SELV/PELV only.

Step 13: Overall safety validation

The goal of this step is to prove that the safety integrated system fulfills its requirements. It executes all the tests and produce the reports define in the step 7 of the “safety lifecycle”. It should include:

- Verify that there is no overrun condition during any of the system state (verification of the system bit %S19 in the MAST, FAST, AUX0 tasks, and of the max and current SAFE task execution time (%SW42 and %SW43).
- Verify the good behavior formula: $\Sigma_{all\ tasks} \frac{Exe_{task}}{Period_{task}} < 80\%$
- Verify the special operating modes (module unlock, maintenance input, invalid channel, wiring defect).

Step 14: Overall operation, maintenance and repair

- Execute the proof tests at the right period.
- Monitor the SAId [see note].
  NOTE: As long as the SAId has not changed, the safety portion of the application has not been changed. Refer to the S_SYST_STAT_MX function block for details on SAId behavior.
- Monitor the configuration lock status of each safety module.
- Record the repair operations.
- If a module is replaced, execute (at minimum) the commissioning operations related to this module.
- Record the deviations.

Step 15: Overall modification and retrofit

Any modification should be treated as a new design. An impact analysis may help to define the part of the former safety system that can be kept and the part that must be designed again.

Chapter 6
M580 Safety I/O Modules

Introduction

This chapter describes the M580 safety I/O modules.

What Is in This Chapter?

This chapter contains the following sections:

Section	Topic	Page
6.1	M580 Safety I/O Module Shared Features	38
6.2	BMXSAI0410 Analog Input Module	43
6.3	BMXSDI1602 Digital Input Module	58
6.4	BMXSDO0802 Digital Output Module	88
6.5	BMXSRA0405 Digital Relay Output Module	104

Section 6.1
M580 Safety I/O Module Shared Features

Introduction

This section describes the shared or common features of M580 safety I/O modules.

What Is in This Section?

This section contains the following topics:

Topic	Page
Introducing the M580 Safety I/O Modules	39
Diagnostics Overview for M580 Safety I/O Modules	41

Introducing the M580 Safety I/O Modules

Introduction

The following four M580 safety I/O modules are certified for use in safety applications:

- BMXSAI0410 (Analog Input)
- BMXSDI1602 (Digital Input)
- BMXSDO0802 (Digital Output)
- BMXSRA0405 (Digital Relay Output)

Use the four safety I/O modules to connect the safety PAC to the sensors and actuators that are part of the safety loop. Each safety I/O module incorporates a dedicated safety processor. You can install these I/O modules in the local backplane or in RIO drops.

Installation and Housing Requirements

Install your M580 safety equipment in a manner that meets:

- The IEC 60950 pollution degree 2 standard for the safety of information technology equipment; and
- IEC 60529 standard for IP54 ingress protection, so that:
  - the presence of dust does not interfere with equipment operation, and
  - splashing water has no harmful effect on the equipment or operations.

Typically these standards are accomplished by placing the safety equipment in a housing, such as a cabinet.

Maximum Operation Altitude

The maximum operating altitude for the M580 safety I/O modules is 2000 m above sea level.

Communication Between the PAC and I/O

The M580 safety CPU and Copro together control all backplane exchanges, while the safety I/O respond to the commands of the CPU and Copro. Safety I/O modules can be installed in either a BMXXBP•••• X Bus rack or a BMEXBP•••• Ethernet rack.

Communications between the safety PAC and safety I/O modules in the local main rack are made via the backplane.

Communications between the safety PAC and safety I/O modules installed in an RIO drop are made through an adapter module installed on the RIO drop, either:

- a BMECRA31210 adapter, for an Ethernet rack, or
- a BMXCRA31210 adapter, for an X Bus rack.

NOTE: A BMXCRA31200 adapter cannot be used to connect safety I/O modules to the M580 safety PAC.

Optionally, you can use BMXNRP0200 or BMXNRP0201 fiber optic repeater modules to extend the physical link between the CPU and Copro in the local rack and the adapter in the RIO drop. Fiber optic repeater modules enhance RIO network noise immunity and increase cabling distance while maintaining the full dynamic range of the network and the safety integrity level.

The communication protocol between the safety I/O and PAC enables their exchanges. It permits both devices to check the accuracy of received data, detect corrupted data, and determine if the transmitting module becomes non-operational. Thus, a safety loop may include any non-interfering *(see page 23)* RIO adapters and backplane.

External Power Supply Used with Digital Safety I/O

The BMXSDI1602 and BMXSDO0802 digital modules require an 24 Vdc protected extra low voltage (PELV) external power supply to provide power to sensors and actuators. The safety I/O modules supervise the non-safety process power supply for overvoltage and undervoltage conditions.

⚠ ⚠ DANGER

PELV OVERVOLTAGE CATEGORY II POWER SUPPLY REQUIRED

Use only a PELV type overvoltage category II power supply, with a maximum output of 60 Vdc, to supply power to sensors and actuators.

Failure to follow these instructions will result in death or serious injury.

Diagnostics Overview for M580 Safety I/O Modules

Introduction

Each M580 safety I/O module presents the following diagnostic features:

- Self-test at module start-up
- Continuous built-in runtime self-test
- Module and channel diagnostic LEDs

In addition, the digital safety I/O modules also perform wiring diagnostics.

Power On Self Test

At power up, the I/O modules perform an extended series of power on self-tests. If the result of these tests are:

- Successful: The modules are considered to be healthy and are operational.
- Unsuccessful: The modules are not considered to be healthy and are not operational. In this case, the inputs are set to 0, and the outputs are de-energized.

NOTE: If the 24 Vdc external power supply is not connected to a digital input or digital output module, the power on self-tests are not performed and the module does not start.

Continuous Built-In Tests

During runtime, the I/O modules continuously perform self-tests. The input modules verify that they are able to read data from the sensors over the complete range. The output modules verify that the actual state of the output is the same as the commanded state.

LEDs

Each safety I/O module provides module and channel LED diagnostics on the front face of the module:

- The top four LEDs (**Run**, **Err**, **I/O**, and **Lck**) together describe the state of the module.
- The bottom two or four (depending on the module) rows of LEDs combine with the top four LEDs to describe the state and health of each input or output channel.

Refer to the LED diagnostics topic for the following safety I/O modules for more information on the how to read the LEDs for that module:

- BMXSAI0410 safety analog input module *(see page 196)*
- BMXSDI1602 safety digital input module *(see page 202)*
- BMXSDO0802 safety digital output module *(see page 208)*
- BMXSRA0405 safety digital relay output module *(see page 214)*

Wiring Diagnostics for Digital Modules

Both the safety digital input module and the safety digital output module can detect the following channel wiring diagnostic conditions:

- Open (or cut) wire.
- Short circuit to the 0 V ground.
- Short circuit to the 24 Vdc (if sensor power is internally provided).
- Crossed circuits between two channels (if sensor power is internally provided).

NOTE: The availability of these diagnostic functions depends on the specific wiring design of the module to its field devices. Refer to the application wiring examples for the following safety digital I/O modules for more information:

- BMXSDI1602 safety digital input module *(see page 65)*
- BMXSDO0802 safety digital output module *(see page 94)*

Section 6.2
BMXSAI0410 Analog Input Module

Introduction

This section describes the BMXSAI0410 M580 safety analog input module.

What Is in This Section?

This section contains the following topics:

Topic	Page
BMXSAI0410 Safety Analog Input Module	44
BMXSAI0410 Wiring Connector	46
BMXSAI0410 Input Application Wiring Examples	49
BMXSAI0410 Data Structure	55

BMXSAI0410 Safety Analog Input Module

Introduction

The BMXSAI0410 safety analog input module presents the following features:

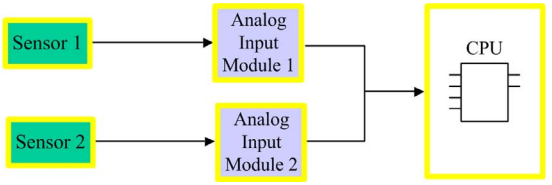
- 4 isolated analog 4...20 mA current input channels.
- 16 bit resolution, spanning the data range of 0...25 mA.
- Current out of range detection, for current values less than 3.75 mA or greater than 20.75 mA.
- Supports the following SIL3 (IEC61508) standards:
  - Category 2 (Cat2) / Performance Level d (PLd) is achieved using 1 input channel (one-out-of-one (1oo1D) evaluation).
  - Cat4/PLe is achieved using 2 input channels (one-out-of-two (1oo2) evaluation).
- LED diagnostic *(see page 196)* display provided for the module and for each input channel.
- Module hot swap during runtime.
- Module CCOTF when operating in maintenance mode *(see page 224)*. (CCOTF is not supported in safety mode *(see page 223)*.)

High Availability

You can design your safety application to varying levels of performance and availability, by using single or redundant input channels and modules, as follows:

Design:	**Safety Function Levels:**			
Input Channels => Modules	**SIL**	**Cat**	**PL**	**High Availability?**
Single input channel to single input module *(see page 50)*	SIL3	Cat 2	PLd	–
Single input channel to redundant input modules *(see page 51)*	SIL3	Cat 2	PLd	✓
Redundant input channels to single input module *(see page 52)*	SIL3	Cat 4	PLe	–
Redundant input channels to redundant input modules *(see page 53)*	SIL3	Cat 4	PLe	✓
✓ : Provided – : Not provided				

The following figure illustrates the redundant analog input configuration:



The analog input current value from sensor 1 and sensor 2 are sent by input module 1 and input module 2, respectively, over a black channel to a safety CPU. The CPU executes a dedicated function block, (`S_AIHA`, in each of two separate, compiled logic programs to manage and select data from the two input modules. This function block operates as follows:

- If the health status of the input data coming from module 1 is OK, the input data from this module is used in the safety function.
- If the health status of the input data coming from module 1 is not OK, but the health status of the input data coming from module 2 is OK, the input data from module 2 is used.
- If the health status of the input data from both module 1 and module 2 is not OK, then the system activates the safety function.

BMXSAI0410 Wiring Connector

Introduction

The BMXSAI0410 analog input module includes 4 analog inputs. The module presents two pair of pins – two positive channel (Ch) pins and two negative common (Com) pins – for each input.

For each input:

- the two channel pins (Ch*n*) are internally connected, and
- the two common pins (Com*n*) also are internally connected.

To connect an analog sensor to an input, you can use either channel pin and either common pin for that input.

Terminal Blocks

You can use the following Schneider Electric 20-point terminal blocks to fit the 20 pin connector on the front of the module:

- screw clamp terminal block BMXFTB2010
- age clamp terminal block BMXFTB2000
- spring type terminal block BMXFTB2020

NOTE: Terminal blocks can be removed only when power to the module is OFF.

Process Power Supply

A 24 Vdc protected extra low voltage (PELV) overvoltage category II process power supply is required. Schneider Electric recommends a power supply that does not automatically restore power after a power interruption.

⚠ DANGER

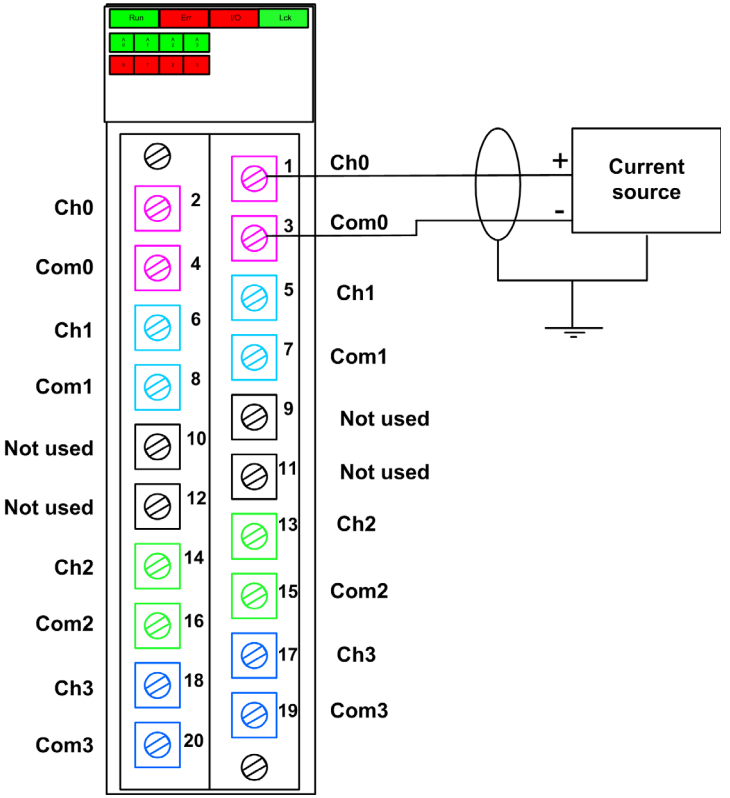
LOSS OF THE ABILITY TO PERFORM SAFETY FUNCTIONS

Use only a PELV type process power supply with a maximum output of 60 V.

Failure to follow these instructions will result in death or serious injury.

Wiring Connector

The following example shows you how to wire a single input on the module:



NOTE: The module detects a cut wire condition and reports it as a current out of range condition (less than 3.75 mA) by setting the `OOR` element of the `T_U_ANA_SIS_CH_IN` *(see page 57)* structure to “1”.

Mapping Inputs to Connector Pins

The following provides a description of each pin on the BMXSAI0410 analog input module:

Pin Description	Pin Number on Terminal Block		Pin Description
Input (+) of channel 0	2	1	Input (+) of channel 0
Input (–) of channel 0	4	3	Input (–) of channel 0
Input (+) of channel 1	6	5	Input (+) of channel 1
Input (–) of channel 1	8	7	Input (–) of channel 1
Not used	10	9	Not used
Not used	12	11	Not used
Input (+) of channel 2	14	13	Input (+) of channel 2
Input (–) of channel 2	16	15	Input (–) of channel 2
Input (+) of channel 3	18	17	Input (+) of channel 3
Input (–) of channel 3	20	19	Input (–) of channel 3

NOTE: Because the two positive pins for each input are internally connected, you need to use only one positive pin for an input channel. Similarly, because the two negative pins for each input are internally connected, you need to use only one negative pin for each input channel.
For example, to connect an analog sensor to input channel 0, you can connect:

- Positive wire of the sensor to either pin 1 or pin 2.
- Negative wire of the sensor to either pin 3 or pin 4.

BMXSAI0410 Input Application Wiring Examples

Introduction

You can wire the BMXSAI0410 safety analog input module to analog sensors to achieve SIL3 compliance in several different ways, depending on:

- the required Category (Cat2 or Cat4) and Performance Level (PLd or PLe) standard
- your application's requirements for high availability

⚠ CAUTION

RISK OF UNINTENDED OPERATION

The maximum safety integrity level (SIL) is determined by the quality of sensor and the length of the proof-test interval to IEC 61508. If you are using sensors that do not meet the quality of the intended SIL standard, always wire these sensors redundantly to two channels.

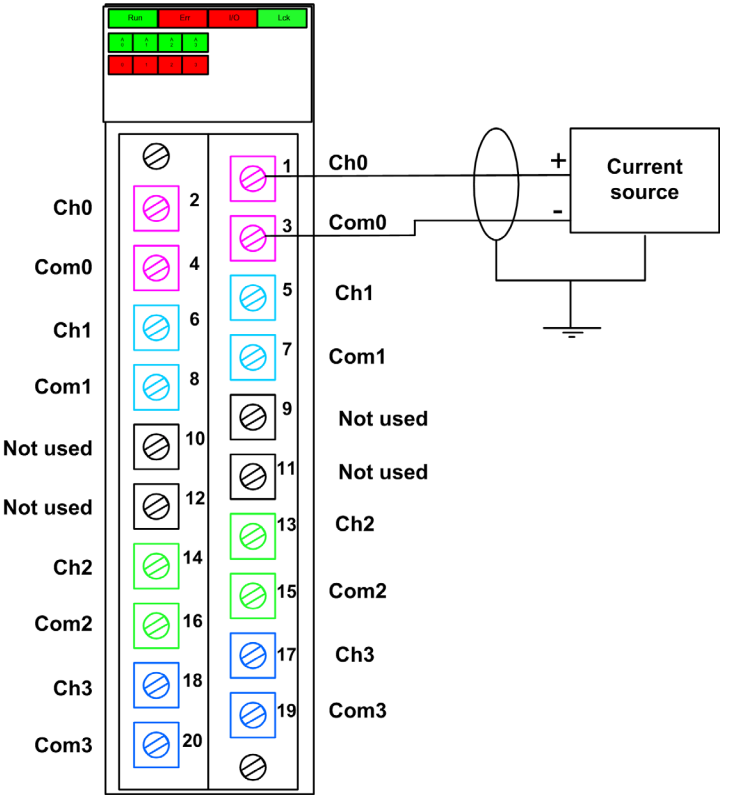
Failure to follow these instructions can result in injury or equipment damage.

The following SIL3 digital input application wiring examples are described, below:

- Cat2/PLd:
  - a single sensor wired to one input.
- Cat2/PLd with high availability:
  - two sensors wired to two input points on different input modules.
- Cat4/PLe:
  - two sensors, each wired to a different input point on the same input module.
- Cat4/PLe with high availability:
  - two pair of sensors (for a total of four sensors): the sensors of the first pair are each wired to a different input point on one module, and the sensors of the second pair are each wired to a different input point on a second module.

SIL3 Cat2/PLd

The following example presents a single sensor wired to one input point on a single input module. The CPU performs 1oo1D evaluation on the single monitored value:



⚠ CAUTION

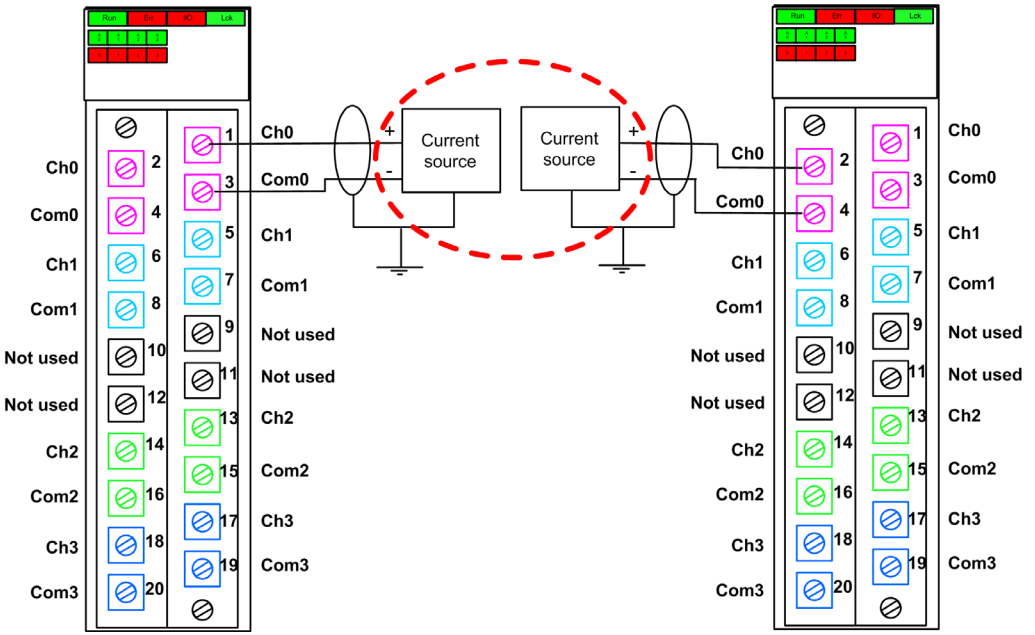
RISK OF UNINTENDED OPERATION

To achieve SIL3 according to IEC61508 and Category 2/Performance Level d according to ISO13849 using this wiring design, you must use a suitable, qualified sensor.

Failure to follow these instructions can result in injury or equipment damage.

SIL3 Cat2/PLd with High Availability

The following example presents two sensors that monitor the same process variable. Each sensor is connected to a single input point on different input modules. The CPU performs 1OO1D evaluation of the single monitored value:



NOTE: In this design, use the `S_AIHA` function block in the SAFE task to manage the two process variable values reported by the two sensors.

⚠ CAUTION

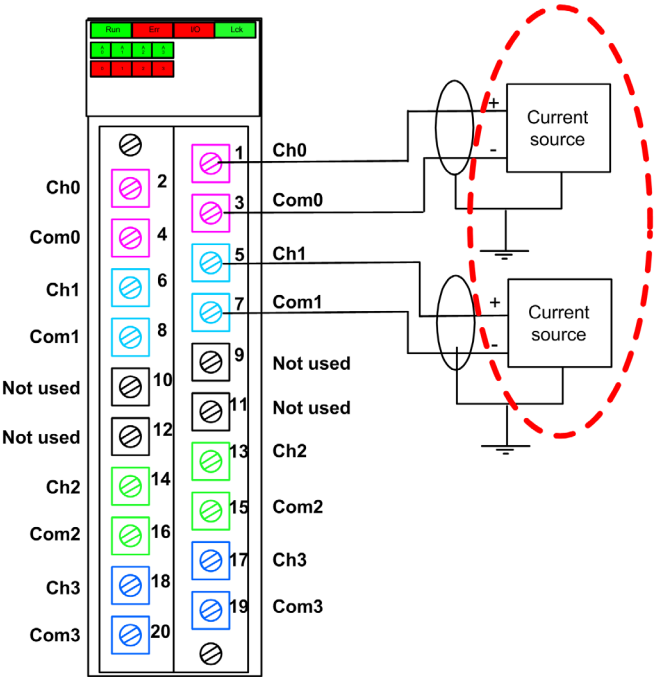
RISK OF UNINTENDED OPERATION

To achieve SIL3 according to IEC61508 and Category 2/Performance Level d according to ISO13849 using this wiring design, you must use a suitable, qualified sensor.

Failure to follow these instructions can result in injury or equipment damage.

SIL3 Cat4/PLe

The following example presents two sensors that monitor the same process variable. Each sensor is connected to a single input point on the same input module. The CPU performs 1oo2 evaluation of the competing values provided by the two sensors for the same process variable:

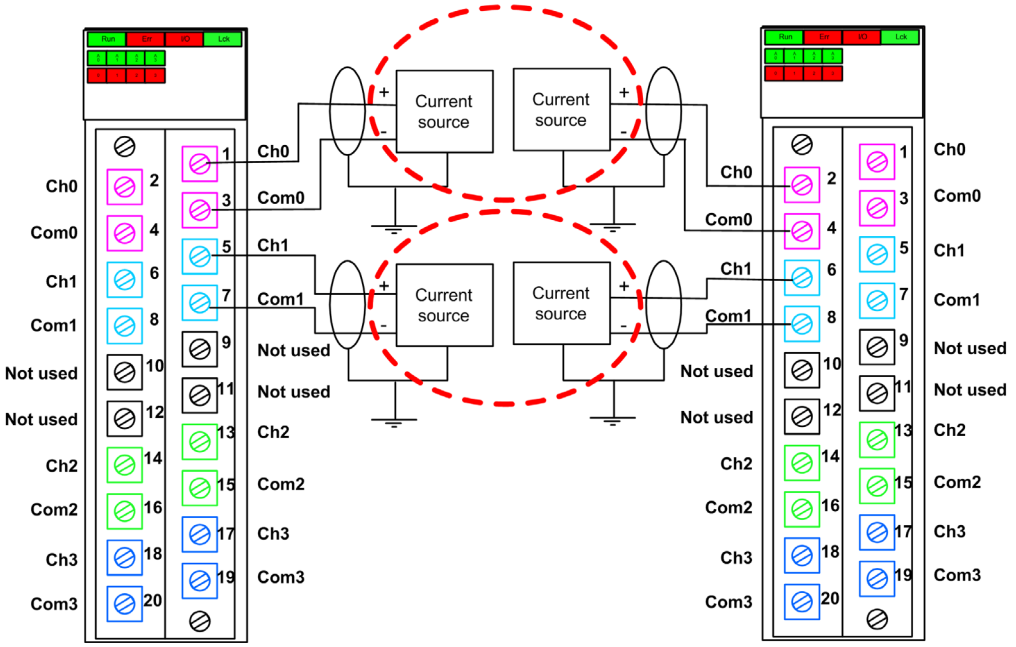


NOTE: In this design, use the `S_AI_COMP` function block in the SAFE task to perform the 1oo2 evaluation of the competing values coming from the two sensors.

⚠ CAUTION
RISK OF UNINTENDED OPERATION To achieve SIL3 according to IEC 61508 and Category 4/Performance Level e according to ISO13849 using this wiring design, you must use a suitable, qualified sensor. **Failure to follow these instructions can result in injury or equipment damage.**

SIL3 Cat4/PLe with High Availability

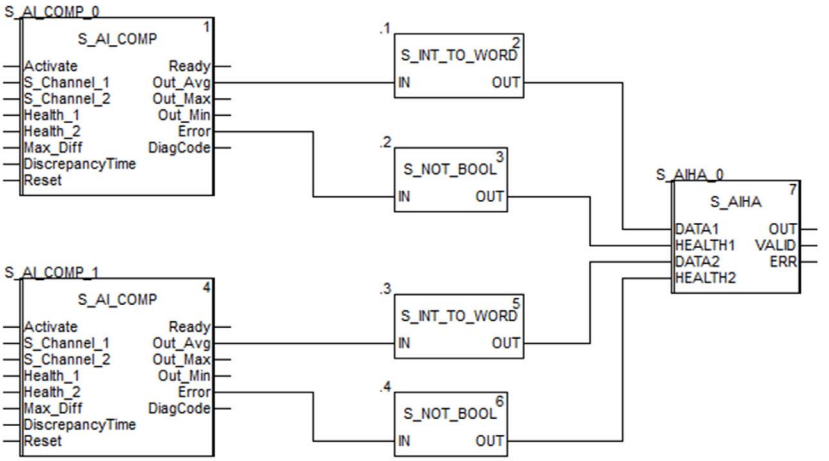
The following example presents two pair of redundant sensors, which monitor the same process variable. Each sensor is connected to a single input point on two different input modules (two inputs on each module). This design makes it possible for the CPU to perform 1oo2 evaluation:



NOTE: In this design, you need to use the `S_AI_COMP` and `S_AIHA` function blocks inside the SAFE task to manage the four input signals:

- `S_AI_COMP` to perform 1oo2 evaluation of two pairs of values coming from both sensors connected to the same module.
- `S_AIHA` to manage the high availability feature.

The following function block diagram depicts the above referenced code segment design:



⚠ CAUTION
RISK OF UNINTENDED OPERATION To achieve SIL3 according to IEC 61508 and Category 4/Performance Level e according to ISO13849 using this wiring design, you must use a suitable, qualified sensor. **Failure to follow these instructions can result in injury or equipment damage.**

BMXSAI0410 Data Structure

Introduction

The T_U_ANA_SIS_IN_4 device derived data type (DDDT) is the interface between the BMXSAI0410 analog input module and the application running in the CPU. The T_U_ANA_SIS_IN_4 DDDT incorporates the data types T_SAFE_COM_DBG_IN and T_U_ANA_SIS_CH_IN.

All of these structures are described, below.

T_U_ANA_SIS_IN_4 DDDT Structure

The T_U_ANA_SIS_IN_4 DDDT structure includes the following elements:

Element	Data Type	Description	Access
MOD_HEALTH[1]	BOOL	• 1: The module is operating correctly. • 0: The module is not operating correctly.	RO
SAFE_COM_STS[1]	BOOL	• 1: Module communication is valid. • 0: Module communication is not valid.	RO
S_COM_DBG	T_SAFE_COM_DBG_IN	Safe communication debug structure.	RO
CONF_LOCKED	BOOL	• 1: Module configuration is locked. • 0: Module configuration is not locked.	RO
CH_IN	ARRAY[0...3] of T_U_ANA_SIS_CH_IN	Array of structure of channel.	–
MUID[2]	ARRAY[0...3] of DWORD	Module unique ID (auto-assigned by Unity Pro)	RO
RESERVED	ARRAY[0...9] of INT	–	–

1. When the SAFE task on CPU is not in running mode, the data exchanged between the CPU and the module are not updated and MOD_HEALTH and SAFE_COM_STS are set to 0.
2. This auto-generated value can be changed by executing the **Build → Renew Ids & Rebuild All** command in the Unity Pro main menu.

T_SAFE_COM_DBG_IN Structure

The T_SAFE_COM_DBG_IN structure includes the following elements:

Element	Data Type	Description	Access[1]
S_COM_EST	BOOL	• 1: Communication with the module is established. • 0: Communication with the module is not established or corrupted.	RO
M_NTP_SYNC	BOOL	• 1: The module is synchronized with the NTP server. • 0: The module is not synchronized with the NTP server.	RO
CPU_NTP_SYNC	BOOL	• 1: The CPU is synchronized with the NTP server. • 0: The CPU is not synchronized with the NTP server.	RO
CHECKSUM	BYTE	Communication frame checksum.	RO
COM_DELAY	UINT	Communication delay between two values received by the module: • 0: No delay. The communication was received on time. • 1...65534: The time, in ms, since the last communication was received by the CPU from the module. • 65535: The CPU did not receive a communication from the module.	RO
COM_TO	UINT	Communication time-out value for communications coming from the module. **NOTE:** You may want to edit this read/write value to equal or exceed the actual communication time for the module (for example, in a remote RIO drop).	R/W
STS_MS_IN	UINT	NTP timestamp value for the fraction of a second, to the nearest ms, of the data received from the module.	RO
S_NTP_MS	UINT	NTP time value for the fraction of a second, to the nearest ms, for the current cycle.	RO
STS_S_IN	UDINT	NTP timestamp value in seconds of the data received from the module.	RO
S_NTP_S	UDINT	NTP time value in seconds for the current cycle.	RO
CRC_IN	UDINT	CRC value for data received from the module.	RO

T_U_ANA_SIS_CH_IN Structure

The T_U_ANA_SIS_CH_IN structure includes the following elements:

Element	Data Type	Description	Access
FCT_TYPE	WORD	• 1: The channel is enabled. • 0: The channel is not enabled.	RO
CH_HEALTH[1]	BOOL	• 1: The channel is operational. • 0: An error has been detected on the channel, which is not operational.	RO
VALUE	INT	Analog input value.	RO
OOR	BOOL	• 1: The channel input current value is out of range, either: ❍ <3.75 mA ❍ >20.75 mA • 0: The channel input current value is not out of range.	RO
IC	BOOL	• 1: Invalid channel detected by the module. • 0: The channel is declared internally operational by the module.	RO
1. When the SAFE task on CPU is not in running mode, the data exchanged between the CPU and the module are not updated and CH_HEALTH is set to 0.			

Section 6.3
BMXSDI1602 Digital Input Module

Introduction

This section describes the BMXSDI1602 M580 safety digital input module.

What Is in This Section?

This section contains the following topics:

Topic	Page
BMXSDI1602 Safety Digital Input Module	59
BMXSDI1602 Wiring Connector	61
BMXSDI1602 Input Application Wiring Examples	65
BMXSDI1602 Data Structure	85

BMXSDI1602 Safety Digital Input Module

Introduction

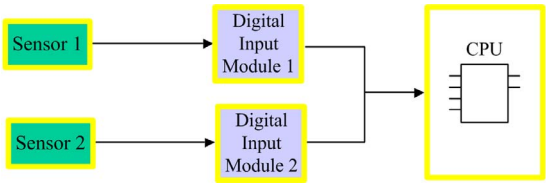
The BMXSDI1602 safety input module presents the following features:

- 16 Type 3 (IEC61131-2) inputs, in two electrically non-isolated groups of 8 inputs.
- 24 Vdc rated input voltage.
- Supports the following SIL3 (IEC61508) standards:
  - Category 2 (Cat2) / Performance Level d (PLd) is achieved using 1 input channel (one-out-of-one (1oo1D) evaluation).
  - Cat4/PLe is achieved using 2 input channels (one-out-of-two (1oo2) evaluation).
- Compatible with 2 or 3 wire proximity sensors.
- Provides optionally two 24 Vdc outputs (VS1 and VS2) for short-circuit to 24 Vdc supervision:
  - VS1 to monitor short-circuit on inputs 0...3 (rank A & B).
  - VS2 to monitor short-circuit on inputs 4...7 (rank A & B).
- Monitor external 24 Vdc sensor supply voltage.
- LED diagnostic display *(see page 202)* provided for the module and for each input channel.
- Configurable (enable/disable) channel wiring diagnostics that can detect the following conditions:
  - Open (or cut) wire.
  - Short circuit to the 0 V ground.
  - Short circuit to the 24 Vdc (if sensor power is internally provided).
  - Crossed circuits between two channels (if sensor power is internally provided).
- Module hot swap during runtime.
- Module CCOTF when operating in maintenance mode *(see page 224)*. (CCOTF is not supported in safety mode *(see page 223)*.)

High Availability

You can use two sensors connected two different input channels located on different input modules to monitor the same physical value, and thereby increase system availability.

The following figure illustrates the redundant digital input configurations:



The input state value from sensor 1 and sensor 2 are sent by input module 1 and input module 2, respectively, over a black channel to a safety CPU. The CPU executes a dedicated function block, S_DIHA, to manage and select the data from the two input modules. This function block operates as follows:

- If the health status of the input data coming from module 1 is OK, the input data from this module is used in the safety function.
- If the health status of the input data coming from module 1 is not OK, but the health status of the input data coming from module 2 is OK, the input data from module 2 is used.
- If the health status of the input data from both module 1 and module 2 is not OK, then the state of the input is set to the safe state (“0”) in order to activate the safety function.

Refer to the description of input application wiring examples *(see page 65)* for details on how to wire the module for high availability.

BMXSDI1602 Wiring Connector

Introduction

The BMXSDI1602 digital input module presents 16 inputs in two groups of 8 inputs. The first group consists of inputs 0...3 (rank A & B), the second group consists of inputs 4...7 (rank A & B). There is no isolation between these two groups.

Power can be provided to the sensors either directly from the external power supply, or internally via the VS1 and VS2 power supplies. Each design is presented, below.

Terminal Blocks

You can use the following Schneider Electric 20-point terminal blocks to fit the 20 pin connector on the front of the module:

- screw clamp terminal block BMXFTB2010
- age clamp terminal block BMXFTB2000
- spring type terminal block BMXFTB2020

NOTE: Terminal blocks can be removed only when power to the module is OFF.

Process Power Supply

A 24 Vdc protected extra low voltage (PELV) overvoltage category II process power supply is required. Schneider Electric recommends a power supply that does not automatically restore power after a power interruption.

⚠ DANGER

LOSS OF THE ABILITY TO PERFORM SAFETY FUNCTIONS

Use only a PELV type process power supply with a maximum output of 60 V.

Failure to follow these instructions will result in death or serious injury.

Fuse

A fast blow fuse is required to help protect the external power supply against short-circuit and over voltage conditions.

NOTICE

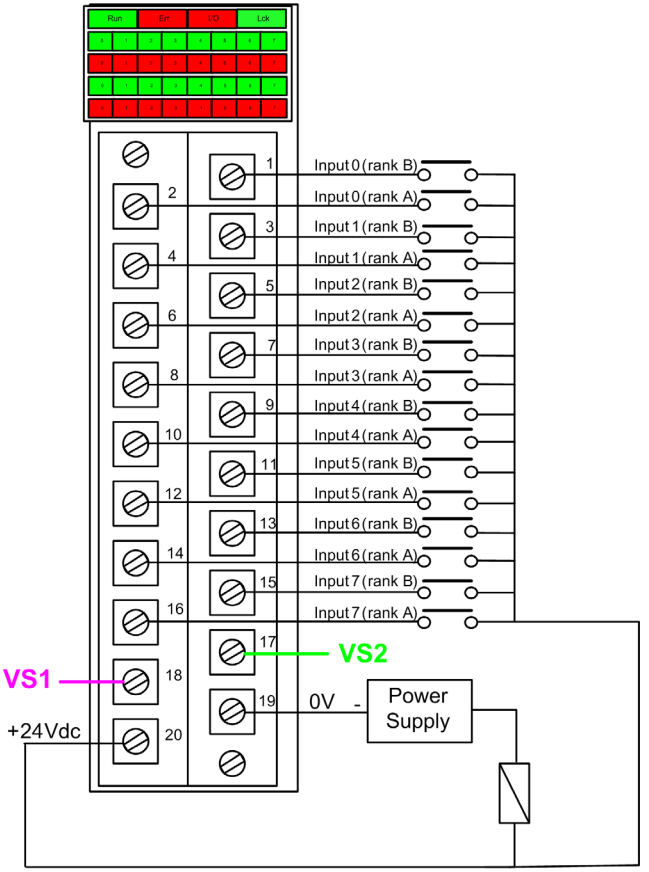
IMPROPER FUSE SELECTION

Use fast acting fuses to help protect the electronic components of the digital input module from an over current condition. Improper fuse selection can result in damage to the input module.

Failure to follow these instructions can result in equipment damage.

Wiring Connector: Sensors Supplied with External Power

In the following design, the sensors are powered directly from an external power supply:



NOTE: Powering the sensors externally limits the channel diagnostics the module can perform. In this wiring design, the module can detect:

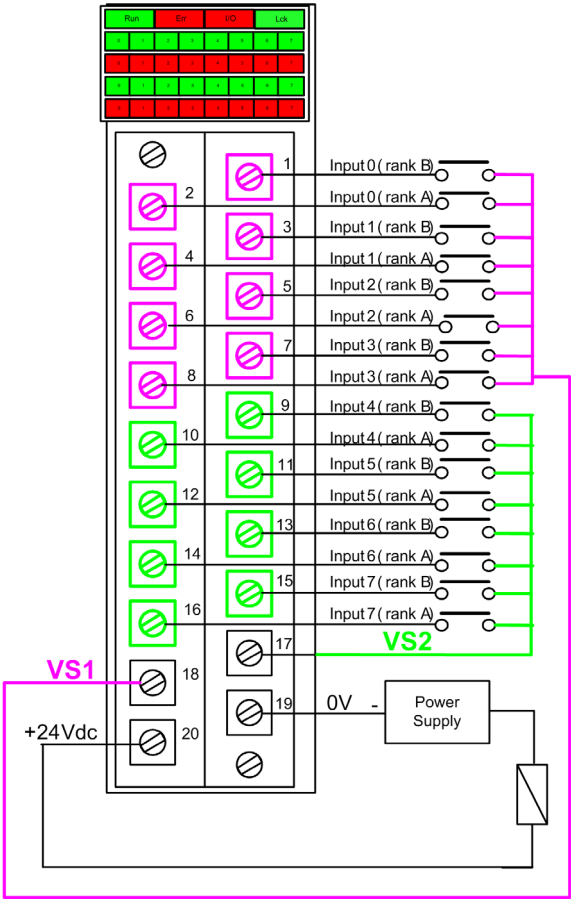
- A cut (or open) wire condition (if enabled for the channel in Unity Pro).
- A short circuit to ground condition.

However, in this design, the module does not detect:

- A short circuit to 24 Vdc condition.
- A cross circuit condition with other wiring input.

Wiring Connector: Sensors Supplied with Internal VS Power

In the following design, sensors for channels 0...3 are supplied by the monitored VS1 power supply and sensors for channels 4...7 are supplied by the monitored VS2 power supply:



If you use this design, apply internal power to the channel groups as follows:

- Use VS1 to supply power to channels 0...3 (rank A and B).
- Use VS2 to supply power to channels 4...7 (rank A and B).

NOTE: In this design, the module can detect:

- A short circuit to 24 Vdc condition (if enabled for the channel in Unity Pro).
- A cross circuit condition with other wiring input.
- A cut (or open) wire condition (if enabled for the channel in Unity Pro).
- A short circuit to ground condition.

Mapping Inputs to Connector Pins and Unity Pro Channels

The following provides a description of each pin on the BMXSDI1602 input module, and maps each pin to the channel for that pin as it appears in the channel **Configuration** tab for the module in Unity Pro XLS:

Unity Pro Channel	Pin Description	Pin Number on Terminal Block		Pin Description	Unity Pro Channel
0	Input 0 (rank A)	2	1	Input 0 (rank B)	8
1	Input 1 (rank A)	4	3	Input 1 (rank B)	9
2	Input 2 (rank A)	6	5	Input 2 (rank B)	10
3	Input 3 (rank A)	8	7	Input 3 (rank B)	11
4	Input 4 (rank A)	10	9	Input 4 (rank B)	12
5	Input 5 (rank A)	12	11	Input 5 (rank B)	13
6	Input 6 (rank A)	14	13	Input 6 (rank B)	14
7	Input 7 (rank A)	16	15	Input 7 (rank B)	15
–	VS1 Power Supply	18	17	VS2 Power Supply	–
–	24 Vdc Process Power Supply	20	19	24 Vdc Process Power Supply	–

BMXSDI1602 Input Application Wiring Examples

Introduction

You can wire the BMXSDI1602 safety digital input module to sensors to achieve SIL3 compliance in several different ways, depending on:

- the required Category (Cat2 or Cat4) and Performance Level (PLd or PLe) standard
- your application's requirements for high availability

⚠ CAUTION

RISK OF UNINTENDED OPERATION

The maximum safety integrity level (SIL) is determined by the quality of sensor and the length of the proof-test interval to IEC 61508. If you are using sensors that do not meet the quality of the intended SIL standard, always wire these sensors redundantly to two channels.

Failure to follow these instructions can result in injury or equipment damage.

The following SIL3 digital input application wiring examples are described, below:

- Cat2/PLd:
  - a single sensor wired to one input
- Cat2/PLd with high availability:
  - a single sensor wired to two input points on different input modules
  - two sensors wired to two input points on different input modules
- Cat4/PLe:
  - a single sensor wired to two input points on the same input module
  - two sensors, each wired to a different input point on the same input module
- Cat4/PLe with high availability:
  - two sensors, each wired to two input different input points on different input modules

Configurable Wiring Diagnostics in Unity Pro

For the BMXSDI1602 safety digital input module, use its **Configuration** page in Unity Pro to:

- Enable **Short circuit to 24V detection** for each energized channel. This test performs the following actuator wiring diagnostics for a channel:
  - Short circuit to 24 Vdc detection.
  - Crossed circuit detection between two output channels.

  The principle of the diagnostic is to measure a leakage current (Ileakage) on the line. If 0.4 mA <Ileakage < 1.3 mA, the external line is considered as cut (or in a short-circuit to ground condition). The diagnostic is performed with a period less than 10 ms.
  - For a dry contact sensor, we recommend that you set in parallel with the sensor a 33 K Ω resistor.
  - Using DDP 2 or 3 wires, the leakage current needs to fall within the limits defined above. You need to define the value of the resistor to set in parallel with the sensor, considering the natural leakage current of the sensor and the internal resistor of the input (7.5 K Ω).
- Enable **Open wire detection** for each of eight channels, which performs the following wiring diagnostics for that channel:
  - Open (or cut) wire detection (i.e. the output channel is not connected to the actuator).
  - Short circuit wiring detection to the 0 Vdc ground.

  The principle is to provide power to the sensors, by group of 8 channels (with VS1 for channels 0 to 3 (Rank A & B) and VS2 for channels 4 to 7 (Rank A & B)). A pulse to OFF is applied to these power outputs periodically with a period less than 1 second and a duration less than 1 ms. During this pulse, if the current read into the input is not null, the module considers the input in short-circuit.

⚠ WARNING

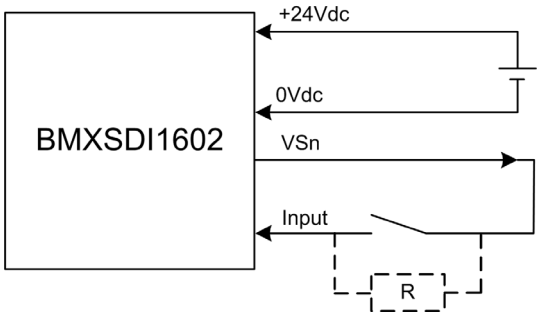
RISK OF UNINTENDED OPERATION

Schneider Electric recommends that you enable the available diagnostics provided in Unity Pro to detect or exclude the conditions listed above. If a diagnostic test is not enabled or is not available in Unity Pro, you will need to apply another safety measure to detect or exclude these conditions.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

SIL3 Cat2/PLd

Single sensor connected with one input, supplied by internal VS:



In this example, if internal power is supplied via:

- VS1, use channels 0...3 ranks A & B.
- VS2, use channels 4...7 ranks A & B.

NOTE: Consider adding a Shottky diode to the input loop, between the sensor and the input point, to reduce the likelihood that a short circuit to 24 Vdc condition on one channel might cause the same condition on an adjacent channel.

Because the sensor is supplied power internally via a VS pin, the following channel wiring diagnostics apply:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc[1]	Yes	< 1 s
Crossed circuits between two channels[1]	Yes	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

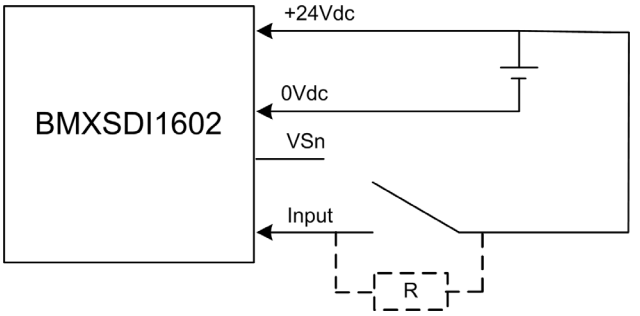
⚠ WARNING

RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS IN THE SAME GROUP

The module cannot detect crossed circuits between two channels in the same channel VS group. You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Single sensor connected with one input, powered by external power:



Because the sensor is supplied power externally, the following channel wiring diagnostics apply:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc	No	–
Crossed circuits between two channels	No	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

⚠ WARNING

RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS

The module cannot detect crossed circuits between two channels (in the case of a single sensor connected with one input, powered by external power, depicted above). You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING

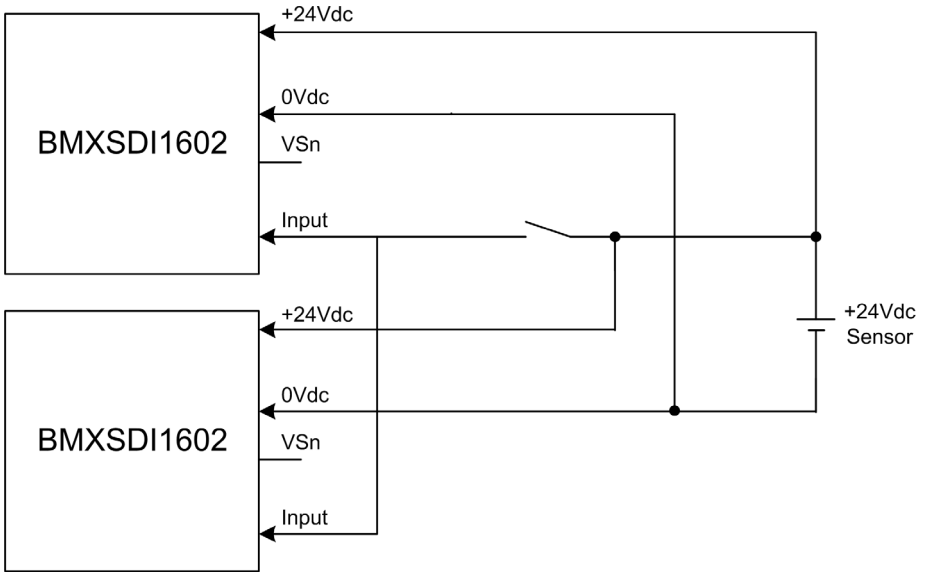
RISK OF SHORT CIRCUIT TO THE 24 Vdc

The module cannot detect a short circuit to the 24 Vdc condition (in the case of a single sensor connected with one input, powered by external power, depicted above). You will need to apply another safety measure to detect or exclude to this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

SIL3 Cat2/PLd with High Availability

Single sensor connected on 2 inputs powered by external power:



Because the single sensor is supplied power externally, the following channel wiring diagnostics apply:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	No	–
Short circuit to the 0 V ground	No	
Short circuit to the 24 Vdc[1]	No	
Crossed circuits between two channels	No	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

⚠ WARNING

RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS

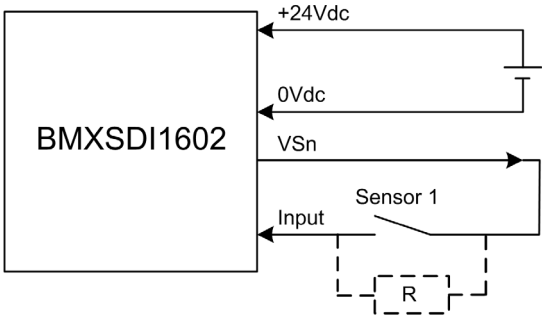
The module cannot detect crossed circuits between two channels (in the case of a single sensor connected on two inputs, powered by external power, depicted above). You will need to apply another safety measure to detect or exclude this condition.

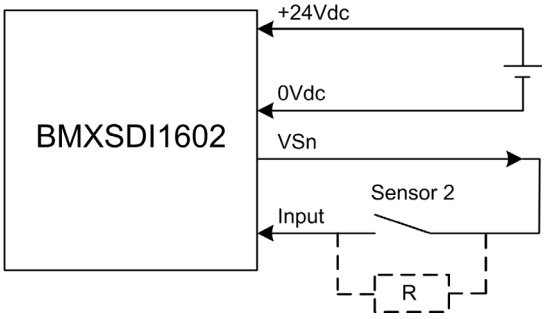
Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING
RISK OF SHORT CIRCUIT TO THE 24 Vdc The module cannot detect a short circuit to the 24 Vdc condition (in the case of a single sensor connected on two inputs, powered by external power, depicted above). You will need to apply another safety measure to detect or exclude this condition. **Failure to follow these instructions can result in death, serious injury, or equipment damage.**

2 redundant sensors connected on single inputs of 2 modules using VS:

The following example presents two redundant sensors (which may or may not be linked mechanically) that are used to acquire the same process variable. Each sensor is wired to a single input point on a different input module, with power supplied by the monitored VS power supply:





In this example, if internal power is supplied via:

- VS1, use channels 0...3 ranks A & B.
- VS2, use channels 4...7 ranks A & B.

NOTE:

- In this design, you could use the `S_DIHA` function block to manage the two input signals.
- Consider adding a Shottky diode to the input loop, between the sensor and the input point, to reduce the likelihood that a short circuit to 24 Vdc condition on one channel might cause the same condition on an adjacent channel.

Because the sensor is supplied power internally via a VS pin, the following channel wiring diagnostics apply:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc[1]	Yes	< 1 s
Crossed circuits between two channels	Yes	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

⚠ WARNING

RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS IN THE SAME GROUP

The module cannot detect crossed circuits between two channels in the same channel VS group. You will need to apply another safety measure to detect and exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

2 redundant sensors connected on single inputs of 2 modules using external power:

NOTE: Alternatively, power could be supplied to the sensors by an external power supply. In this case, a short circuit to the 24 Vdc condition and a crossed circuits between two channels condition would not be detectable.

Because the sensor is supplied power internally via a VS pin, the following channel wiring diagnostics apply:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc	No	–
Crossed circuits between two channels	No	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

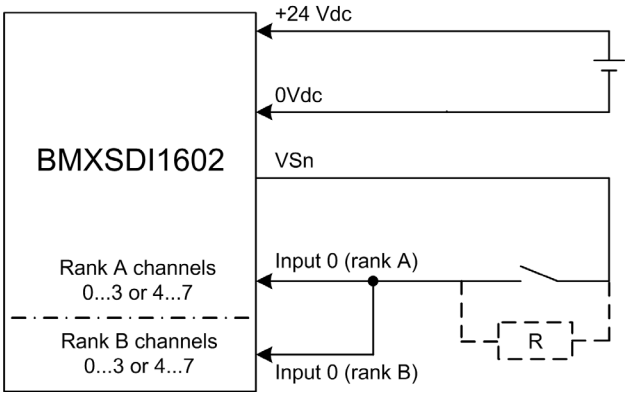
⚠ WARNING
RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS The module cannot detect crossed circuits between two channels (in the case of two redundant sensors connected on single inputs of two modules using external power). You will need to apply another safety measure to detect or exclude this condition. **Failure to follow these instructions can result in death, serious injury, or equipment damage.**

⚠ WARNING
RISK OF SHORT CIRCUIT TO THE 24 Vdc The module cannot detect a short circuit to the 24 Vdc condition (in the case of two redundant sensors connected on single inputs of two modules using external power). You will need to apply another safety measure to detect or exclude this condition. **Failure to follow these instructions can result in death, serious injury, or equipment damage.**

Cat4/PLe

Single sensor connected on 2 inputs of same module using Vs:

The following example presents a single sensor wired to two input points on the same input module, with power supplied by the monitored VS power supply:



In this example, if internal power is supplied via:

- VS1, use channels 0...3 ranks A & B.
- VS2, use channels 4...7 ranks A & B.

NOTE:

- In this design, you could use the `S_EQUIVALENT` function block to manage the two input signals.
- Consider adding a Shottky diode to the input loop, between the sensor and the input point, to reduce the likelihood that a short circuit to 24 Vdc condition on one channel might cause the same condition on an adjacent channel.

Wiring diagnostic with single sensor connected on two inputs, using power from the VS pin:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc[1]	Yes	< 1 s
Crossed circuits between two channels	Yes	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

⚠ WARNING

RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS IN THE SAME GROUP

The module cannot detect crossed circuits between two channels in the same channel VS group. You will need to apply another safety measure to detect and exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Single sensor connected on 2 inputs of same module using external power supply:

NOTE: Alternatively, power could be supplied to the sensors by an external power supply. In this case, a short circuit to the 24 Vdc condition and a crossed circuits between two channels condition would not be detectable.

Wiring diagnostic with single sensor connected on two inputs, using external power:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc[1]	No	–
Crossed circuits between two channels	No	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

⚠ WARNING

RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS

The module cannot detect crossed circuits between two channels (in the case of a single sensor connected on two inputs of same module using an external power supply). You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING

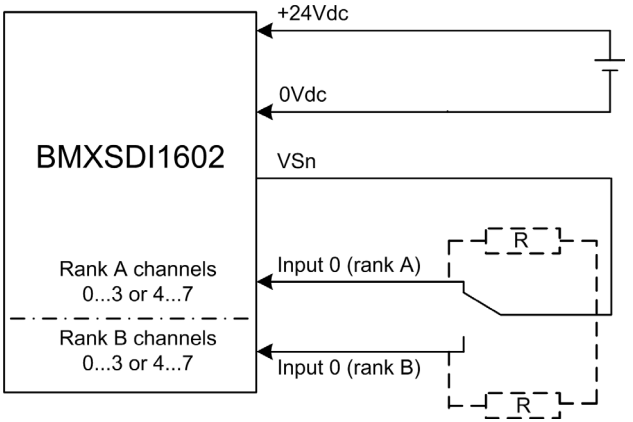
RISK OF SHORT CIRCUIT TO THE 24 Vdc

The module cannot detect a short circuit to the 24 Vdc condition (in the case of a single sensor connected on two inputs of same module using an external power supply). You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Non-equivalent sensor connected on 2 non-equivalent inputs of same module using Vs:

The following example presents a single non-equivalent sensor wired to two input points on the same input module, with power supplied by the monitored VS power supply. The module performs 1oo2 evaluation:



In this example, if internal power is supplied via:

- VS1, use channels 0...3 ranks A & B.
- VS2, use channels 4...7 ranks A & B.

NOTE:

- In this design, you could use the `S_ANTIIVALENT` function block to manage the two input signals.
- Consider adding a Shottky diode to the input loop, between the sensor and the input point, to reduce the likelihood that a short circuit to 24 Vdc condition on one channel might cause the same condition on an adjacent channel.

Wiring diagnostic with single non-equivalent sensors connected on two inputs, using power from the VS pin:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc[1]	Yes	< 1 s
Crossed circuits between two channels	Yes	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

Non-equivalent sensor connected on 2 non-equivalent inputs of same module using external power:

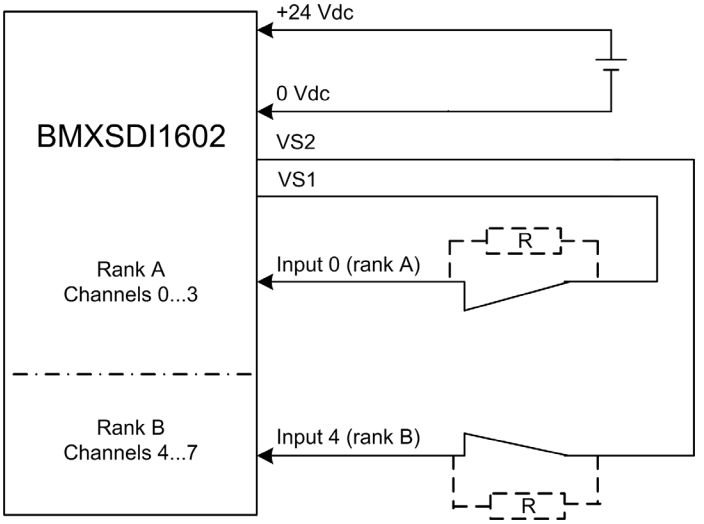
NOTE: Alternatively, power could be supplied to the sensors by an external power supply. In this case, a short circuit to the 24 Vdc condition and a crossed circuits between two channels condition would not be detectable.

Wiring diagnostic with single non-equivalent sensors connected on two inputs, using external power:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc[1]	No	–
Crossed circuits between two channels	No	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

Acquisition of the same process variable using two separated sensors (linked mechanically or not) using VS:

The following example presents two redundant sensors (which may or may not be linked mechanically) that are used to acquire the same process variable. Each sensor is wired to a single input point on the same input module, with power supplied by the monitored VS power supply:



NOTE:

- Inputs 0...3 from Rank A are used with Inputs 4...7 from Rank B.
- Inputs 0...3 from Rank B are used with Inputs 4...7 from Rank A.

⚠ WARNING
RISK OF UNINTENDED OPERATION To achieve SIL3 according to IEC 61508 and Category 4/Performance Level e according to ISO13849 using this wiring design, you must use suitable, qualified sensors. **Failure to follow these instructions can result in death, serious injury, or equipment damage.**

In this example, if internal power is supplied via:

- VS1, use channels 0...3 ranks A & B.
- VS2, use channels 4...7 ranks A & B.

NOTE:

- In this design, you could use the `S_EQUIVALENT` function block to manage the two input signals.
- Consider adding a Shottky diode to the input loop, between the sensor and the input point, to reduce the likelihood that a short circuit to 24 Vdc condition on one channel might cause the same condition on an adjacent channel.

Wiring diagnostic with single sensor connected to two inputs, using power from the VS pin:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc[1]	Yes	< 1 s
Crossed circuits between two channels	Yes	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

⚠ WARNING

RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS IN THE SAME GROUP

The module cannot detect crossed circuits between two channels in the same channel VS group (in the case of the acquisition of the same process variable using two separated sensors using VS supplied power). You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING

RISK OF SHORT CIRCUIT TO THE 24 Vdc

The module cannot detect a short circuit to the 24 Vdc condition (in the case of the acquisition of the same process variable using two separated sensors using VS supplied power). You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Acquisition of the same process variable using two separated sensors (linked mechanically or not) using external power:

NOTE: Alternatively, power could be supplied to the sensors by an external power supply. In this case, a short circuit to the 24 Vdc condition and a crossed circuits between two channels condition would not be detectable.

Wiring diagnostic with single sensor connected to two inputs, using external power:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc[1]	No	–
Crossed circuits between two channels	No	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

⚠ WARNING

RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS

The module cannot detect crossed circuits between two channels (in the case of the acquisition of the same process variable using two separated sensors using external power). You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING

RISK OF UNINTENDED OPERATION

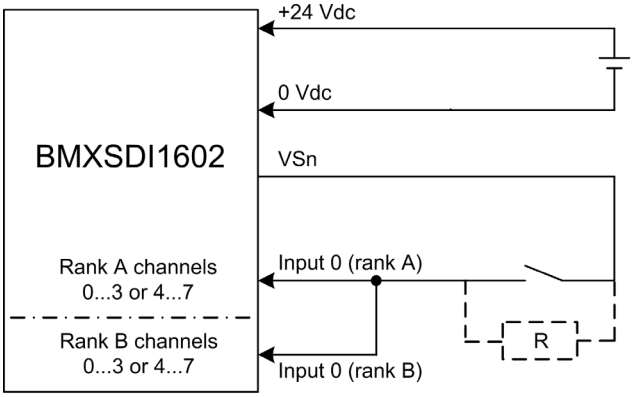
To achieve SIL3/Cat4/PLe using this wiring, you must use a suitably qualified sensor.

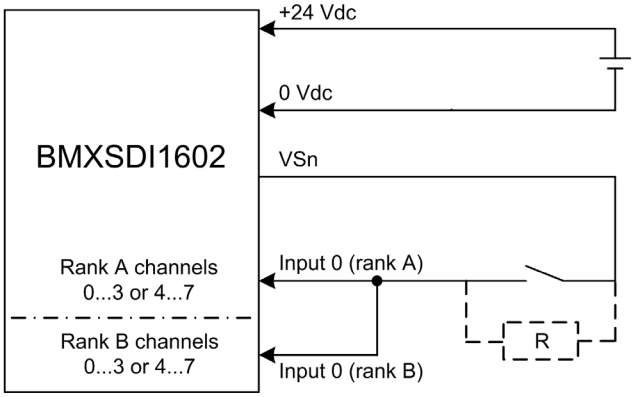
Failure to follow these instructions can result in death, serious injury, or equipment damage.

Cat4/PLe with High Availability

Wiring scheme with single-channel connection of two redundant single-channel sensors using Vs:

The following example presents two redundant single-channel sensors (which may or may not be mechanically linked), each wired to two input points on two different input modules, with power supplied by the monitored VS power supply:





In this example, if internal power is supplied via:

- VS1, use channels 0...3 ranks A & B.
- VS2, use channels 4...7 ranks A & B.

NOTE:

- In this design, you could use the `S_EQUIVALENT` and `S_DIHA` function blocks to manage the four input signals.
- Consider adding a Shottky diode to the input loop, between the sensor and the input point, to reduce the likelihood that a short circuit to 24 Vdc condition on one channel might cause the same condition on an adjacent channel.

Wiring diagnostic with single sensor connected to two inputs, using power from the VS pin:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc[1]	Yes	< 1 s
Crossed circuits between two channels	Yes	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

⚠ WARNING

RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS IN THE SAME GROUP

The module cannot detect crossed circuits between two channels in the same channel VS group. You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Wiring scheme with single-channel connection of two redundant single-channel sensors using external power:

NOTE: Alternatively, power could be supplied to the sensors by an external power supply. In this case, a short circuit to the 24 Vdc condition and a crossed circuits between two channels condition would not be detectable.

Wiring diagnostic with single sensor connected to two inputs, using external power:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc[1]	No	–
Crossed circuits between two channels	No	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

⚠ WARNING

RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS

The module cannot detect crossed circuits between two channels (in the case of a single-channel connection of two redundant single-channel sensors using external power). You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING

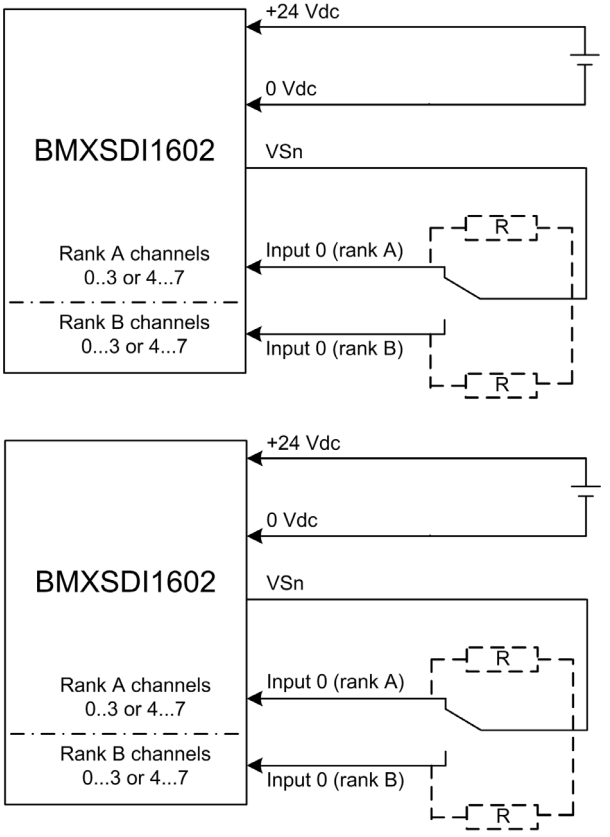
RISK OF SHORT CIRCUIT TO THE 24 Vdc

The module cannot detect a short circuit to the 24 Vdc condition (in the case of a single-channel connection of two redundant single-channel sensors using external power). You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Non-equivalent sensor (linked mechanically or not) connected on 2 non-equivalent inputs of two different modules using Vs:

The following example presents two pair of redundant non-equivalent sensors (which may or may not be mechanically linked), each wired to a single input point on two different input modules (two on each module), with power supplied by the monitored VS power supply:



In this example, if internal power is supplied via:

- VS1, use channels 0...3 ranks A & B.
- VS2, use channels 4...7 ranks A & B.

NOTE:

- In this design, you need to use the `S_ANTIVALENT` and `S_DIHA` function blocks to manage the four input signals.
  - `S_ANTIVALENT` to perform 1oo2 evaluation of two pairs of values coming from both sensors connected to the same module.
  - `S_DIHA` to manage the high availability feature.
- Consider adding a Shottky diode to the input loop, between the sensor and the input point, to reduce the likelihood that a short circuit to 24 Vdc condition on one channel might cause the same condition on an adjacent channel.

Because the sensor is supplied power internally via a VS pin, the following channel wiring diagnostics apply:

Condition	Detectable?	Typical Detection Time
Open (or cut) wire[1]	Yes	< 10 ms
Short circuit to the 0 V ground	Yes	
Short circuit to the 24 Vdc[1]	Yes	< 1 s
Crossed circuits between two channels	Yes	
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.		

⚠ WARNING

RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS IN THE SAME GROUP

The module cannot detect crossed circuits between two channels in the same channel VS group. You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Non-equivalent sensor (linked mechanically or not) connected on 2 non-equivalent inputs of two different modules using external power:

NOTE: Alternatively, power could be supplied to the sensors by an external power supply (in the case of a non-equivalent sensor connected on two non-equivalent inputs off two different modules using external power). In this case, a crossed circuits between two channels condition would not be detectable.

⚠ WARNING

RISK OF CROSSED CIRCUITS BETWEEEN CHANNELS

The module cannot detect crossed circuits between two channels. You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING

RISK OF UNINTENDED OPERATION

To achieve SIL3 according to IEC 61508 and Category 4/Performance Level e according to ISO13849 using this wiring design, you must use suitable, qualified sensors.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

BMXSDI1602 Data Structure

Introduction

The T_U_DIS_SIS_IN_16 device derived data type (DDDT) is the interface between the BMXSDI1602 digital input module and the application running in the CPU. The T_U_DIS_SIS_IN_16 DDDT incorporates the data types T_SAFE_COM_DBG_IN and T_U_DIS_SIS_CH_IN.

All of these structures are described, below.

T_U_DIS_SIS_IN_16 DDDT Structure

The T_U_DIS_SIS_IN_16 DDDT structure includes the following elements:

Element	Data Type	Description	Access
MOD_HEALTH[1]	BOOL	• 1: The module is operating correctly. • 0: The module is not operating correctly.	RO
SAFE_COM_STS[1]	BOOL	• 1: Module communication is valid. • 0: Module communication is not valid.	RO
PP_STS	BOOL	• 1: The process power supply is operational. • 0: The process power supply is not operational.	RO
CONF_LOCKED	BOOL	• 1: Module configuration is locked. • 0: Module configuration is not locked.	RO
S_COM_DBG	T_SAFE_COM_DBG_IN	Safe communication debug structure.	RO
CH_IN_A	ARRAY[0...7] of T_U_DIS_SIS_CH_IN	Array of structure of channel from rank A.	–
CH_IN_B	ARRAY[0...7] of T_U_DIS_SIS_CH_IN	Array of structure of channel from rank B.	–
MUID[2]	ARRAY[0...3] of DWORD	Module unique ID (auto-assigned by Unity Pro)	RO
RESERVED	ARRAY[0...9] of INT	–	–

1. When the SAFE task on CPU is not in running mode, the data exchanged between the CPU and the module are not updated and MOD_HEALTH and SAFE_COM_STS are set to 0.
2. This auto-generated value can be changed by executing the **Build → Renew Ids & Rebuild All** command in the Unity Pro main menu.

T_SAFE_COM_DBG_IN Structure

The T_SAFE_COM_DBG_IN structure includes the following elements:

Element	Data Type	Description	Access
S_COM_EST	BOOL	• 1: Communication with the module is established. • 0: Communication with the module is not established or corrupted.	RO
M_NTP_SYNC	BOOL	• 1: The module is synchronized with the NTP server. • 0: The module is not synchronized with the NTP server.	RO
CPU_NTP_SYNC	BOOL	• 1: The CPU is synchronized with the NTP server. • 0: The CPU is not synchronized with the NTP server.	RO
CHECKSUM	BYTE	Communication frame checksum.	RO
COM_DELAY	UINT	Communication delay between two values received by the module: • 0: No delay. The communication was received on time. • 1...65534: The time, in ms, since the last communication was received by the CPU from the module. • 65535: The CPU did not receive a communication from the module.	RO
COM_TO	UINT	Communication time-out value for communications coming from the module.	R/W
STS_MS_IN	UINT	NTP timestamp value for the fraction of a second, to the nearest ms, of the data received from the module.	RO
S_NTP_MS	UINT	NTP time value for the fraction of a second, to the nearest ms, for the current cycle.	RO
STS_S_IN	UDINT	NTP timestamp value in seconds of the data received from the module.	RO
S_NTP_S	UDINT	NTP time value in seconds for the current cycle.	RO
CRC_IN	UDINT	CRC value for data received from the module.	RO

T_U_DIS_SIS_CH_IN Structure

The T_U_DIS_SIS_CH_IN structure includes the following elements:

Element	Data Type	Description	Access
CH_HEALTH[1]	BOOL	• 1: The channel is operational. • 0: An error has been detected on the channel, which is not operational.	RO
VALUE[2]	EBOOL	• 1: The input is energized. • 0: The input is de-energized.	RO
OC	BOOL	• 1: The channel is open or short circuited to ground. • 0: The channel is connected and not short circuited to ground.	RO
SC	BOOL	• 1: The channel is short circuited to a 24 V source, or cross circuited between two channels. • 0: The channel is not short circuited to a 24 V source or cross circuited between two channels.	RO
IC	BOOL	• 1: Invalid channel detected by the module. • 0: The channel is declared internally operational by the module.	RO
V_OC	BOOL	Configuration status of the open or short circuit to ground test: • 1: Enabled. • 0: Disabled.	RO
V_SC	BOOL	Configuration status of the short circuit to 24 V source test: • 1: Enabled. • 0: Disabled.	RO

1. When the SAFE task on CPU is not in running mode, the data exchanged between the CPU and the module are not updated and CH_HEALTH is set to 0.
2. The VALUE element can be time-stamped by the BMX CRA or the BME CRA.

Section 6.4
BMXSDO0802 Digital Output Module

Introduction

This section describes the BMXSDO0802 M580 safety digital output module.

What Is in This Section?

This section contains the following topics:

Topic	Page
BMXSDO0802 Safety Digital Output Module	89
BMXSDO0802 Wiring Connector	91
BMXSDO0802 Output Application Wiring Examples	94
BMXSDO0802 Data Structure	100

BMXSDO0802 Safety Digital Output Module

Introduction

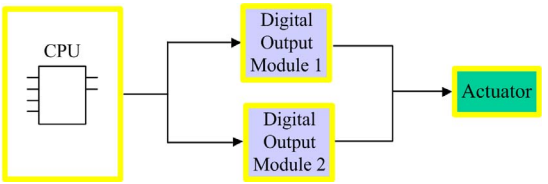
The BMXSDO0802 safety digital output module presents the following features:

- 8 non-electrically isolated 0.5 A outputs.
- 24 Vdc rated output voltage.
- Achieves SIL3 (IEC61508) Category 4 (Cat4) / Performance Level e (PLe) evaluation.
- Monitors the external pre-actuator power supply.
- LED diagnostic display *(see page 208)* provided for the module and for each output channel.
- Automatically provided channel wiring diagnostics that can detect the following conditions when the output is *energized*:
  - Overload current
  - Short circuit to the 0 Vdc ground
- Configurable (enable/disable) channel wiring diagnostics that can detect the following conditions:
  - Open (or cut) wire.
- Configurable (enable/disable) channel wiring diagnostics that can detect the following conditions when the output is *de-energized*:
  - Short circuit to the 0 V ground.
- Configurable (enable/disable) channel wiring diagnostics that can detect the following conditions when the output is *energized* or *de-energized*:
  - Short circuit to the 24 Vdc.
  - Crossed circuits between two channels (if sensor power is internally provided).
- Configurable fallback settings for each channel that are applied if communication between the CPU and output module are lost.
- Module hot swap during runtime.
- Module CCOTF when operating in maintenance mode *(see page 224)*. (CCOTF is not supported in safety mode.) *(see page 223)*

High Availability

You can connect the CPU to two output modules via a black channel, then connect each output module to a single actuator. No function block is needed, because the signal from the CPU is connected to both output channels.

The following figure illustrates the redundant digital output configuration for high availability:



The health of each output module can be read from the elements of its `T_U_DIS_SIS_OUT_8` *(see page 100)* DDDT structure. You can use this data to determine if a module needs to be replaced. If a module ceases to be operational and needs to be replaced, the system continues to run in a SIL3 compliant configuration while the module exchange takes place.

Refer to the high availability output wiring example *(see page 97)* for details on this design.

BMXSDO0802 Wiring Connector

Introduction

The BMXSDO0802 digital output module presents a single group of 8 outputs.

- Both common +24 Vdc power supply pins (18 and 20) are internally connected.
- All common 0 V pins (1, 3, 5, 7, 9, 11, 13, 15, 17 and 19) are internally connected.

Terminal Blocks

You can use the following Schneider Electric 20-point terminal blocks to fit the 20 pin connector on the front of the module:

- screw clamp terminal block BMXFTB2010
- age clamp terminal block BMXFTB2000
- spring type terminal block BMXFTB2020

NOTE: Terminal blocks can be removed only when power to the module is OFF.

Process Power Supply

A 24 Vdc protected extra low voltage (PELV) overvoltage category II process power supply is required. Schneider Electric recommends a power supply that does not automatically restore power after a power interruption.

Fuse

A fast blow fuse, maximum of 6 A, is required to help protect the external power supply against short-circuit and over voltage conditions.

⚠ CAUTION

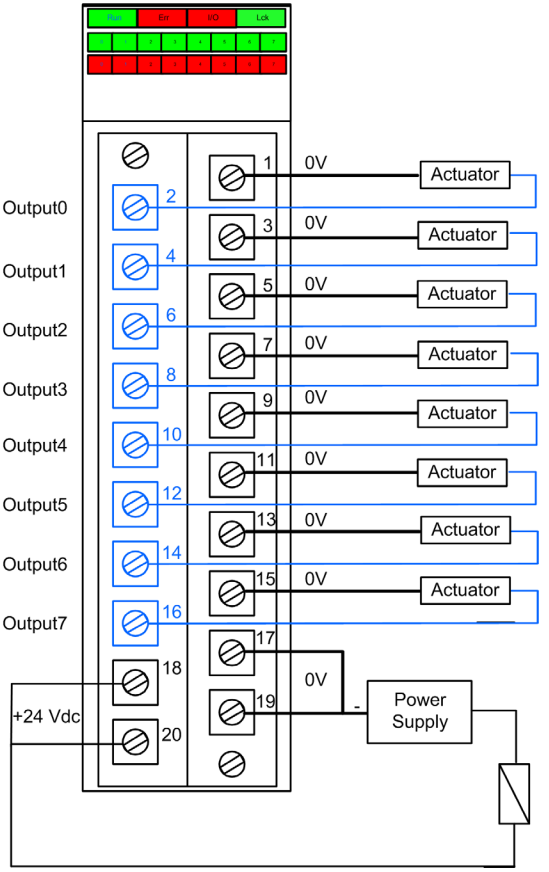
IMPROPER FUSE SELECTION

Use fast acting fuses to help protect the electronic components of the digital output module from an over current condition. Improper fuse selection can result in damage to the module.

Failure to follow these instructions can result in injury or equipment damage.

Wiring Connector Pins

The following wiring diagram presents a single output module connected to 8 actuators:



Mapping Outputs to Connector Pins

The following provides a description of each pin on the BMXSDO0802 output module:

Pin Description	Pin Number on Terminal Block		Pin Description
Output 0	2	1	Common 0V
Output 1	4	3	Common 0V
Output 2	6	5	Common 0V
Output 3	8	7	Common 0V
Output 4	10	9	Common 0V
Output 5	12	11	Common 0V
Output 6	14	13	Common 0V
Output 7	16	15	Common 0V
24 Vdc Process Power Supply	18	17	Common 0V
24 Vdc Process Power Supply	20	19	Common 0V

BMXSDO0802 Output Application Wiring Examples

Introduction

You can wire the BMXSDO0802 safety digital output module to actuators to achieve SIL3 Category 4 (Cat4) / Performance Level e (PLe) compliance in different ways, depending on your requirements for high availability.

⚠ CAUTION

RISK OF UNINTENDED OPERATION

The maximum safety integrity level (SIL) is determined by the quality of actuator and the length of the proof-test interval to IEC 61508. If you are using actuators that do not meet the quality of the intended SIL standard, always wire these actuators redundantly to two channels.

Failure to follow these instructions can result in injury or equipment damage.

The following SIL3 Cat4/PLe digital output application wiring examples are described, below:

- Cat4/PLe:
  - a single output module channel commanding one process variable. A single actuator is employed in this design.
- Cat4/PLe with high availability:
  - two redundant output modules, each with a channel connected to a separate actuator, but commanding the same process variable.

⚠ CAUTION

RISK OF UNINTENDED OPERATION

When the equipment is used in a fire and gas application, or when the demand state of the output is energized:

- Your proof test procedure must include a test that the cut wire detection is effective by removing the terminal block and verifying that the corresponding error bits are set.
- Verify the short circuit to ground detection effectiveness, either by enabling this **Pulse test to energized** diagnostic function in the module's **Configuration** tab, or by implementing another procedure (for example, by setting the output to 1 and checking the diagnostics, and so forth.)
- Avoid using lamp-like actuators, because their impedance is very low when they are switched on, which can result in a risk of detecting a false short-circuit or overload condition.

Failure to follow these instructions can result in injury or equipment damage.

Configurable Wiring Diagnostics in Unity Pro

For the BMXSDO0802 safety digital output module, use its **Configuration** page in Unity Pro to:

- Enable **Short circuit to 24V detection** for each energized channel. This test performs the following actuator wiring diagnostics for a channel:
  - Short circuit to 24 Vdc detection
  - Crossed circuit detection between two output channels
- Enable **Open wire detection** for each of eight channels, which performs the following wiring diagnostics for that channel:
  - Open (or cut) wire detection (i.e. the output channel is not connected to the actuator)
  - Short circuit wiring detection to the 0 Vdc ground
- Enable the **Pulse test to energized** for each output channel. This test is performed periodically when the output is in the de-energized state, and applies a pulse (duration less than 1 ms) to the output to determine if it can transition to the energized state. If the current exceeds a threshold of 0.7 A, the output is reported as being in a short circuit condition with the 0 Vdc ground. The test period is less than 1 s.

⚠ WARNING

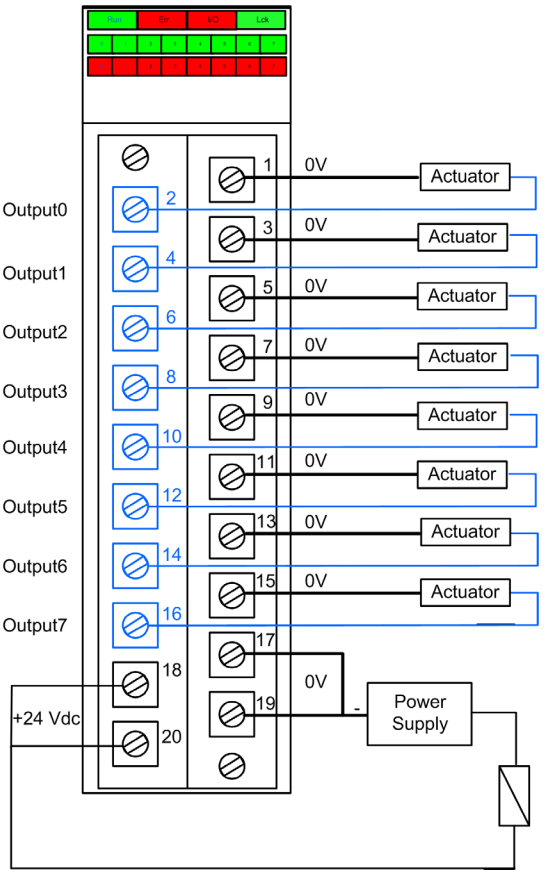
RISK OF UNINTENDED OPERATION

Schneider Electric recommends that you enable the available diagnostics provided in Unity Pro to detect and respond to the conditions listed above. If a diagnostic test is not enabled or is not available in Unity Pro, you will need to apply another safety measure to detect or exclude these conditions.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

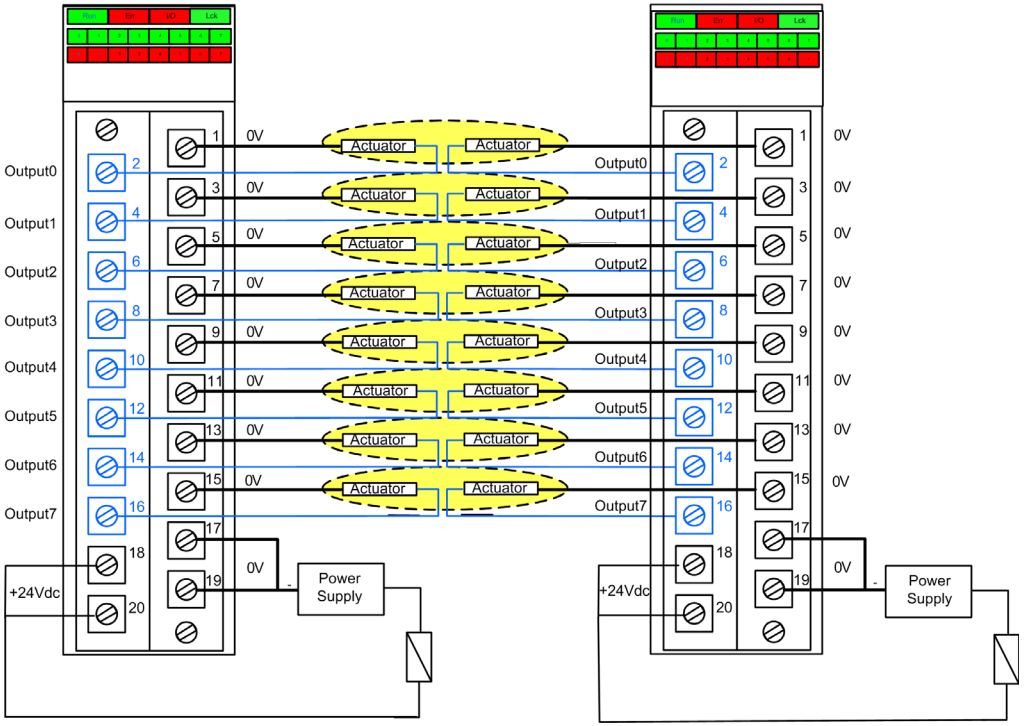
SIL 3 Cat4/PLe - Single Digital Output Module Example

The following example presents one exclusive actuator wired to each output on a single output module. Each loop is SIL 3 Cat4/PLe:



IL 3 Cat4/PLe - High Availability Example:

In the following wiring diagram, two redundant outputs command the same process variable. As displayed below, each output is connected to separate actuators, then each actuator executes the same command sent over different channels. Alternatively, the two redundant outputs could be wired together to command the same actuator.



Output Wiring Diagnostic Summary

The two designs provide the following wiring diagnostics:

Condition	Diagnostic Provided in Output State?	
	Energized	De-energized
Open (or cut) wire[1]	Yes. Diagnosed each cycle.	Yes. Diagnosed each cycle.
Output in overload[2]	Yes. Diagnosed each cycle.	No.
Short circuit to the 0 V ground	Yes. Diagnosed each cycle.	Yes. Diagnostic period < 1 s.

1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro.
2. After the condition is resolved, re-arm the output by de-energizing it.

Condition	Diagnostic Provided in Output State?	
	Energized	De-energized
Short circuit to the 24 Vdc[1]	Yes. Diagnostic period < 1 s.	Yes. Diagnosed each cycle.
Crossed circuits between two channels	Yes. Diagnostic period < 1 s.	Yes. Diagnosed each cycle.
1. This diagnostic function is performed if enabled in the module **Configuration** tab in Unity Pro. 2. After the condition is resolved, re-arm the output by de-energizing it.		

⚠ WARNING

RISK OF SHORT CIRCUIT TO 0 Vdc GROUND

For the short circuit to the 0 V ground condition with the output state de-energized, it is recommended that you enable the **Open wire detection** option in the module's **Configuration** tab. Alternatively, you will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING

RISK OF SHORT CIRCUIT TO THE 24 Vdc

For the short circuit to the 24 Vdc condition with the output state either energized or de-energized, it is recommended that you enable the **Short circuit to 24V detection** option in the module's **Configuration** tab. Alternatively, you will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING

RISK OF CROSSED CIRCUITS

The module cannot detect the crossed circuits between two channels condition with the output state de-energized and the other channel de-energized. You will need to apply another safety measure to detect or exclude this condition, if it occurs when the output state changes to energized.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING

RISK OF CROSSED CIRCUITS

For the crossed circuits between two channels condition with the output state de-energized and the other channel energized, it is recommended that you enable the **Short circuit to 24V detection** option in the module's **Configuration** tab. Alternatively, you will need to apply another safety measure to detect or exclude this condition when the output state changes to energized.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING

RISK OF CROSSED CIRCUITS

The module cannot detect the crossed circuits between two channels condition with the output state energized and the other channel de-energized. You will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING

RISK OF CROSSED CIRCUITS

For the crossed circuits between two channels condition with the output state energized and the other channel energized, it is recommended that you enable the **Short circuit to 24V detection** option in the module's **Configuration** tab. Alternatively, you will need to apply another safety measure to detect or exclude this condition.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

BMXSDO0802 Data Structure

Introduction

The T_U_DIS_SIS_OUT_8 device derived data type (DDDT) is the interface between the BMXSDO0802 digital output module and the application running in the CPU. The T_U_DIS_SIS_OUT_8 DDDT incorporates the data types T_SAFE_COM_DBG_OUT and T_U_DIS_SIS_CH_OUT.

All of these structures are described, below.

T_U_DIS_SIS_OUT_8 DDDT Structure

The T_U_DIS_SIS_OUT_8 DDDT structure includes the following elements:

Element	Data Type	Description	Access
MOD_HEALTH[1]	BOOL	• 1: The module is operating correctly. • 0: The module is not operating correctly.	RO
SAFE_COM_STS[1]	BOOL	• 1: Module communication is valid. • 0: Module communication is not valid.	RO
PP_STS	BOOL	• 1: The process power supply is operational. • 0: The process power supply is not operational.	RO
CONF_LOCKED	BOOL	• 1: Module configuration is locked. • 0: Module configuration is not locked.	RO
S_COM_DBG	T_SAFE_COM_DBG_OUT	Safe communication debug structure.	RO
CH_OUT	ARRAY[0...7] of T_U_DIS_SIS_CH_OUT	Array of structure of channel.	RO
S_TO	UINT	Safety control timeout before module enters fallback state.	RO
MUID[2]	ARRAY[0...3] of DWORD	Module unique ID (auto-assigned by Unity Pro)	RO
RESERVED_1	ARRAY[0...8] of INT	–	–
RESERVED_2	ARRAY[0...6] of INT	–	–

1. When the SAFE task on CPU is not in running mode, the data exchanged between the CPU and the module are not updated and MOD_HEALTH and SAFE_COM_STS are set to 0.
2. This auto-generated value can be changed by executing the **Build → Renew Ids & Rebuild All** command in the Unity Pro main menu.

T_SAFE_COM_DBG_OUT Structure

The T_SAFE_COM_DBG_OUT structure includes the following elements:

Element	Data Type	Description	Access
S_COM_EST	BOOL	• 1: Communication with the module is established. • 0: Communication with the module is not established or corrupted.	RO
M_NTP_SYNC	BOOL	• 1: The module is synchronized with the NTP server. • 0: The module is not synchronized with the NTP server.	RO
CPU_NTP_SYNC	BOOL	• 1: The CPU is synchronized with the NTP server. • 0: The CPU is not synchronized with the NTP server.	RO
CHECKSUM	BYTE	Communication frame checksum.	RO
COM_DELAY	UINT	Communication delay between two values received by the module: • 0: No delay. The communication was received on time. • 1...65534: The time, in ms, since the last communication was received by the CPU from the module. • 65535: The CPU did not receive a communication from the module.	RO
COM_TO	UINT	Communication time-out value for communications coming from the module.	R/W
STS_MS_IN	UINT	NTP timestamp value for the fraction of a second, to the nearest ms, of the data received from the module.	RO
S_NTP_MS	UINT	NTP time value for the fraction of a second, to the nearest ms, for the current cycle.	RO
STS_S_IN	UDINT	NTP timestamp value in seconds of the data received from the module.	RO
S_NTP_S	UDINT	NTP time value in seconds for the current cycle.	RO
CRC_IN	UDINT	CRC value for data received from the module.	RO
STS_MS_OUT	UINT	NTP timestamp value for the fraction of a second, to the nearest ms, of the data to be sent to the module.	RO

Element	Data Type	Description	Access
STS_S_OUT	UDINT	NTP timestamp value in seconds of the data to be sent to the module.	RO
CRC_OUT	UDINT	CRC value for data to be sent to the module.	RO

T_U_DIS_SIS_CH_OUT Structure

The T_U_DIS_SIS_CH_OUT structure includes the following elements:

Element	Data Type	Description	Access
CH_HEALTH[1]	BOOL	• 1: The channel is operational. • 0: An error has been detected on the channel, which is not operational.	RO
VALUE	EBOOL	Safe command of output channel: • 1: Command the output closed (energized). • 0: Command the output open (de-energized).	R/W
TRUE_VALUE[2]	BOOL	Read back value of the output relay channel: • 1: The output is closed (energized). • 0: The output is open (de-energized).	RO
OC	BOOL	• 1: The channel is open or short circuited to ground. • 0: The channel is connected and not short circuited to ground.	RO
SC	BOOL	• 1: The channel is short circuited to a 24 V source, or cross circuited with another channel. • 0: The channel is not short circuited to a 24 V source or cross circuited.	RO
OL	BOOL	• 1: The channel is overloaded or short circuited to 0V. • 0: The channel is not overloaded or short circuited to 0V.	RO
IC	BOOL	• 1: Invalid channel detected by the module. • 0: The channel is declared internally operational by the module.	RO
1. When the SAFE task on CPU is not in running mode, the data exchanged between the CPU and the module are not updated and CH_HEALTH is set to 0. 2. The TRUE_VALUE element can be time-stamped by the BMX CRA or the BME CRA.			

Element	Data Type	Description	Access
V_OC	BOOL	Configuration status of the open circuit test: ● 1: Enabled. ● 0: Disabled.	RO
V_SC	BOOL	Configuration status of the short circuit to 24 V source test: ● 1: Enabled. ● 0: Disabled.	RO
V_PULSE_ON	BOOL	Configuration status of the pulse test to energized: ● 1: Enabled. ● 0: Disabled.	RO
CH_FBC	BOOL	Configuration status of the channel fallback configuration: ● 1: User defined value. ● 0: Hold last value.	RO
CH_FBST	BOOL	Configuration status of the user-defined channel fallback state configuration: ● 1: Enabled. ● 0: Disabled.	RO
1. When the SAFE task on CPU is not in running mode, the data exchanged between the CPU and the module are not updated and CH_HEALTH is set to 0. 2. The TRUE_VALUE element can be time-stamped by the BMX CRA or the BME CRA.			

Section 6.5
BMXSRA0405 Digital Relay Output Module

Introduction

This section describes the BMXSRA0405 M580 safety digital relay output module.

What Is in This Section?

This section contains the following topics:

Topic	Page
BMXSRA0405 Safety Digital Relay Output Module	105
BMXSRA0405 Wiring Connector	106
BMXSRA0405 Output Application Wiring Examples	109
BMXSRA0405 Data Structure	117

BMXSRA0405 Safety Digital Relay Output Module

Introduction

The BMXSRA0405 safety digital relay output module presents the following features:

- 4 relay outputs with 5 A current.
- Rated output voltage of 24 Vdc and 24...230 Vac (over voltage category II).
- Achieves up to SIL3 (IEC61508) Category 4 (Cat4) / Performance Level e (PLe) evaluation.
- Support for 8 pre-defined application wiring configuration selections.
- Configurable automatic self-test monitoring of the relay capacity to execute the commanded output state (depending on the selected application wiring configuration).
- Configurable module settings for fallback mode and fallback timeout (in ms).
- LED diagnostic display *(see page 214)* provided for the module and for each output channel.
- Module hot swap during runtime.
- Module CCOTF when operating in maintenance mode *(see page 224)*. (CCOTF is not supported in safety mode. *(see page 223)*)

BMXSRA0405 Wiring Connector

Introduction

The BMXSRA0405 digital relay output module includes 4 relays and supports up to 4 outputs. The module presents a pair of *a* and *b* pins for each relay. Note that for each relay:

- the two *a* pins are internally connected, and
- the two *b* pins also are internally connected.

Terminal Blocks

You can use the following Schneider Electric 20-point terminal blocks to fit the 20 pin connector on the front of the module:

- screw clamp terminal block BMXFTB2010
- age clamp terminal block BMXFTB2000
- spring type terminal block BMXFTB2020

NOTE: Terminal blocks can be removed only when power to the module is OFF.

Process Power Supply

You need to install the appropriate 24 Vdc or 24 Vac to 230 Vac process power supply.

Fuse

A fast blow fuse, maximum of 6 A, that is suitable for the selected application and relay design is required. Always install an external fuse in series with the external power supply, the relay, and the load.

⚠ WARNING

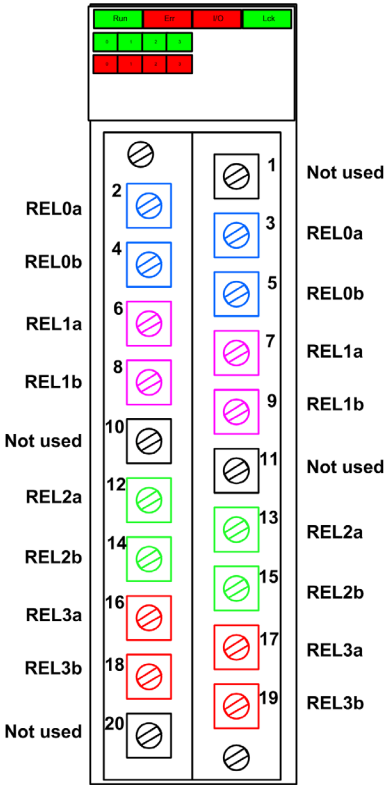
RISK OF UNINTENDED OPERATION

It is your responsibility to implement appropriate wiring diagnostics to detect and prevent the occurrence of dangerous faults on the external wiring.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Wiring Connector

The following example presents the pins on the relay module:



Mapping Inputs to Connector Pins

The following provides a description of each pin on the BMXSRA0405 digital relay output module:

Pin Description	Pin Number on Terminal Block		Pin Description
NO contact, Relay 0a	2	1	Not used
NO contact, Relay 0b	4	3	NO contact, Relay 0a
NO contact, Relay 1a	6	5	NO contact, Relay 0b
NO contact, Relay 1b	8	7	NO contact, Relay 1a
Not used	10	9	NO contact, Relay 1b
NO contact, Relay 2a	12	11	Not used
NO contact, Relay 2b	14	13	NO contact, Relay 2a
NO contact, Relay 3a	16	15	NO contact, Relay 2b

Pin Description	Pin Number on Terminal Block		Pin Description
NO contact, Relay 3b	18	17	NO contact, Relay 3a
Not used	20	19	NO contact, Relay 3b

NOTE: Because the two *a* pins for each relay are internally connected, you need to use only one *a* pin for each relay. Similarly, because the two *b* pins for each relay are internally connected, you need to use only one *b* pin for each relay.

BMXSRA0405 Output Application Wiring Examples

Introduction

You can configure the BMXSRA0405 safety digital output relay module to achieve either SIL2 Category 2 (Cat2) / Performance Level c (PLc) or SIL3 Cat4 / PLe compliance in different ways, depending on:

- the number of outputs the module will support, and
- how you want to test the ability of the module to place the actuator into the intended demand state, either:
  - automatically by the module (in this case, there is no state transition for the actuator) or
  - by means of a procedure that performs and checks a daily transition of the signal from the module to the actuator (in this case, the transition impacts the actuator state).

Make this configuration by selecting an application number (described in the tables, below) in the **Function** list in the module **Configuration** tab in Unity Pro.

SIL2 Cat2 / PLc wiring design applications:

Function	Demand State	Relays	Outputs	Signal Test?		Wiring Diagram (see below)
				Automatic Signal Test?[1]	Daily Signal Transition?	
Application_1	De-energized	1	4	No	Yes	A
Application_2	De-energized	2	2	Yes	No	B
Application_3	Energized	1	4	No	Yes	A
Application_4	Energized	2	2	Yes	No	C
1. The automatic signal test does not impact the actuator state.						

SIL3 Cat4 / PLe wiring design applications:

Function	Demand State	Relays	Outputs	Signal Test?		Wiring Diagram (see below)
				Automatic Signal Test?[1]	Daily Signal Transition?	
Application_5	De-energized	2	2	No	Yes	C
Application_6	De-energized	4	1	Yes	No	D
Application_7	Energized	2	2	No	Yes	C
Application_8	Energized	2	2	Yes	No	C
1. The automatic signal test does not impact the actuator state.						

Each of these eight application selections are described in the following wiring examples.

Application_1: 4 Outputs, SIL2 / Cat2 / PLc, De-energized State, No Automatic Signal Test

The demand state for this application design is de-energized. If the module detects an internal error for an output, it de-energizes that output.

⚠ CAUTION

LOSS OF THE ABILITY TO PERFORM SAFETY FUNCTIONS

To achieve SIL2 according to IEC61508 and Category 2 / Performance Level c according to ISO 13849 using this wiring design, you need to perform a daily signal transition from the energized state to the de-energized state.

Failure to follow these instructions can result in injury or equipment damage.

Refer to wiring diagram A *(see page 113)*, below, for a depiction of the wiring design for Application_1.

Application_2: 2 Outputs, SIL2 Cat2 / PLc, De-energized State, Automatic Signal Test

The demand state for this application design is de-energized. If the module detects an internal output error on one of the relays used for an output, it de-energizes both relays (Relay 0 and Relay 1 or Relay 2 and Relay 3) for that output.

Your application program needs to command the same output state to all relays that activate the same actuator.

The module sequentially performs an automatic periodic pulse test on each relay. The duration of the test is less than 50 ms. Because of the configuration of the two relays used (in parallel), the test has no impact on the output load (normally *energized*). You can configure the frequency of the test by setting the **Monitoring period** in the **Configuration** tab of the module. Valid test frequency values range from1...1440 minutes.

Refer to wiring diagram B *(see page 114)*, below, for a depiction of the wiring design for Application_2.

Application_3: 4 Outputs, SIL2 / Cat2 / PLc, Energized State, No Automatic Signal Test

The demand state for this application design is energized. If the module detects an internal error for an output, it de-energizes that output, which is the safe state.

⚠ CAUTION

LOSS OF THE ABILITY TO PERFORM SAFETY FUNCTIONS

To achieve SIL2 according to IEC61508 and Category 2 / Performance Level c according to ISO 13849 using this wiring design, you need to perform a daily signal transition from the energized state to the de-energized state.

Failure to follow these instructions can result in injury or equipment damage.

Refer to wiring diagram A *(see page 113)*, below, for a depiction of the wiring design for Application_3.

Application_4: 2 Outputs, SIL2 Cat2 / PLc, Energized State, Automatic Signal Test

The demand state for this application design is energized. If the module detects an internal output error on one of the relays used for an output, it de-energizes both relays (Relay 0 and Relay 1 or Relay 2 and Relay 3) for that output.

Your application program needs to command the same output state to all relays that activate the same actuator.

The module sequentially performs a periodic pulse test on each relay. The duration of the test is less than 50 ms. Because of the configuration of the two relays used (in parallel), the test has no impact on the output load (normally *energized*). You can configure the frequency of the test by setting the **Monitoring period** in the **Configuration** tab of the module. Valid test frequency values range from1...1440 minutes.

Refer to wiring diagram C *(see page 115)*, below, for a depiction of the wiring design for Application_4.

Application_5: 2 Outputs, SIL3 / Cat4 / PLe, De-energized State, No Automatic Signal Test

The demand state for this application design is de-energized. If the module detects an internal output error on one of the relays used for an output, it de-energizes both relays (Relay 0 and Relay 1 or Relay 2 and Relay 3) for that output.

Your application program needs to command the same output state to all relays that activate the same actuator.

⚠ CAUTION

LOSS OF THE ABILITY TO PERFORM SAFETY FUNCTIONS

To achieve SIL3 according to IEC61508 and Category 4 / Performance Level e according to ISO 13849 using this wiring design, you need to perform a daily signal transition from the energized state to the de-energized state.

Failure to follow these instructions can result in injury or equipment damage.

Refer to wiring diagram C *(see page 115)*, below, for a depiction of the wiring design for Application_5.

Application_6: 1 Output, SIL3 / Cat4 / PLe, De-energized State, Automatic Signal Test

The demand state for this application design is de-energized. If the module detects an internal output error on one of the relays used for an output, it de-energizes all relays (Relay 0, Relay 1, Relay 2, and Relay 3) for the module.

Your application program needs to command the same output state to all relays that activate the same actuator.

The module sequentially performs a periodic pulse test on each relay. The duration of the test is less than 50 ms. Because of the configuration of the four relays used (2 pair of two serial relays set in parallel), the test has no impact on the output load (normally *energized*). You can configure the frequency of the test by setting the **Monitoring period** in the **Configuration** tab of the module. Valid test frequency values range from1...1440 minutes.

Refer to wiring diagram D *(see page 116)*, below, for a depiction of the wiring design for Application_6.

Application_7: 4 Outputs, SIL2 / Cat2 / PLc, Energized State, No Automatic Signal Test

The demand state for this application design is energized. If the module detects an internal output error on one of the relays used for an output, it de-energizes both relays (Relay 0 and Relay 1 or Relay 2 and Relay 3) for that output.

Your application program needs to command the same output state to all relays that activate the same actuator.

⚠ CAUTION

LOSS OF THE ABILITY TO PERFORM SAFETY FUNCTIONS

To achieve SIL3 according to IEC61508 and Category 4 / Performance Level e according to ISO 13849 using this wiring design, you need to perform a daily signal transition from the energized state to the de-energized state.

Failure to follow these instructions can result in injury or equipment damage.

Refer to wiring diagram C *(see page 115)*, below, for a depiction of the wiring design for Application_7.

Application_8: 2 Outputs, SIL3 Cat4 / PLe, Energized State, Automatic Signal Test

The demand state for this application design is energized. If the module detects an internal output error on one of the relays used for an output, it de-energizes both relays (Relay 0 and Relay 1 or Relay 2 and Relay 3) for that output.

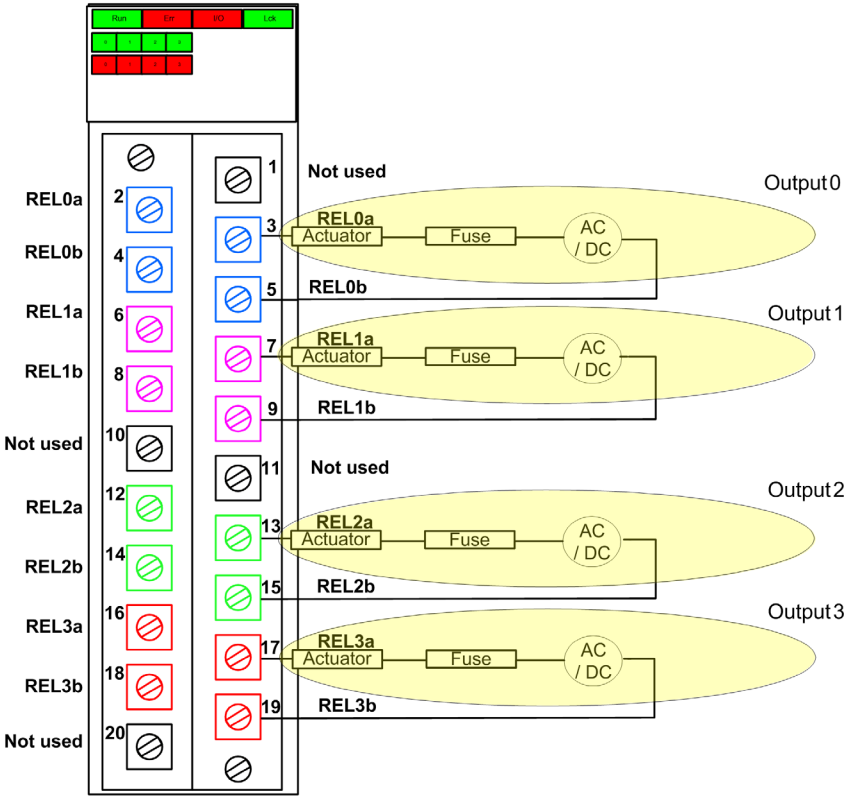
Your application program needs to command the same output state to all relays that activate the same actuator.

The module sequentially performs a periodic pulse test on each relay. The duration of the test is less than 50 ms. Because of the configuration of the two relays used (in serial), the test has no impact on the output load (normally *de-energized*). You can configure the frequency of the test by setting the **Monitoring period** in the **Configuration** tab of the module. Valid test frequency values range from1...1440 minutes.

Refer to wiring diagram C *(see page 115)*, below, for a depiction of the wiring design for Application_8.

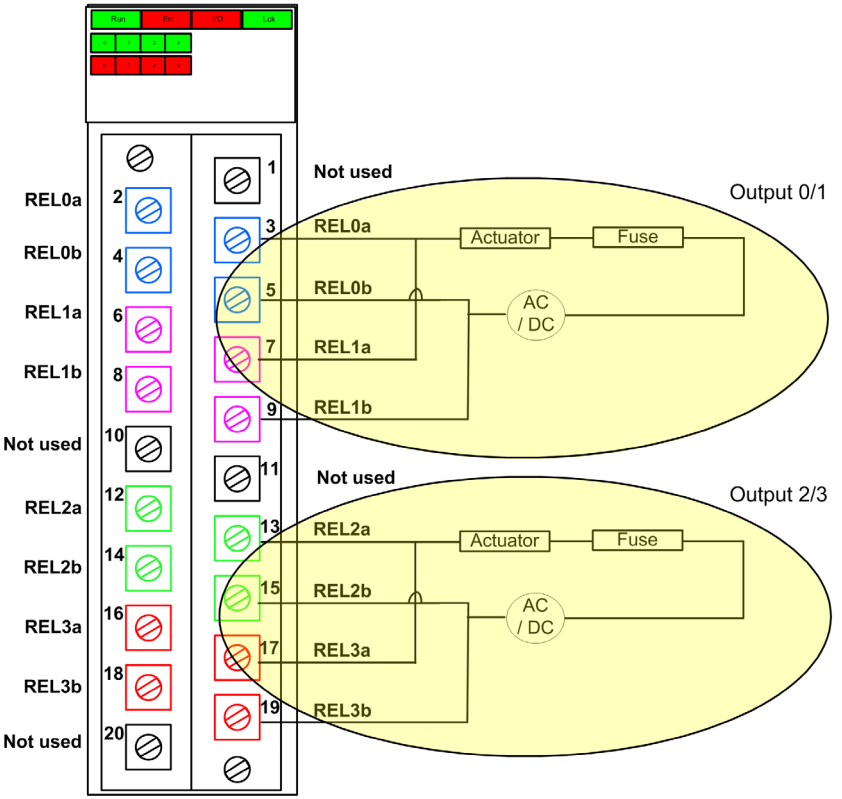
Wiring Diagram A

This wiring diagram applies to Application_1 and Application_3:



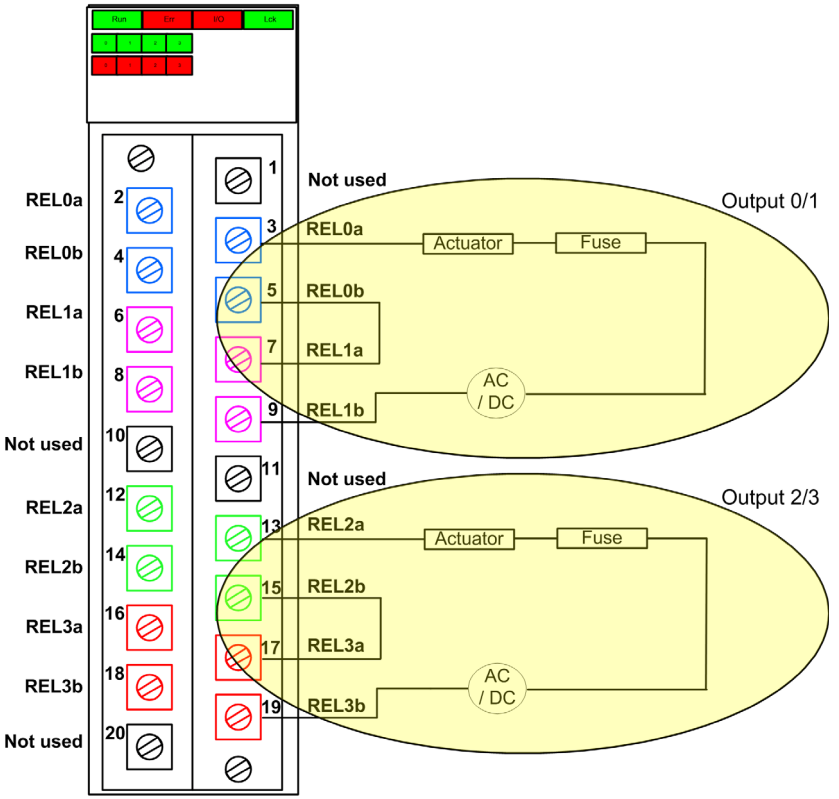
Wiring Diagram B

This wiring diagram applies to Application_2:



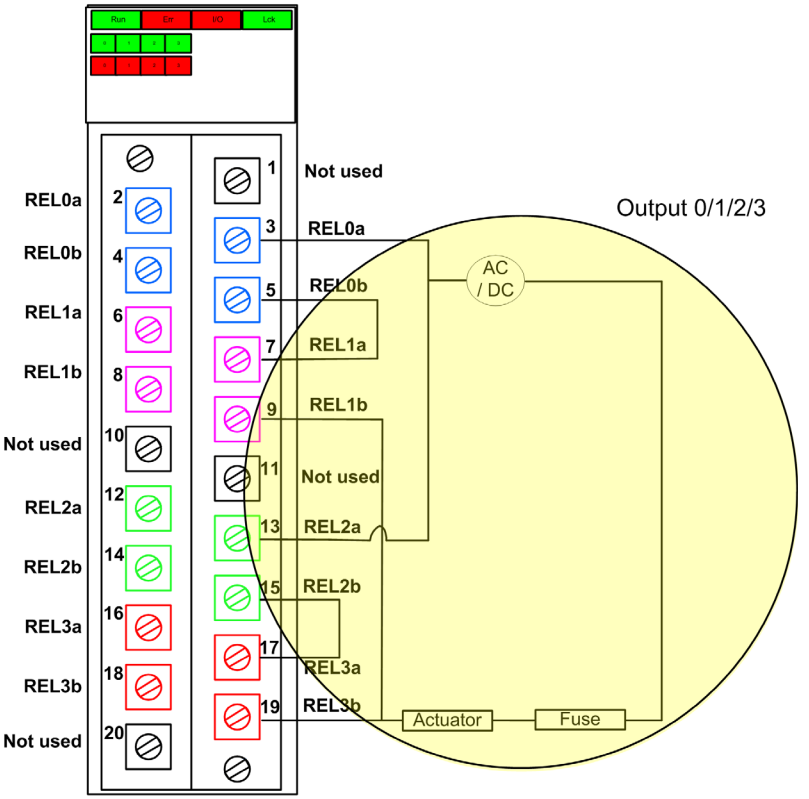
Wiring Diagram C

This wiring diagram applies to Application_4, Application_5, Application_7 and Application_8:



Wiring Diagram D

This wiring diagram applies to Application_6:



BMXSRA0405 Data Structure

Introduction

The T_U_DIS_SIS_OUT_4 device derived data type (DDDT) is the interface between the BMXSRA0405 relay output module and the application running in the CPU. The T_U_DIS_SIS_OUT_4 DDDT incorporates the data types T_SAFE_COM_DBG_OUT and T_U_DIS_SIS_CH_ROUT.

All of these structures are described, below.

T_U_DIS_SIS_OUT_4 DDDT Structure

The T_U_DIS_SIS_OUT_4 DDDT structure includes the following elements:

Element	Data Type	Description	Access
MOD_HEALTH[1]	BOOL	• 1: The module is operating correctly. • 0: The module is not operating correctly.	RO
SAFE_COM_STS[1]	BOOL	• 1: Module communication is valid. • 0: Module communication is not valid.	RO
CONF_LOCKED	BOOL	• 1: Module configuration is locked. • 0: Module configuration is not locked.	RO
APPLI	UINT	Relay application configuration: 1, 2, 3, 4, 5, 6, 7 or.	RO
TIME_PERIOD	UINT	Timer period for relay automatic monitoring (in minutes).	RO
S_COM_DBG	T_SAFE_COM_DBG_OUT	Safe communication debug structure.	RO
CH_OUT	ARRAY[0...3] of T_U_DIS_SIS_CH_ROUT	Array of structure of channel.	–
S_TO	UINT	Safety control timeout before module enters fallback state.	RO
MUID[2]	ARRAY[0...3] of DWORD	Module unique ID (auto-assigned by Unity Pro)	RO
RESERVED_1	ARRAY[0...7] of INT	–	–
RESERVED_2	ARRAY[0...6] of INT	–	–

1. When the SAFE task on CPU is not in running mode, the data exchanged between the CPU and the module are not updated and MOD_HEALTH and SAFE_COM_STS are set to 0.
2. This auto-generated value can be changed by executing the **Build → Renew Ids & Rebuild All** command in the Unity Pro main menu.

T_SAFE_COM_DBG_OUT Structure

The T_SAFE_COM_DBG_OUT structure includes the following elements:

Element	Data Type	Description	Access
S_COM_EST	BOOL	• 1: Communication with the module is established. • 0: Communication with the module is not established or corrupted.	RO
M_NTP_SYNC	BOOL	• 1: The module is synchronized with the NTP server. • 0: The module is not synchronized with the NTP server.	RO
CPU_NTP_SYNC	BOOL	• 1: The CPU is synchronized with the NTP server. • 0: The CPU is not synchronized with the NTP server.	RO
CHECKSUM	BYTE	Communication frame checksum.	RO
COM_DELAY	UINT	Communication delay between two values received by the module: • 0: No delay. The communication was received on time. • 1...65534: The time, in ms, since the last communication was received by the CPU from the module. • 65535: The CPU did not receive a communication from the module.	RO
COM_TO	UINT	Communication time-out value for communications coming from the module.	R/W
STS_MS_IN	UINT	NTP timestamp value for the fraction of a second, to the nearest ms, of the data received from the module.	RO
S_NTP_MS	UINT	NTP time value for the fraction of a second, to the nearest ms, for the current cycle.	RO
STS_S_IN	UDINT	NTP timestamp value in seconds of the data received from the module.	RO
S_NTP_S	UDINT	NTP time value in seconds for the current cycle.	RO
CRC_IN	UDINT	CRC value for data received from the module.	RO
STS_MS_OUT	UINT	NTP timestamp value for the fraction of a second, to the nearest ms, of the data to be sent to the module.	RO

Element	Data Type	Description	Access
STS_S_OUT	UDINT	NTP timestamp value in seconds of the data to be sent to the module.	RO
CRC_OUT	UDINT	CRC value for data to be sent to the module.	RO

T_U_DIS_SIS_CH_ROUT Structure

The T_U_DIS_SIS_CH_ROUT structure includes the following elements:

Element	Data Type	Description	Access
CH_HEALTH[1]	BOOL	• 1: The channel is operational. • 0: An error has been detected on the channel, which is not operational.	RO
VALUE	EBOOL	Safe command of output channel: • 1: Command the output closed (energized). • 0: Command the output open (de-energized).	R/W
TRUE_VALUE[2]	BOOL	Read back value of the relay output channel: • 1: The output is closed (energized). • 0: The output is open (de-energized).	RO
IC	BOOL	• 1: Invalid channel detected by the module. • 0: The channel is declared internally operational by the module.	RO
CH_FBC	BOOL	Configuration status of the channel fallback configuration: • 1: User defined value. • 0: Hold last value.	RO
CH_FBST	BOOL	Configuration status of the user-defined channel fallback state configuration: • 1: Enabled. • 0: Disabled.	RO
1. When the SAFE task on CPU is not in running mode, the data exchanged between the CPU and the module are not updated and CH_HEALTH is set to 0. 2. The TRUE_VALUE element can be time-stamped by the BMX CRA or the BME CRA.			

Chapter 7
M580 Safety Power Supply

Introduction

This chapter describes the BMXCPS4002S safety power supply module.

What Is in This Chapter?

This chapter contains the following topics:

Topic	Page
BMXCPS4002S Power Supply Safety	122
BMXCPS4002S Safety Power Supply Module Diagnostics	123

BMXCPS4002S Power Supply Safety

Introduction

The BMXCPS4002S power supply is certified for use in the M580 safety PAC.

⚠ WARNING

LOSS OF THE ABILITY TO PERFORM SAFETY FUNCTIONS

Use only the BMXCPS4002S power supply in any rack that includes an M580 safety module. Check both your physical installation and your project in Unity Pro to confirm that only BMXCPS4002S power supply modules are used.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Power Supply Functionality

The BMXCPS4002S power supply module converts 110...240 Vac power into two output voltages, 24 Vdc and 3.3 Vdc, which are distributed over the backplane.

The power detects over voltage, overload, and short-circuit conditions on both the 3.3 Vdc and 24 Vdc backplane lines, and permits up to a maximum voltage of 36 Vdc.

If the 36 Vdc upper threshold is detected, the module takes the following responsive actions:

- a reset is performed, causing the modules that receive power from the power supply to be re-initialized.
- if the upper voltage threshold was detected on the:
  - 24 Vdc backplane line: the PAC is powered down.
  - 3.3 Vdc backplane line: the PAC operation stops, but the PAC continues to receive power.

Refer to the topic *Diagnostics for the 24 Vdc and 3.3 Vdc Backplane Voltages* *(see page 123)* for information on how to respond to these conditions.

Redundant Power Supply Modules

The BMXCPS4002S is a redundant power supply module. Two BMXCPS4002S power supply modules can be installed - one as master one as slave - in a redundant Ethernet rack. For more information on redundant power supplies, refer to the topic *Redundant Power Supply Modules* in the *Modicon M580 Hardware Reference Manual*

BMXCPS4002S Safety Power Supply Module Diagnostics

Diagnostics for the 24 Vdc and 3.3 Vdc Backplane Voltages

The BMXCPS4002S safety power supply automatically provides detection for an overvoltage, overload, or short-circuit condition that may occur with respect to both the 24 VDC and 3.3 VDC Backplane Voltages.

If the power supply detects one of these conditions on the 24 Vdc voltage, the following occurs:

- The power conversion function is shut down for the entire backplane.
- A RESET command is issued for all modules in the rack.
- The power supply **OK** LED is turned OFF.
- The entire PAC is powered down.

If the power supply detects one of these conditions on the 3.3 Vdc voltage, the following occurs:

- The power conversion function is shut down for the 3.3 Vdc backplane voltage.
- A RESET command is issued for all modules in the rack.
- The power supply **OK** LED is turned OFF.
- Operation of the entire PAC program is stopped, although some PAC circuits may continue to receive power.

In either case, to recover from these conditions, take the following steps:

1. Power down the primary power line.
2. Check the compatibility between the estimated power supply consumption of the PAC against the capacity of the BMXCPS4002S safety power supply module on the 24 Vdc and 3.3 Vdc backplane lines.
3. Eliminate the cause of the underlying condition.
4. Wait for 1 minute after power down.
5. Apply power on the primary line to restart the BMXCPS4002S safety power supply module.

Alarm Relay Contact Diagnostics

The BMXCPS4002S safety power supply presents a two-pin alarm relay contact that you can use to obtain the following information:

- If the relay is activated (i.e. closed):
  - Both the 24 Vdc and 3.3 Vdc backplane voltages are OK; and
  - RESET is not active.; and
  - If the power supply is placed in the main local rack:
    - the CPU is operational, and
    - the CPU is in RUN mode.
- If the relay is de-activated (i.e. open):
  - Either or both of the 24 Vdc and 3.3 Vdc backplane voltages are not OK; or
  - RESET is active; or
  - If the power supply is placed in the main local rack:
    - the CPU is not operational, or
    - the CPU is in STOP mode.

Chapter 8
Validating an M580 Safety System

Introduction

This chapter shows you how to perform calculations that validate your M580 safety system.

What Is in This Chapter?

This chapter contains the following sections:

Section	Topic	Page
8.1	M580 Safety Module Architectures	126
8.2	M580 Safety Module SIL & MTTF Values	133
8.3	M580 Safety System Performance and Timing Calculations	139

Section 8.1
M580 Safety Module Architectures

Introduction

This section presents the internal architectures of the safety modules.

What Is in This Section?

This section contains the following topics:

Topic	Page
M580 Safety CPU and Coprocessor Safety Architecture	127
BMXSAI0410 Analog Input Module Safety Architecture	129
BMXSDI1602 Digital Input Module Safety Architecture	130
BMXSDO0802 Digital Output Module Safety Architecture	131
BMXSRA0405 Digital Relay Output Module Safety Architecture	132

M580 Safety CPU and Coprocessor Safety Architecture

Introduction

The BMEP584040S CPU and the BMEP58CPROS3 Coprocessor (Copro) are certified by the TÜV Rheinland Group for use in Safety Integrity Level 3 (SIL3) M580 safety solutions.

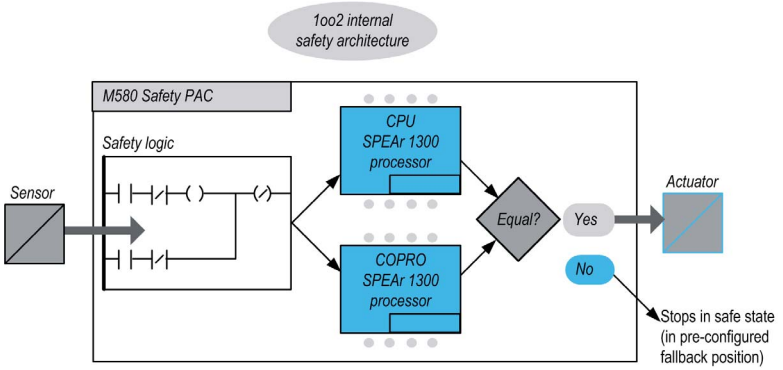
Working together, the CPU and Copro provide the following SIL3 safety level functions:

- Independent double execution of the safety task code.
- Comparison of the results of the double code execution.
- Periodic self-tests.
- Support for a 1oo2 (“one out of two”) architecture.

NOTE: In addition to the safety functionality, the BMEP584040S CPU also provides comparable features of equivalent non-safety M580 CPUs. Refer to the *M580 Hardware Reference Guide* for information regarding the non-safety features of the BMEP584040S CPU.

Description of the Internal CPU & Copro Architecture

The M580 safety CPU and Copro each contains a SPEAr 1300 processor. Each processor executes the safety logic in its own memory area, and compares the results of the execution at the end of the safe task. The following figure shows the internal architecture of the M580 Safety CPU:



Double Code Generation and Execution

The two processors inside the M580 safety PAC provide for double code generation and execution. This diversity provides the following advantages in error detection:

- Two executable code programs are generated independently. The use of two independent code compilers aids in the detection of systemic errors in code generation.
- The two generated code programs are executed by two separate processors. Thus, the CPU can detect both systematic errors in the code execution and random errors in the PAC.
- Each of the two processors uses its own independent memory area. Thus, the PAC can detect random errors in the RAM, and a full RAM test is not necessary at every scan.

1oo2 Architecture

1oo2 (“one out of two”) architecture means that two independent channels execute the safety process and, if an error is detected on either channel, the safety function is activated and a shut-down occurs.

Watchdog

A hardware and a firmware watchdog check the PAC activity and the time required to execute the safety program logic.

NOTE: Configure the software watchdog (in the **Properties of SAFE** dialog) to allow for:

- application execution time
- filtering of any detected I/O communication errors
- process safety time.

For more information, refer to the topic *Process Safety Time* *(see page 139)*.

Memory Check

The integrity of the content of static memory is tested via cyclic redundancy check (CRC) and the double code execution. The integrity of the content of dynamic memory is tested by double code execution, by a periodic memory test, and by an error correcting code (ECC) mechanism that detects and corrects the most common instances of corrupted internal data. At cold start, these tests are re-initialized and fully performed before the CPU goes into Stop or Run mode.

Over Voltage Monitoring

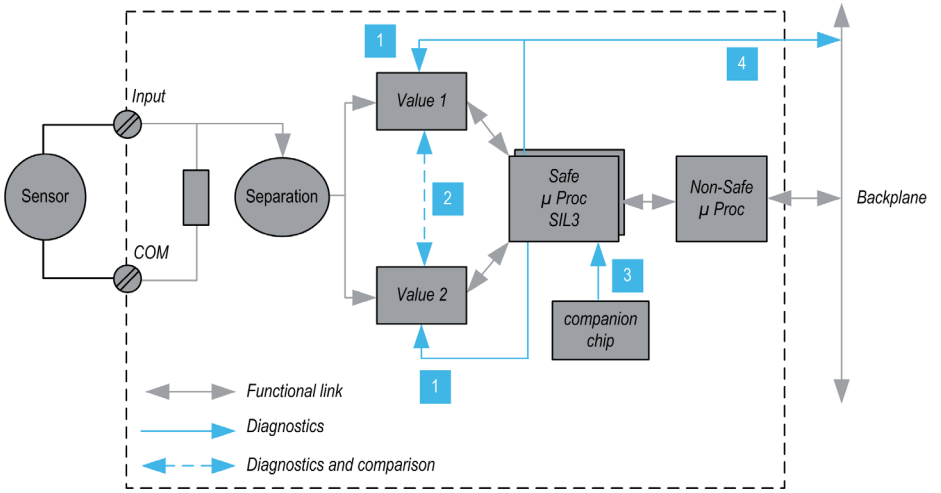
The CPU receives power from the dedicated BMXCPS4002S safety power supply module over the backplane line. The power supply module provides a regulated 24V with an absolute maximum voltage in the range 0...36V.

Embedded in the CPU is an embedded function that checks the internal power supplies. If an undervoltage or overvoltage condition is detected, the PAC shuts down.

BMXSAI0410 Analog Input Module Safety Architecture

Safety Function Architecture

The BMXSAI0410 module internal architecture performs its safety function as follows:

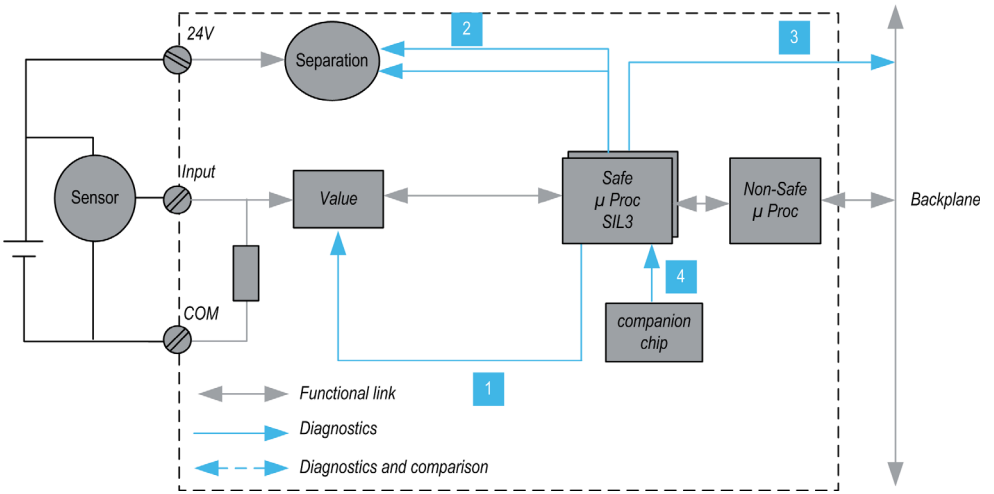


1. The measuring devices are regularly monitored for their ability to measure, without a detected error, 10 analog values between 4 and 20 mA. The linearity of the measuring stages is verified at the same time.
2. Each input value is acquired by 2 identical circuits. The measuring values are compared by the safety processor. If values are different, that channel is determined to be not valid. A maximum discrepancy of 0.35% of the 20 mA full scale range is tolerable between the two values.
3. The companion chip supplies the safety processor, continuously diagnoses the safety processor, and monitors backplane voltage.
4. The supply voltage from the backplane is monitored to detect if an over voltage or under voltage condition occurs.

BMXSDI1602 Digital Input Module Safety Architecture

Safety Function Architecture

The BMXSDI1602 module internal architecture performs the safety function as follows:

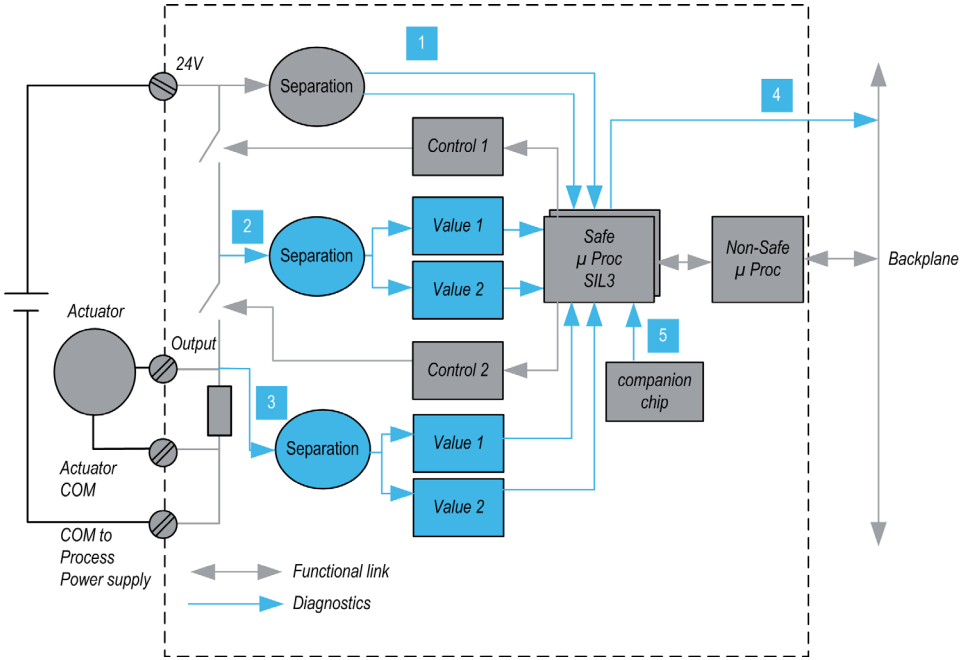


1. The measuring devices are continuously monitored for their ability to measure a "1" and a "0".
2. The external 24 Vdc power supply is continuously monitored by the safety processor. Each input value is acquired by two identical circuits. The acquired values are compared by the safety processor. If values are different, the channel is declared not valid.
3. The supply voltage from the backplane is monitored to detect an over or under voltage condition.
4. The companion chip supplies the safety processor, continuously diagnoses the safety processor, and monitors backplane voltage.

BMXSDO0802 Digital Output Module Safety Architecture

Safety Function Architecture

The BMXSDO0802 module internal architecture performs the safety function as follows:



1 The external 24 Vdc power supply is continuously monitored by the safety processor.

2 Each output consists of 2 switches in series between the external +24 Vdc power supply and the ground. The mid-point value (2) is redundantly read and sent to safety processor. The mid-points measuring values are compared by the safety processor. If the values are not the expected ones, the channel is declared not valid.

3 The low-point value (3) is also supervised in order to perform external wiring diagnostic.

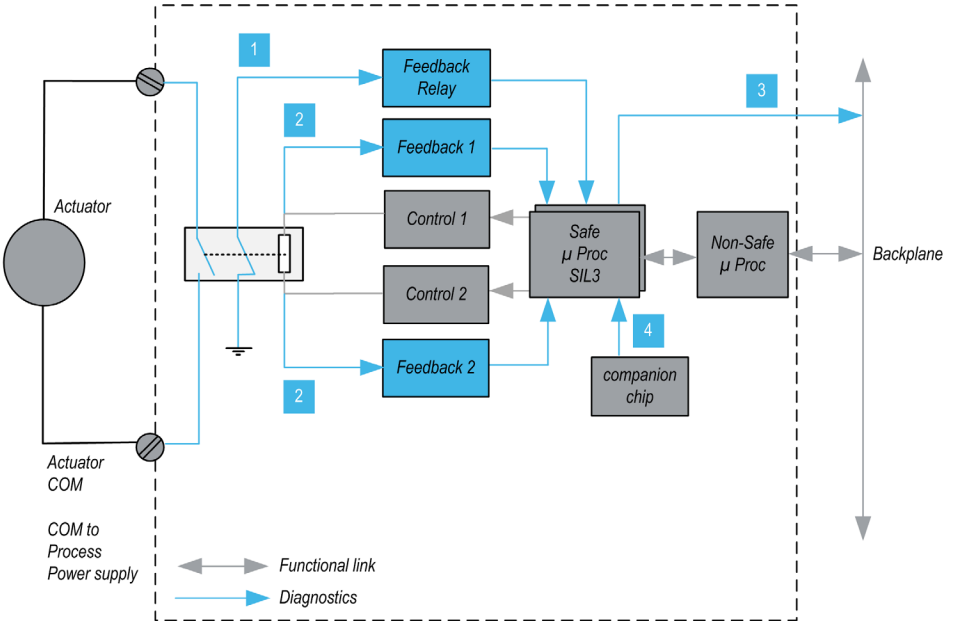
4 The supply voltage from the backplane is monitored to determine if there is an over voltage or under voltage condition.

5 The companion chip supplies the safety processor, continuously diagnoses the safety processor, and monitors backplane voltage.

BMXSRA0405 Digital Relay Output Module Safety Architecture

Safety Function Architecture

The BMXSRA0405 module internal architecture performs the safety function as follows:



1 The state of the relay is continuously monitored by the safety processor, which reads the state of an NC contact mechanically linked to the NO contact, and which in turn is linked to the actuator.

2 The state of the relay command is continuously monitored. Each input is received by 2 identical circuits. The measured values are compared by the safety processor. If values are different, the channel is declared to be not valid.

3 The supply voltage from the backplane is monitored to determine if there is an over voltage or under voltage condition.

4 The companion chip supplies the safety processor, continuously diagnoses the safety processor, and monitors backplane voltage.

Section 8.2
M580 Safety Module SIL & MTTF Values

Safety Integrity Level Calculations

Classification of the Schneider Electric Products

The M580 safety PAC can consist of:

- Safety modules, which can perform safety functions, including:
  - CPU and coprocessor
  - I/O modules
  - power supply
- Non-interfering modules *(see page 23)*, which do not perform safety functions, but enable you to add non-safety elements to your safety project.

NOTE:

- Because non-interfering modules are not part of the safety loop, they are not part of safety integrity level calculations.
- An error detected in a non-interfering module does not negatively impact the execution of the safety functions.
- The BMXCPS4002S power supply is certified. Because it presents a negligible dangerous failure rate (<1% of the SIL3 target), it is not included in safety integrity level calculations for the safety loop. As a consequence, neither PFH nor PFD are provided for the module.

PFD/PFH Values for M580 Safety Modules

Schneider Electric offers the following safety modules certified for use in safety applications. The safety modules are listed with their corresponding probabilities of failure *(see page 136)* (PFD/PFH) values for different proof test intervals *(see page 138)* (PTIs). The PFD/PFH are expressed as values that contribute to the overall PFD/PFH of the entire safety loop *(see page 16)*.

The tables below list the safety modules and their PFD/PFH values for SIL2 and SIL3 applications, where applicable:

Product Type	Product Reference	SIL	PTI = 1 year	
			PFD_G	PFH_G
CPU with Copro	BMEP584040S with BMEP58CPROS3	SIL3[1]	2.38E-06	5.44E-10
Analog input	BMXSAI0410	SIL3[2]	5.76E-06	1.31E-09
Digital input	BMXSDI1602	SIL3[2]	6.81E-06	1.56E-09
Digital output	BMXSDO0802	SIL3[1]	5.75E-06	1.31E-09
Digital relay output	BMXSRA0405	SIL2 [3]	5.85E-06	1.58E-09
		SIL3[4]	5.84E-06	1.34E-09
		SIL3[5]	–	1.35E-09
Power supply	BMXCPS4002S	SIL3	–	–
1. 1 output @ 80° C 2. 1 input @ 80° C 3. 1 relay per output @ 80° C 4. 2 relays per output @ 80° C 5. 4 relays per output @ 80° C				

Product Type	Product Reference	SIL	PTI = 5 years	
			PFD_G	PFH_G
CPU & Copro	BMEP584040S & BMEP58CPROS3	SIL3 [1]	1.22E-05	5.69E-10
Analog input	BMXSAI0410	SIL3[2]	2.88E-05	1.31E-09
Digital input	BMXSDI1602	SIL3[2]	3.41E-05	1.56E-09
Digital output	BMXSDO0802	SIL3[1]	2.88E-05	1.31E-09
Digital relay output	BMXSRA0405	SIL2 [3]	2.92E-05	1.68E-09
		SIL3[4]	2.92E-05	1.34E-09
		SIL3[5]	–	1.35-09
Power supply	BMXCPS4002S	SIL3	–	–
1. 1 output @ 80° C 2. 1 input @ 80° C 3. 1 relay per output @ 80° C 4. 2 relays per output @ 80° C 5. 4 relays per output @ 80° C				

Product Type	Product Reference	SIL	PTI = 10 years	
			PFD_G	PFH_G
CPU & Copro	BMEP584040S & BMEP58CPROS3	SIL3[1]	2.54E-05	6.00E-10
Analog input	BMXSAI0410	SIL3[2]	5.76E-05	1.31E-09
Digital input	BMXSDI1602	SIL3[2]	6.81E-05	1.56E-09
Digital output	BMXSDO0802	SIL3[1]	5.75E-05	1.31E-09
Digital relay output	BMXSRA0405	SIL2 [3]	5.84E-05	1.68E-09
		SIL3[4]	5.84E-05	1.34E-09
		SIL3[5]	–	1.35E-09
Power supply	BMXCPS4002S	SIL3	–	–
1. 1 output @ 80° C 2. 1 input @ 80° C 3. 1 relay per output @ 80° C 4. 2 relays per output @ 80° C 5. 4 relays per output @ 80° C				

Product Type	Product Reference	SIL	PTI = 20 years	
			PFD_G	PFH_G
CPU & Copro	BMEP584040S & BMEP58CPROS3	SIL3[1]	5.44E-05	6.63E-10
Analog input	BMXSAI0410	SIL3[2]	1.15E-04	1.31E-09
Digital input	BMXSDI1602	SIL3[2]	1.36E-04	1.56E-09
Digital output	BMXSDO0802	SIL3[1]	1.15E-04	1.31E-09
Digital relay output	BMXSRA0405	SIL2 [3]	1.17E-04	1.68E-09
		SIL3[4]	1.17E-04	1.34E-09
		SIL3[5]	–	1.35E-09
Power supply	BMXCPS4002S	SIL3	–	–
1. 1 output @ 80° C 2. 1 input @ 80° C 3. 1 relay per output @ 80° C 4. 2 relays per output @ 80° C 5. 4 relays per output @ 80° C				

MTTF Values for M580 Safety Modules

The M580 safety modules present the following MTTF characteristics:

Product Type	Product Reference	MTTF (years)		
		25° C	55° C	>60° C
CPU	BMEP584040S	87.2	44.9	24.4
Coprocessor	BMEP58CPROS3	108.0	56.0	30.1
Analog input	BMXSAI0410	54.2	26.1	14.2
Digital input	BMXSDI1602	31.5	13.0	6.1
Digital output	BMXSDO0802	45.8	25.0	13.4
Digital relay output	BMXSRA0405	36.9	26.5	17.9
Power supply	BMXCPS4002S	88.0	47.5	25.4

Probabilities of Failure for SIL3 Applications

For SIL3 applications, the IEC 61508 defines the following probabilities of failure on demand (PFD) and probabilities of failure per hour (PFH) for each safety loop, depending on the mode of operation:

- PFD ≥ 10^{-4} to < 10^{-3} for low demand mode of operation
- PFH ≥ 10^{-8} to < 10^{-7} for high demand mode of operation

The M580 Safety PAC is certified for use in low and high demand systems.

Safety Integrity Level Sample Calculation

This sample calculation shows you how to determine:

- The risk contribution of the Schneider Electric safety modules to your safety application; and
- The remaining amount of risk that other devices in the safety loop (for example, sensors and actuators) can contribute to your safety application for a given safety integrity level and mode of operation.
  NOTE: When calculating the risk contribution of sensors and actuators to your safety application, contact the manufacturers of these devices and obtain the PFD/PFH values for the appropriate proof test interval.

The following Schneider Electric safety modules are included in this example:

- 1: BMEP584040S CPU
- 1: BMEP58CPROS3 Copro
- 1: BMXSAI0410 Analog input
- 1: BMXSDO0802 Digital output
- 1: BMXCPS4002S Power supply

The following calculation employs PFH_G values for a high demand mode of operation for a SIL3 safety loop with a PTI of 20 years. The maximum permissible PFH value for this safety application is 10^{-7} (or 1.0E-7):

Safety module		Contribution (Scientific Notation)	Remaining Contribution for Sensors & Actuators
CPU with Copro		6.63E-10	–
Analog input		1.31E-09	
Digital output		1.31E-09	
Power Supply		–	
Total	numeric	**3.28E-9**	**96.72E-9**
	% max	**3.28%**	**96.72%**
note 1: The relay output uses four relays to support one output.			

Values for M580 Safety Modules for Machinery

Schneider Electric offers the following safety modules certified for use in safety machinery applications according to ISO13849-1 standard. The table below list the safety modules and their values, category and level where applicable:

Product Type	Product Reference	Configuration	Category	Performance Level	MTTFd (years)	DCav
CPU with Copro	BMEP584040S & BMEP58CPROS3	NA	4	e	4228	High (>99%)
Analog input	BMXSAI0410	using 1 channel	2	c	255	99.66%
		using 2 channels	4	e	255	99.66%
Digital input	BMXSDI1602	using 1 channel	2	c	231	99.69%
		using 2 channels	4	e	231	99.69%
Digital output	BMXSDO0802	NA	4	e	253	99.63%
Digital relay output	BMXSRA0405	using 1 channel	2	e	156	99.77%
		using 2 channels	4	c	156	99.77%

Safety Times Description

The M580 safety PAC has a minimum PAC cycle time of 10 ms, which is necessary for processing the signals from the I/O modules, executing the program logic, and setting the outputs. For calculating the maximum PAC reaction time, you need to know the maximum reaction time of the sensors and actuators that are being used. In addition, the maximum PAC reaction time depends on the process safety time (PST) *(see page 139)* required for your process.

Proof Test Interval

The proof text is a periodic test you need to perform to detect failures in a safety-related system so that, if necessary, the system can be restored to a like new condition or as close as practical to this condition. The time period between these tests is the proof test interval.

The proof test interval depends on the targeted safety integrity level, the sensors, actuators and the PAC application. The M580 safety system is suitable for use in a SIL3 application and a proof test interval of 20 years.

Section 8.3
M580 Safety System Performance and Timing Calculations

Process Safety Time

Description of the Process Safety Time

The process safety time (PST) is an essential measure of a process executed by a safety loop. It is defined as the period between the occurrence of a failure in equipment under control (EUC) and the occurrence of a hazardous event if the safety function is not performed (i.e. if the safe state is not achieved).

NOTE: The process safety time is determined by your specific safety process. You need to verify that your safety-related system can perform its safety functions within the process safety time.

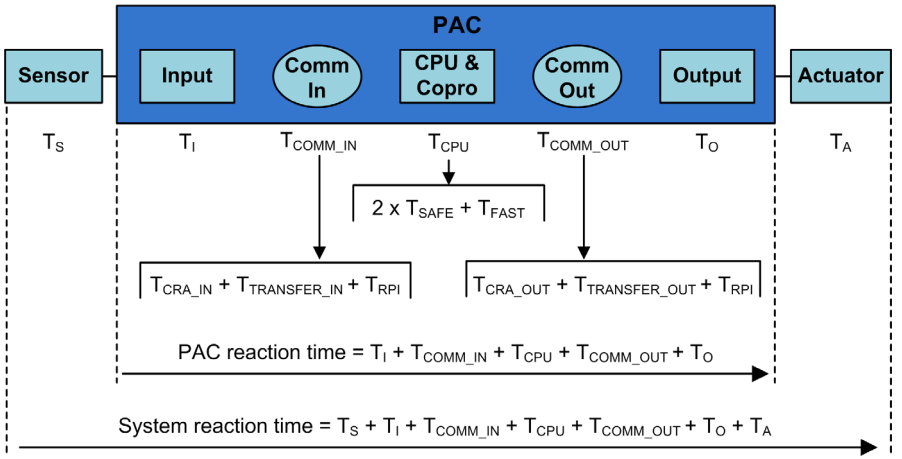
Description of the System Reaction Time

The system reaction time is the sum of the PAC reaction time, plus the reaction times for both the selected sensor (T_S) and the selected actuator (T_A).

NOTE: T_S and T_A are device specific.

For each safety loop, verify that the system reaction time is less than the process safety time.

System reaction time is illustrated below:



The components of the system reaction time can include the following:

Component	Description	Estimated Worst Case Value
T_S	Reaction time required by the selected sensor to react to a process event.	Device specific.
T_I	Time required by the input module to sample and confirm a sensor event.	6 ms
T_{COMM_IN}	Input communication delay. Equals the sum of the following times: $T_{CRA_IN} + T_{TRANSFER_IN} + T_{RPI}$ **NOTE:** Applies primarily to an input module located in a remote rack. For an input module in the same rack as the CPU, this value is so small as to be negligible.	–
T_{CRA_IN}	Time required by a CRA to sample a safety input module.	2 ms
$T_{TRANSFER_IN}$	Time required to transfer the input value to the CPU, including: ● The delay in switches and CRA. ● The delay to transfer and analyze a full assembly. ● The jitter introduced by the multi tasking (wait to finish a lower priority transfer, wait to perform all higher priority transfers, and so forth). **NOTE:** This time is not known. It is assumed to be a measurable value in addition to all the other worse case conditions.	–
T_{RPI}	The refresh period for the current assembly. **NOTE:** By default, this value is equal to one-half (0.5) of the configured SAFE task period.	–
T_{CPU}	The CPU and coprocessor reaction time. Equals the sum of the following times: $T_{SAFE} + T_{JITTER}$ where: $T_{JITTER} = T_{SAFE} + T_{FAST}$. Thus, maximum (i.e. worst case) $T_{CPU} = 2 \times T_{SAFE} + T_{FAST}$	–
T_{FAST}	This value includes the time spent executing higher priority tasks, i.e. the FAST task. **NOTE:** To simplify the formula, it is assumed that no system task is in an overrun condition. Thus, this value equals the configured FAST task period, or 0 if the FAST task is not configured.	–
T_{SAFE}	The configured SAFE task period.	–

Component	Description	Estimated Worst Case Value
T_{COMM_OUT}	Output communication delay. Equals the sum of the following times: $T_{CRA_OUT} + T_{TRANSFER_OUT} + T_{RPI}$ **NOTE:** Similar to T_{COMM_IN}, this value applies only to an output module located in a remote rack, but not to an output module in the same local rack as the CPU. For an input module in the same rack as the CPU, this value is so small as to be negligible	–
$T_{TRANSFER_OUT}$	Time required to transfer the output value from the CPU to the output module, including: ● The delay in switches and CRA. ● The delay to transfer and analyze a full assembly. ● The jitter introduced by the multi tasking (wait to finish a lower priority transfer, wait to perform all higher priority transfers, and so forth). **NOTE:** This time is not known. It is assumed to be a measurable value in addition to all the other worse case conditions.	0 ms
T_{CRA_OUT}	Time required by a CRA to write to a safety output module.	2 ms
T_O	Equal to the sum of the following times: ● Delay time between reading and applying the CPU output value (0...3 ms). ● Time required by the safety output module to modify the physical output, i.e. to propagate the change from X ram to the physical output (between 0...3 ms).	6 ms
T_A	Reaction time for the selected actuator.	Device specific.

Description of the PAC Reaction Time

For I/O placed in the local main rack (with the CPU) the PAC reaction time is the sum of the related reaction times for both the selected input module (T_I) and the selected output module (T_O), plus the CPU & Copro reaction time (T_{CPU}):

PAC reaction time (local) = $T_{CPU} + T_I + T_O$

If the I/O are located in a remote rack, the PAC reaction time also includes input communication delay (T_{COMM_IN}) and output communication delay (T_{COMM_OUT}) times:

PAC reaction time (remote) = $T_{CPU} + T_{COMM_IN} + T_I + T_{COMM_OUT} + T_O$

Description of the CPU & Copro Reaction Time

The CPU & Copro reaction time is directly impacted by both the SAFE task period and the FAST task period. Verify that safety logic will be executed within the SAFE task period. Because the FAST task takes priority over the SAFE task, you also need to consider the time to execute the FAST task when estimating CPU & Copro reaction time.

Because a signal may appear just at the beginning of the execution cycle when the signals have already been processed, two cycles may be necessary to react to the signal.

This leads to the following equation for the maximum (i.e. worst case) reaction time:

CPU & Copro reaction time = $2 \times T_{SAFE} + T_{FAST}$.

NOTE: If you are using peer-to-peer safe communication *(see page 159)* to perform the safety function, the CPU reaction time estimation is different.

Description of the Time for Input Modules

The maximum times (worst case) for the safety digital input module and for the safety analog input module T_I are 6 ms

Description of the Time for Output Modules

The maximum time T_O for the safety digital output module is estimated to be 6 ms.

For the safety digital output module, a fallback timeout T_{OUT} needs to be configured. The module timeout value needs to be greater than the CPU SAFE task period *(see page 142)*.

Computation of System Reaction Time

Knowing the required process safety time (PST) and the maximum reaction time of the sensors and actuators, you are able to calculate the maximum system reaction time (SRT) tolerable in your process.

The maximum (i.e. worst case) system reaction time can be computed as follows:

For systems with I/O in remote drops:

Max SRT = $T_S + T_I + 2 \times (T_{CRA} + T_{RPI}) + 2 \times T_{SAFE} + T_{FAST} + T_O + T_A$.

or

Max SRT = 16 ms + $T_S + 2 \times T_{SAFE} + T_{FAST} + T_A$.

For systems with local I/O:

Max SRT = $T_S + T_I + 2 \times T_{SAFE} + T_{FAST} + T_O + T_A$.

or

Max SRT = 15 ms + $T_S + 2 \times T_{SAFE} + T_{FAST} + T_A$.

Configuring the Maximum CPU SAFE and FAST task Periods

The M580 safety PAC can perform only periodic execution for the SAFE and FAST tasks (cyclic execution is not supported for these tasks).

The SAFE task **Period** and the maximum allowed CPU **Watchdog** settings are configured in the **General** tab of the **Properties of SAFE** dialog. The safety digital output **Fallback Timeout** settings are configured in the **Configuration** tab for the output module.

Similarly, the FAST task **Period** and the maximum allowed CPU **Watchdog** settings are configured in the **General** tab of the **Properties of FAST** dialog.

NOTE:

- Permissible SAFE task period settings range is 10...255 ms, with a default value of 20 ms.
- Permissible FAST task period settings range is 1...255 ms, with a default value of 5 ms.
- Permissible watchdog settings range is 10...500 ms, with a default value of 250 ms.
- Permissible digital output fallback timeout settings range is 0...65535 ms, with a default value of 100 ms.

Verify that the watchdog setting is greater than the SAFE task period.

Check your CPU SAFE task period setting when commissioning your project. At this time, Unity Pro XLS provides the real time values from the PAC.

You can find this information in Unity Pro XLS in the **Task** tab using the menu entry **Tools → PLC Screen**.

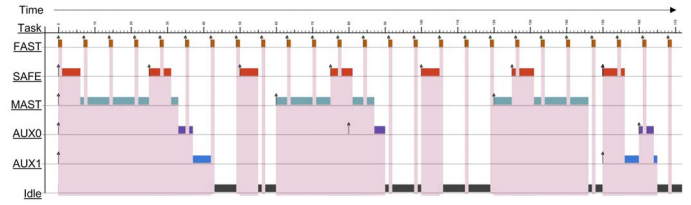
⚠ WARNING

RISK OF EXCEEDING THE PROCESS SAFETY TIME

Set the maximum CPU SAFE task period by taking into account your process safety time. Your CPU SAFE task period must be less than your project process safety time.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

The following drawing illustrates the execution of each task in a multi-task system, and depicts the preemption of CPU resources depending on the task priority:



FAST			SAFE			MAST			AUX0			AUX1			IDLE
Period 7 ms	Exe 1 ms	Bandwidth 14%	Period 25 ms	Exe 5 ms	Bandwidth 20%	Period 60 ms	Exe 18 ms	Bandwidth 30%	Period 80 ms	Exe 3 ms	Bandwidth 4%	Period 150 ms	Exe 5 ms	Bandwidth 3%	Free Bandwidth 29%

NOTE: For optimal CPU performance, Schneider Electric recommends that the 20% of CPU bandwidth remain idle.

Calculating the Impact of Task Execution Periods on CPU Bandwidth

Each configured task consumes a portion of CPU processing time, or bandwidth. The estimated percentage of CPU bandwidth consumed by a task is the result (or quotient) of the estimated execution time required by a task (E_{TASK}) divided by the configured execution period for that task (T_{TASK}), and can be presented as follows:

Task bandwidth = E_{TASK} / T_{TASK}.

Thus, the total percentage of CPU bandwidth consumed by an application is the sum of consumed CPU bandwidth percentages for all tasks.

NOTE: For optimal CPU performance, Schneider Electric recommends that the total percentage of CPU bandwidth consumed by an application not exceed 80%.

The following table presents two applications, and indicates the impact of high priority tasks (FAST and SAFE) on total CPU bandwidth usage:

#	FAST			SAFE			MAST			AUX0			Total
	Per	Exe	BW%	Per	Exe	BW%	Per	Exe	BW%	Per	Exe	BW%	
1	5 ms	1 ms	20%	20 ms	5 ms	25%	50 ms	18 ms	35%	200 ms	30 ms	15%	96%
2	7 ms	1 ms	14%	25 ms	5 ms	20%	60 ms	18 ms	30%	200 ms	30 ms	15%	79%
Per = Task period (T_{TASK}) Exe = Execution time required for the task (E_{TASK}) BW% = Task bandwidth.													

Chapter 9
Safety Library

Safety Library

Introducing the Safety Library

When you install Unity Pro XLS, a safety library of elementary functions (EFs), elementary function blocks (EFBs), and derived function blocks, (DFBs) are automatically included. These EFs, EFBs, and DFBs are identified by the prefix “S_” and are reserved for use in code sections managed by the SAFE task.

NOTE: Also installed is an additional collection of EFs, EFBs and DFBs. This is the same collection of data objects used by non-safety M580 PACs. These EFs, EFBs, and DFBs can be used only in code sections managed by process namespace tasks (MAST, FAST, AUX0, and AUX1).

For a description of the blocks included in the M580 safety library, refer to the *Unity Pro Safety Block Library* document.

Certified Safety Functions and Function Blocks

These are the subset of EFs and Functions Blocks, which can be used inside safety logic. These are provided in the Safety Library:

Family	Group or Name	Type	Description	Same as Safety Quantum library in Unity-Pro V7.0 XLS ?
Logic	S_AND_*, S_OR_*, S_XOR_*, S_NOT_*, S_SHL_*, S_SHR_*, S_ROR_*, S_ROL_*	EF	Type specific, e.g. S_AND with 2 to 32 inputs (inline code)	Yes
Logic	S_RS, S_SR, S_F_TRIG, S_R_TRIG	EFB	–	Yes
Mathematics	S_ADD_*, S_MUL_*, S_SUB_*, S_DIV_*, S_ABS_*, S_SIGN_*, S_NEG_*, S_MOVE, S_SQRT_REAL	EF	Type specific detected error handling (e.g. overflow) to be considered (inline code)	Yes
Comparison	S_GT_*, S_GE_*, S_LT_*, S_LE_*, S_NE_*, S_EQ_*	EF	Type specific (inline code)	Yes
Statistical	S_LIMIT_*, S_MAX_*, S_MIN_*, S_MUX_*, S_SEL	EF	Type specific (inline code)	Yes

Family	Group or Name	Type	Description	Same as Safety Quantum library in Unity-Pro V7.0 XLS ?
Type To Type	S_BOOL_TO_*, S_BYTE_TO_*, S_WORD_TO_*, S_DWORD_TO_*, S_INT_TO_*, S_UNIT_TO_*, S_UDINT_TO_*, S_BIT_TO_BYTE, S_BIT_TO_WORD, S_BYTE_TO_BIT, S_WORD_TO_BIT, S_TIME_TO_UDINT, S_REAL_TO_*	EF	Type specific (inline code)	Yes
Timers & Counters	S_CTU_*, S_CTD_*, S_CTUD_*	EFB	Type specific	Yes
Timers & Counters	S_TON, S_TOF, S_TP	EFB	–	Yes
Peer to peer	S_RD_ETH_MX, S_WR_ETH_MX	DFB	Functions to perform a Safety peer to peer communication	No
Actuator Connection	S_EDM, S_ENABLE_SWITCH, S_ESPE, S_OUTCONTROL, S_GUARD_LOCKING, S_GUARD_MONITORING, S_MODE_SELECTOR	DFB	Machine Safety Function Blocks linked to actuators	No
Sensor Connection	S_EQUIVALENT, S_ANTIVALENT, S_EMERGENCYSTOP, S_TWO_HAND_CONTROL_TYPE_II, S_TWO_HAND_CONTROL_TYPE_III, S_MUTING_SEQ, S_MUTING_PAR, S_AI_COMP	DFB	Machine Safety Function Blocks linked to sensors	No
System	S_SYST_STAT_MX, S_SYST_TIME_MX, S_SYST_CLOCK_MX, S_SYST_RESET_TASK_BIT_MX, S_SYST_READ_TASK_BIT_MX	EFB	System function blocks	No

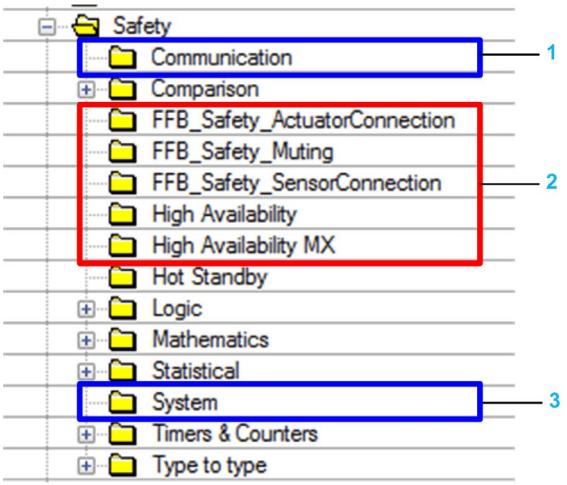
Non-Certified Safety Functions and Function Blocks

These are the subset of Derived Functions Blocks (DFBs), which can be used inside safety logic. These function blocks are not certified. Their purpose is to provide you sample safety function blocks that can be easily reused and adapted. You can copy and paste these function blocks into your application and change them to meet the requirements of your application.

Family	Group or Name	Type	Description
High Availability MX	S_DIHA, S_AIHA	DFB	Function for high availability SIL2 or SIL3 digital input modules (inline code)
Sensor Connection	AI_COMP	DFB	Machine Safety Function Blocks linked to sensors

Viewing the Safety Library in Unity Pro

You can access the safety library only from the SAFE task. When you open the safety library in the **FBD-Editor**, the safety library presents groups of EFs, EFBs, and DFBs. Some of these groups include safety versions of functions and blocks found in non-safety tasks. Others groups, noted below, contain functions and blocks unique to the SAFE task:



1 Blocks for reading and writing safety data values.
2 Blocks for performing safety-specific tasks.
3 Blocks for reading and writing safety system values.

For an example of how some of the safety blocks are implemented, refer to the PAC-to-PAC communication configuration example *(see page 162)*, which includes S_RD_ETH_MX and S_WR_ETH_MX.

Also refer to the *Unity Pro Safety Block Library* for a description of each available safety function and block.

Chapter 10
Data Separation in an M580 Safety System

Introduction

This chapter presents the division of data in an M580 safety system.

What Is in This Chapter?

This chapter contains the following topics:

Topic	Page
Data Separation in an M580 Safety Project	150
How to Transfer Data Between Namespace Areas	153

Data Separation in an M580 Safety Project

Data Separation and Scope

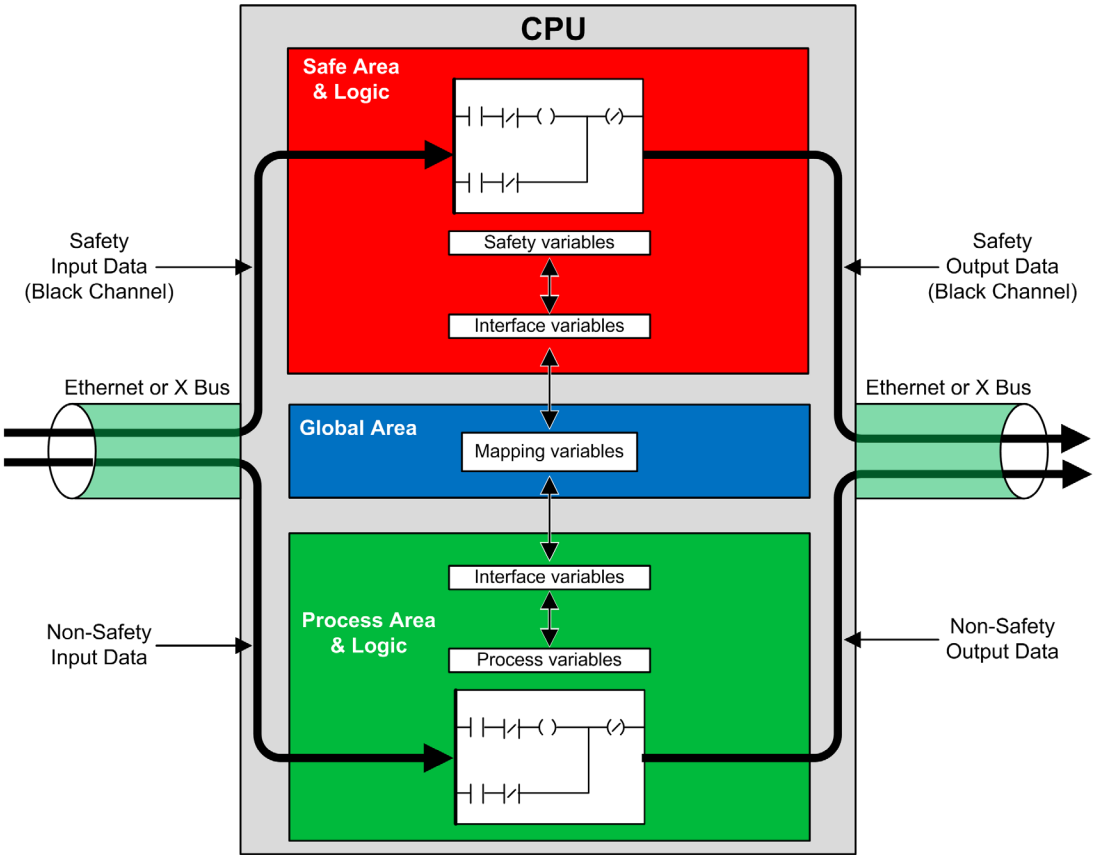
An M580 safety project includes both a safety program and a process (non-safety) program. Unity Pro isolates the logic and data used by the safety program from the logic and data used by the process program. Unity Pro accomplishes this by placing each part of the project into its own namespace (also called an area), either *safe* or *process*.

As a result of this design, the scope of a safety variable is restricted to the safe area, and the scope of a process variable is restricted to the process area. This becomes apparent when you add program logic to your application:

- When you configure an EF or EFB in the SAFE task, only variables created in the safe area are visible. Variables created in the process area are not visible.
- When you configure an EF or EFB in a non-safe (MAST, FAST, AUX0 or AUX1) task, only variables created in the process area are visible. Variables created in the safe area are not visible.

To permit communication between the safe area and the process area, Unity Pro also provides a *global* area. The global area serves as a pass-through for data transmissions between the safe area and the process area. This is accomplished by declaring interface variables in both the safety and process areas, then linking these interface variables to mapping variables declared in the global area.

This data separation in the M580 safety CPU and coprocessor is graphically described below:



Safe, Process and Global Area Properties

The three data areas of an M580 safety project present the following properties:

Area	Supported Variable Types	Scope	External Access
Global	Unlocated variables only. **NOTE:** Located variables cannot be used to map to a safety or process interface variable.	Can access: ● Safety variables, via namespace addressing. ● Process variables, via namespace addressing. ● Other global variables.	Variables from all three areas can be accessed by HMI, SCADA, or FactoryCast applications. (See Note, below.)
Safe	Unlocated variables only.	Can access only other safety variables.	
Process	Both: ● Located variables ● Unlocated variables	Can access only other process variables.	

When an external viewer seeks to read a process variable, the addressing format depends on whether the **Usage of Process Namespace** setting has been selected in the **Scope → common** area of the **Tools → Project Settings...** window. If the **Usage of Process Namespace** setting is

- Selected: the operator screen can read process area variables only by using the format “PROCESS.<variable name>”.
- De-selected: the operator screen can read process area variables only by using the format “<variable name>” without the PROCESS prefix. In this case, verify that each process variable name is unique, and is not the same as any global variable name.

NOTE: If the **Usage of Process Namespace** setting is de-selected, verify that each process variable name is unique, and is not the same as any global variable name. If a variable name is common to both the global and process areas, an error will be detected by Unity Pro when you build the project.

How to Transfer Data Between Namespace Areas

Introduction

The M580 safety PAC includes three different data editors:

- a **Safety Data Editor** to manage data used in the safe namespace.
- a **Process Data Editor** to manage data used in the process namespace.
- a **Global Data Editor** to manage global variables and data types used throughout the application.

Both the **Safety Data Editor** and the **Process Data Editor** include an **Interface** tab. Use the **Interface** tab to create unlocated variables in that process namespace. The **Interface** tab presents of two groups of unlocated variables:

- <inputs>: A variable created in this group can be linked to, and receive data from, a globally scoped pass-through variable in the **Global Data Editor**.
- <outputs>: A variable in this group can be linked to, and send data to, a globally scoped pass-through variable in the **Global Data Editor**.

NOTE: A variable created in either **Interface** tab needs to be all of the following:

- An unlocated variable.
- An EDT or DDT category variable.
- Of the same data type as the global variable to which it is linked.

Unlocated variables created in the **Interface** tab groups of the **Safety Data Editor** and **Process Data Editor** can be linked as follows:

A process variable in this group in the Process Data Editor...	Can be linked to a safety variable in this group in the Safety Data Editor...
<inputs>	<outputs>
<outputs>	<inputs>

Using these three data editors, you can configure the transfer of data between the safe namespace and the process namespace.

Transferring Data Between Namespaces

To pass data between the safe and process namespaces, follow these steps:

Step	Action
1	In the **Variables** tab of the **Global Data Editor**, create a new variable: ● Enter the new variable **Name**. ● Select a data **Type**. **NOTE:** This variable is an unlocated variable.
2	In the **Interface** tab of the **Process Data Editor**, create a new variable. If this variable will be the: ● Sender of data, create it in the **<outputs>** part of the data editor. ● Receiver of data, create it in the **<inputs>** part of the data editor.

Step	Action
3	For the new variable created in the **Interface** tab of the **Process Data Editor**: ● Enter the new variable **Name**. ● Select the same data **Type** as the variable created in the **Global Data Editor** in step 1, above. ● In the **Effective Parameter** column, select the name of the globally scoped variable you created in step 1, above. **NOTE:** If the **Effective Parameter** column is not visible, right-click on the column header bar, select **Customize Columns...**, then select **Effective Parameter** and click **OK**.
4	In the **Interface** tab of the **Safety Data Editor**, create a new variable. If this variable will be the: ● Sender of data, create it in the **<outputs>** part of the data editor. ● Receiver of data, create it in the **<inputs>** part of the data editor. **NOTE:** If you selected **<outputs>** in step 2, select **<inputs>** here; if you selected, **<inputs>** in step 2, select **<outputs>** here.
5	For the new variable created in the **Interface** tab of the **Safety Data Editor**: ● Enter the new variable **Name**. ● Select the same data **Type** as the variable created in the **Global Data Editor** in step 1, above. ● In the **Effective Parameter** column, select the name of the globally scoped variable you created in step 1, above.

After you compile, download and run the edited application program, the value is transferred as follows:

- The exchange of data from the global to the safe area is performed at the beginning of the SAFE task execution.
- The exchange of data from the safe to the global area is performed at the end of the SAFE task execution.
- he exchange of data from the global to the process area is performed at the beginning of the MAST task execution.
- The exchange of data from process to the global area is performed at the end of the MAST task execution.

Chapter 11
M580 Safety System Communications

Introduction

This chapter describes communications within the M580 safety system.

What Is in This Chapter?

This chapter contains the following sections:

Section	Topic	Page
11.1	NTP Service	156
11.2	Peer to Peer Communications	159
11.3	M580 CPU to Safety I/O Communication	176

Section 11.1
NTP Service

Configuring the NTP Service

Introduction

If you are installing safety I/O modules in an RIO drop, the current time needs to be configured for the PAC. To do this, enable the NTP service, then configure the safety CPU as either:

- An NTP server (then configure its time settings), or
- An NTP client (with another system device configured as the NTP server).

If one of these two designs is not implemented, the time settings of the safety I/O modules and CPU will not be synchronized and black channel communication will not operate properly.

⚠ CAUTION

RISK OF UNINTENDED OPERATION

If you are placing safety I/O modules in an RIO drop, the current time needs to be configured for the PAC. Enable the NTP service for your M580 system and configure the safety CPU as an NTP server or an NTP client. Failure to do so will cause the inputs and outputs of safety I/O modules in RIO drops to enter the safe, or de-energized state.

Failure to follow these instructions can result in injury or equipment damage.

Schneider Electric recommends that you configure your safety system to include two redundant NTP servers, so that if the connection to the primary NTP server is lost, the safety modules automatically connect to the backup NTP server.

⚠ CAUTION

LOSS OF TIME SYNCHRONIZATION

During operation of the safety system, do not change the time setting in the NTP server or the CPU. Changing the time during operations can cause a loss of communication and a safety system shut-down.

Failure to follow these instructions can result in injury or equipment damage.

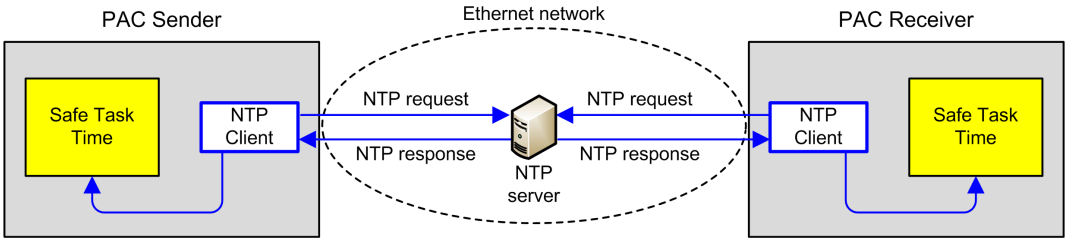
Refer to the topic *NTP Tab* in the *Modicon M580 Hardware Reference Manual* for information on how to configure the NTP service for an M580 CPU.

NTP Service for Peer-to-Peer Communication

The safe Ethernet PAC-to-PAC communication requires the synchronization of the time base of both the sender and receiver PACs.

NOTE: Schneider Electric recommends that you configure each PAC in the role of NTP client, and configure another network device as NTP server.

The following figure describes the sender and receiver PACs time base synchronization principle:



In Unity Pro, configure the NTP service parameters for each client as follows:

- Select **NTP Client**.
- Set the **Primary NTP Server IP address** to the IP address setting for the remote NTP server.
- Schneider Electric recommends a **Polling period** value set to 20 seconds.

NTP Server Time Consistency and System Bits

NTP server time consistency:

- If the NTP server time is consistent with the internal PAC time displayed by the EF `S_SYST_CLOCK` with less than 2 seconds difference, then the time value in the EF `S_SYST_CLOCK` is updated with the last NTP server time received filtered with a slope of 1ms/s.
- If the received NTP server time differs from the internal PAC time displayed by the EF `S_SYST_CLOCK` by more than 2 seconds, then:
  - the last received NTP server time is ignored by the PAC,
  - the time value displayed by the EF `S_SYST_CLOCK` is refreshed internally,
  - the `status` parameter of `S_SYST_CLOCK` is set to 0, and
  - the output parameter `SYNCHRO_NTP` from the `S_RD_ETH_MX` and the `S_WR_ETH_MX` DFBs is set to 0 to indicate this condition.

  In this case, you can reset the internal PAC time by taking one of the following actions:
  - reinitialize the application by a cold start
  - download the application
  - restart the PAC
  - set the system word `%SW128` to #E51A (from #1AE5). In this case, the CPU will accept the next NTP server time received without filtering (1ms/s) and without consistency check.

NOTE: If the NTP synchronization is lost on one of the two PACs (`SYNCHRO_NTP` parameter set to 0), both sender and receiver PACs time base can be de-synchronized. In this case, the safe peer-to-peer communication may cease to be operational and (`S_RD_ETH_MX` DFB `health` output parameter is set to 0).

Section 11.2
Peer to Peer Communications

Introduction

This section describes peer-to-peer communications between M580 safety PACs.

What Is in This Section?

This section contains the following topics:

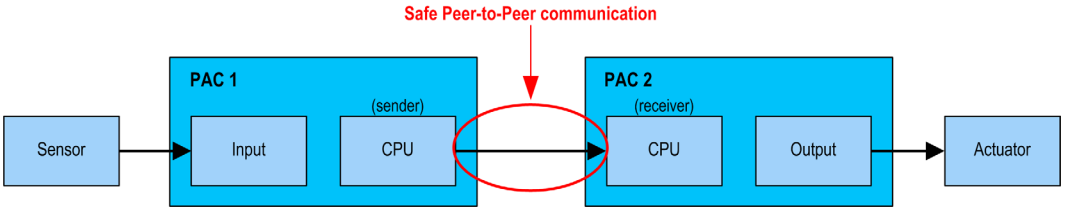
Topic	Page
Peer-to-Peer Communication	160
Peer-to-Peer Architecture	161
M580 Black Channel Communications	168
Configuring the `S_WR_ETH_MX` DFB in the Program Logic of the Sender PAC	170
Configuring the `S_RD_ETH_MX` DFB in the Program Logic of the Receiver PAC	172

Peer-to-Peer Communication

Introduction

You can configure two M580 safety PACs to perform peer-to-peer safe communications over Ethernet. The configuration is based on Modbus TCP scanner communication, embedded in a black channel.

The safety peer-to-peer communication functional overview is as follows:



The communication is performed by two elementary function blocks from the M580 safety block library, that manage the safety loop at a SIL3 level. The protocol detects transmission errors – including omissions, insertions, disordered sequence, delays, inaccurate addressing, and masquerade bits – and manages retransmissions.

This safe communication is possible only between two M580 safety PACs.

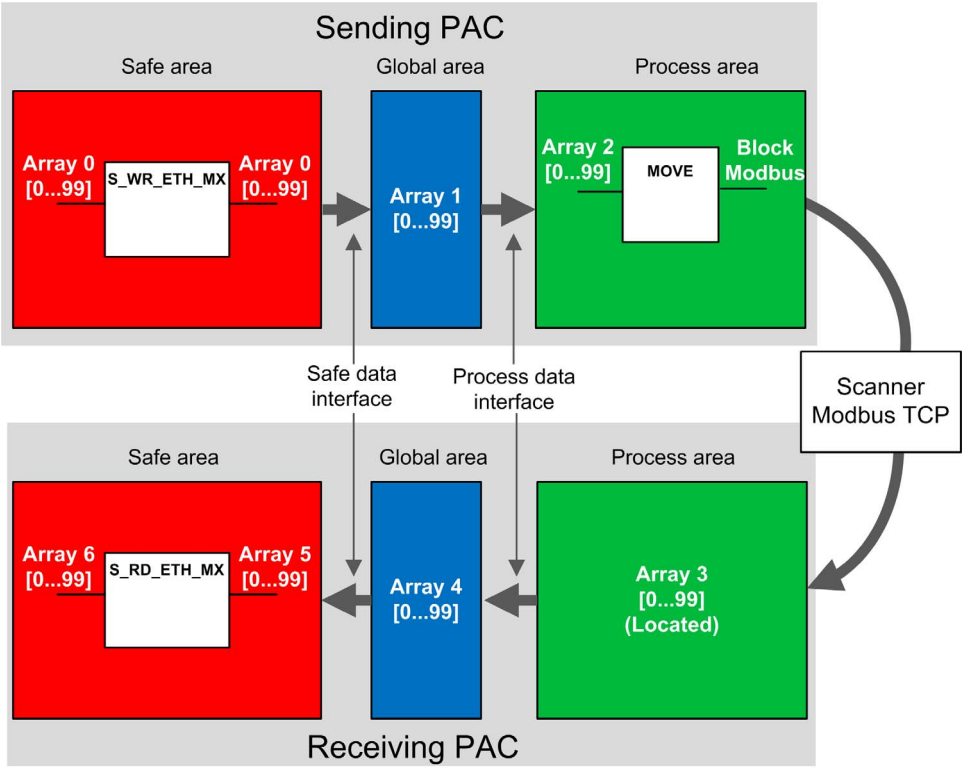
Peer-to-Peer Architecture

Architecture Design

The solution architecture is based on:

- NTP service for time base synchronization.
- Execution of 2 DFBs (`S_WR_ETH_MX` and `MOVE` in the sender PAC and 1 DFB (`S_RD_ETH_MX`) in the receiver PAC).
- Scanning via Modbus TCP, for data transportation.

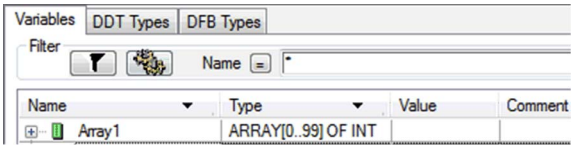
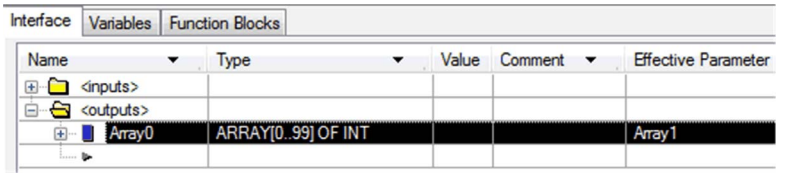
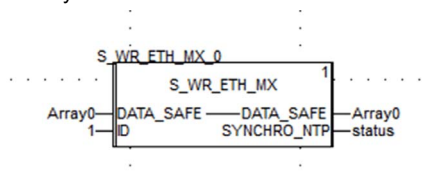
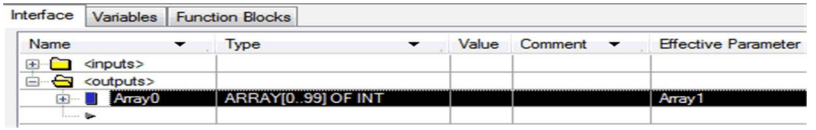
The following figure shows the overview of the process required to perform the safe peer-to-peer communication:

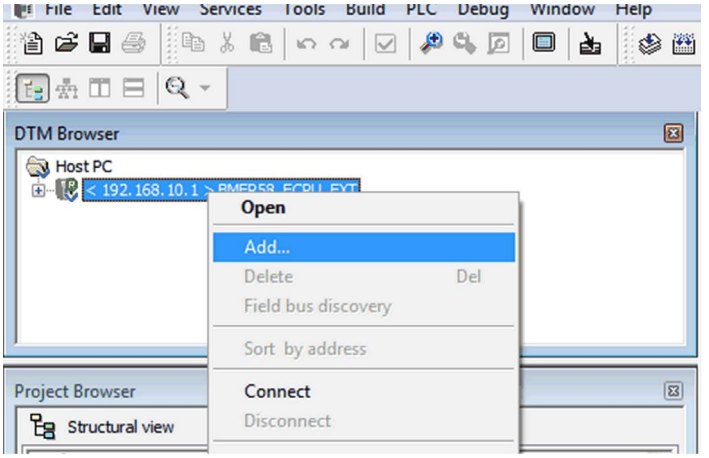
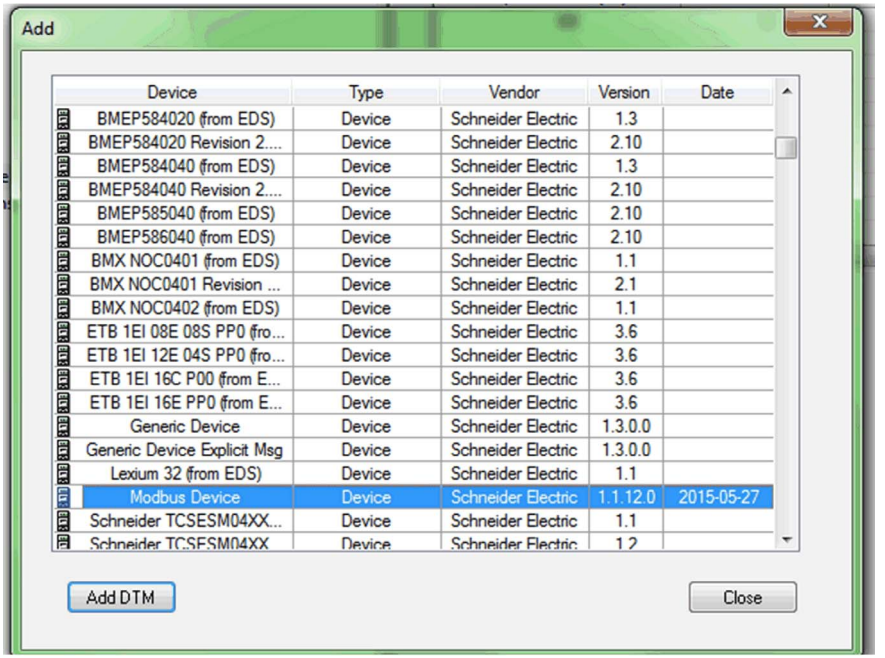


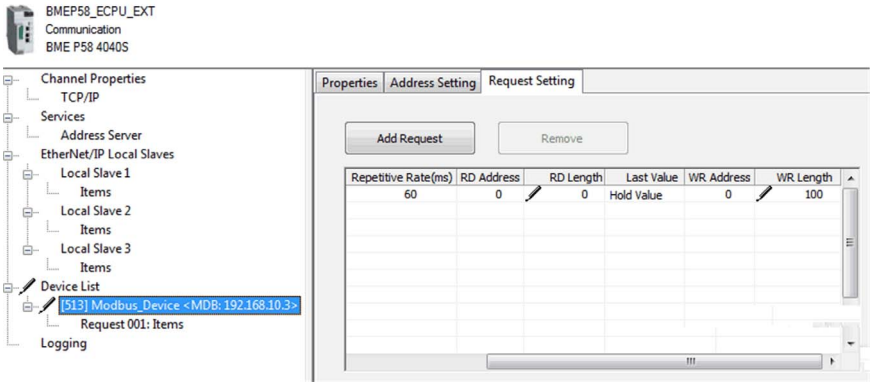
NOTE: On the Ethernet network, you are allowed to mix safety related data and non-safety related data without impact on the integrity level of the safety related data. There is no restriction on the Ethernet network when using the safe peer-to-peer communication.

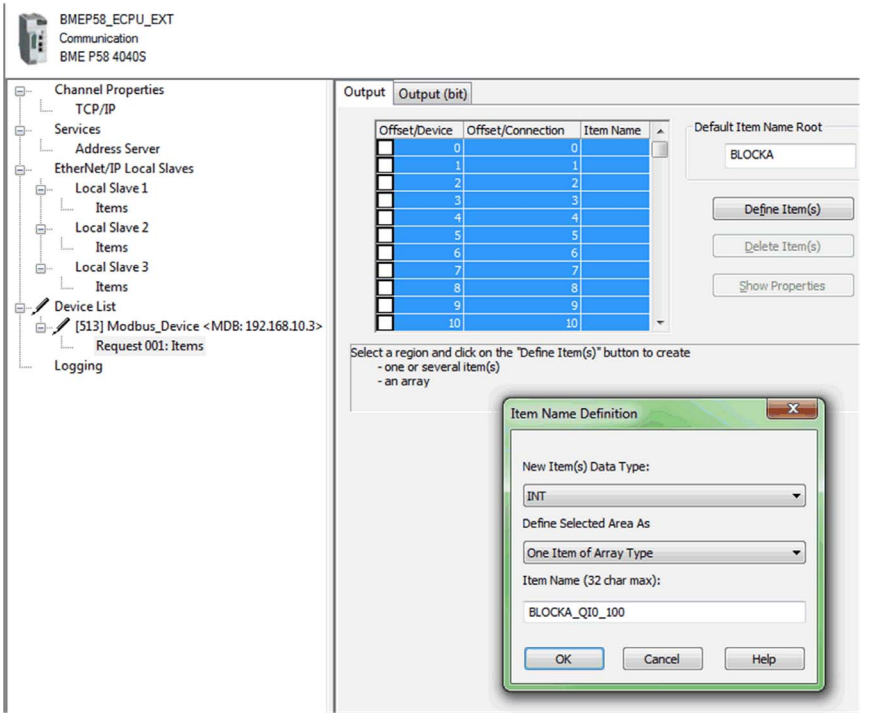
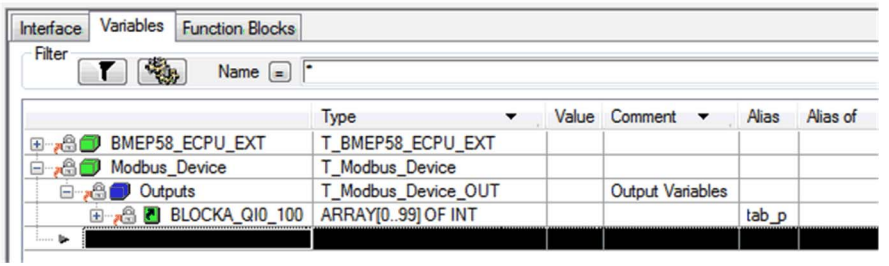
Peer-to-Peer Data Transfer Configuration Details

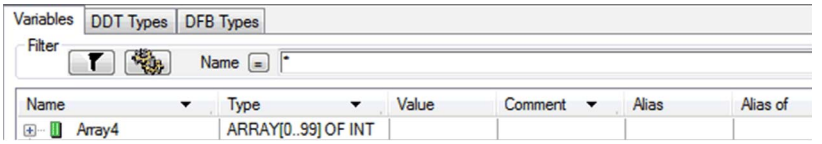
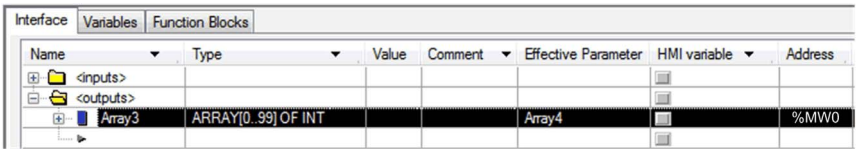
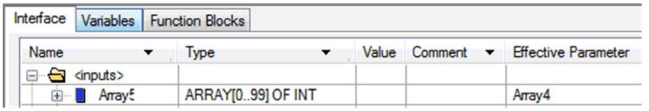
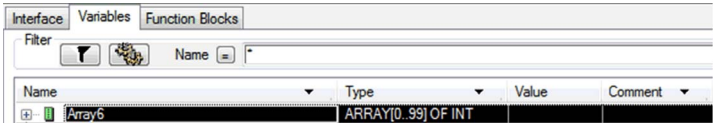
The following example shows you how to configure a peer-to-peer transfer of data between two safety PACs:

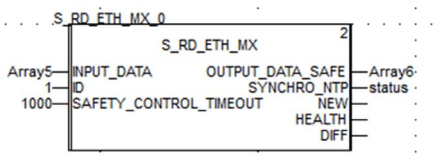
Step	Action
1	On the sending PAC, use the **Data Editor** to create an array of 100 integers in the global data scope area. In this example, the array name is Array1: 
2	On the sending PAC, create another array of 100 integers as an output in the **Interface** tab of the **Safety Data Editor** and link it to the global scope array created in step1, above, in the **Effective Parameter** column. In this example, the array name is Array0:  **NOTE:** The integer variables from index 0 to 90 of the array contain the safety variable values you want to exchange with the receiving PAC. The remaining area is reserved for auto-generated diagnostic data, including a CRC and time-stamp. This diagnostic data is used by the receiving PAC to determine if the transferred data is safe.
3	On the sending PAC, configure the DFB `S_WR_ETH_MX` in a section of the SAFE task. Link the DFB to Array0: 
4	On the sending PAC, create an array of 100 integers as an input in the **Interface** tab of the **Process Data Editor**. Then, link it to the array (Array1, created in the global scope area in step 1, above) in the **Effective Parameter** column: 

Step	Action
5	In the **DTM Browser** of the sending PAC, select the CPU (in this example) or an NOC communications module (if any) then click **Add...** to create a Modbus scanner that can send data via Modbus TCP from the sending PAC to the receiving PAC: 
6	Select **Modbus Device** and click **Add DTM** to add the Modbus scanner: 

Step	Action
7	Open the newly added Modbus device, and in the **Request Setting** tab: ● Set the **WR Length** column, which is the length of the data to be written, to a value of 100, then ● Set the **WR Address** column, which is the address where the table on the receiving PAC will write the data it receives (in this example: 0, which means the sending PAC will write to the table starting to %MW0 in the receiving PAC. 

Step	Action
8	Select the **Request 001: Items** node, then in the **Output** tab define an array type of INT (that is ≥ 100 integers). This is the sending PAC table that will be written to the receiving PAC: 
9	After saving and building the configuration, the block (BLOCKA_QI0_100 in his example) is automatically created as a process variable: 

Step	Action
10	On the sending PAC, in a process code section, use a MOVE DFB to copy the contents of the "tab_p" array to the array defined above in the Modbus device structure: 
11	On the receiving PAC, use the **Data Editor** to create a 100 integer array (Array4) in the global data scope area: 
12	On the receiving PAC, in the **Process Data Editor**, create an array (Array3) of 100 INT in the <outputs> section of the **Interface** tab. Link this array to the global data scope array (Array4, created in step 11) in the **Effective Parameter** column. The data sent by the sending PAC will be written into this array via the Modbus scanner, provided that this variable is located at the address defined in scanner of the sending PAC (in this example %MW0): 
13	On the receiver PAC, use the **Safety Data Editor** to create a 100 integer array (Array5) in the <inputs> section of the **Interface** tab. Then link this array to the global data scope array (Array4, created in step 11) in the **Effective Parameter** column: 
14	On the receiving PAC, use the **Safety Data Editor** to create a 100 integer array (Array6): 

Step	Action
15	On the receiving PAC, in a section of code in the SAFE task, instantiate the `S_RD_ETH_MX` DFB with the array created in step 13 (Array5) as input parameter and with the array created in step 14 (Step6) as output parameter: 

Black Channel Peer-to-Peer

Each peer-to-peer data transmission consists of both *User Safety Data*, which contains the application-related content being transmitted, and *Reserved Data*. The *Reserved Data* is used by the safety PAC to test the reliability of the transmission so that it satisfies the requirements of SIL3. The *Reserved Data* consists of the following elements:

- A CRC calculated by the sending PAC from the data to be transmitted. The receiving PAC checks the CRC before using the transmitted data.
- A communication identifier, which is included in the CRC calculation to help prevent masquerade and insertion attacks on the transmission of safety data.
- A time stamp, containing the time of the transmission in ms. This stamped time is based on the time value provided by the NTP service and is used to synchronize both the sender PAC and receiver PAC. The data sender PAC adds a time value to the data sent to the receiver PAC. The receiver PAC compares the received time stamp with its own time value, and uses it to:
  - Check the age of the data.
  - Reject duplicate transmissions.
  - Determine the chronological order of received transmissions.
  - Determine the elapsed time between receipt of data transmissions.

M580 Black Channel Communications

Black Channel

Black channel is the mechanism used to encrypt and validate transmitted safety data:

- Only Schneider Electric safety equipment can encrypt and decrypt the data sent via the black channel in an M580 safety system.
- The health of each safety data transmission is tested by both the transmitting and receiving safety module for each transmitted message.

The effect of using the black channel is to permit the transmission of safety data through non-safe intermediate equipment, such as backplanes, Ethernet cabling, communication adapters, and so forth. Because black channel transmissions are encrypted, the intermediate equipment cannot read or alter the content of the transmitted safety data without being detected.

Black channel transmissions operate independently of the communication protocol used for the transmission:

- X Bus is the carrier for backplane transmissions between safety devices on the same rack (e.g. from the CPU to local I/O, or from a communication remote adapter (CRA) to local I/O).
- EtherNet/IP is carrier for data transmissions between racks (e.g. from the CPU to a CRA).

Safety I/O modules and the CPU can send and receive black channel communications. For each transmission, the transmitting device (CPU or I/O) adds the following information to the message:

- a CRC tag to enable testing of the message content
- a time stamp to enable testing of the timeliness of the message
- other information– including the application version and the I/O configuration used – that identifies the I/O module in the transmission

When using safety I/O modules on a remote rack, configure the safety CPU as either an NTP server, or as an NTP client (with another system device configured as the NTP server). If one of these two designs is not implemented, the time settings of the safety I/O modules and CPU will not be synchronized and black channel communication will not operate properly.

⚠ CAUTION

RISK OF UNINTENDED OPERATION

If you are placing safety I/O modules in an RIO drop, the current time needs to be configured for the PAC. Enable the NTP service for your M580 system and configure the safety CPU as an NTP server or an NTP client. Failure to do so will cause the inputs and outputs of safety I/O modules in RIO drops to enter the safe, or de-energized state.

Failure to follow these instructions can result in injury or equipment damage.

The receiving device (I/O or CPU) decrypts the message and tests the accuracy of its content. The following conditions can be detected:

Condition	Description
Transmit errors	Error detected in the message address or routing.
Repeats	Message sent multiple times.
Deleted data	Part of the message is missing, or the message is lost.
Inserted data	Extra data is added to the message.
Out of sequence data	The message order is changed.
Corrupted data	One or more bit errors detected in the message.
Delays	The message delivery time is excessively long.
Masquerade	The source of the message is not permitted to send data.

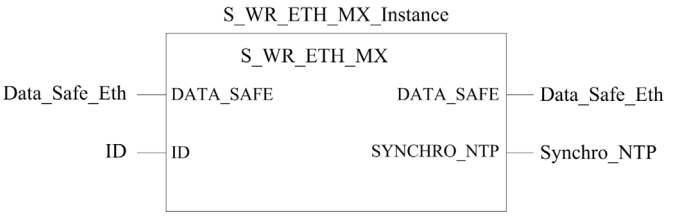
When any of these errors are detected, the channel is determined to be unhealthy and the appropriate safety function is executed:

- If the CPU detects that a transmission from an input module is unhealthy, the CPU sets input values for that module to the safe or de-energized state (value = “0”).
- If an output module detects a transmission from the CPU is unhealthy, it places its outputs into their pre-configured fallback state.

Configuring the S_WR_ETH_MX DFB in the Program Logic of the Sender PAC

Representation

DFB representation:



For an extended description of this DFB, refer to the *Unity Pro Safety Block Library*.

Description

This DFB calculates data (reserved data containing a CRC and a time stamp) required by the receiver to check and manage errors detected during the safe peer-to-peer communication.

The S_WR_ETH_MX DFB function block has to be called at each cycle in the sender PAC. Within the cycle, it has to be executed in the logic after all required modifications have been performed on the data to be sent. This means that the data to be sent may not be modified within the cycle after the execution of the DFB, otherwise the CRC information used in the reserved data area will not be correct and the safe peer-to-peer communication will not succeed.

You have to assign the ID parameter a unique value that identifies the safe peer-to-peer communication between a sender and a receiver.

⚠ WARNING

LOSS OF ABILITY TO PERFORM SAFETY FUNCTIONS

The ID parameter value must be unique and fixed in the network for a sender/receiver pair.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Input Parameter

Description of the input parameter:

Parameter	Data Type	Meaning
ID	INT	Communication identifier. The ID value is used to calculate the CRC. It is unique and has the same value as the value used by the sender. **NOTE:** You have to assign the ID parameter a unique value that identifies the safe peer-to-peer communication between a sender and a receiver.

Input/Output Parameter

Description of the input/output parameter:

Parameter	Data Type	Meaning
DATA_SAFE	ARRAY[0...99] of INT	Safety data variables array structure, which is modified by S_WR_ETH_MX. Composed of “User safety data” (from index 0 to 90) and “Reserved data” (from index 91 to 99).

Output Parameter

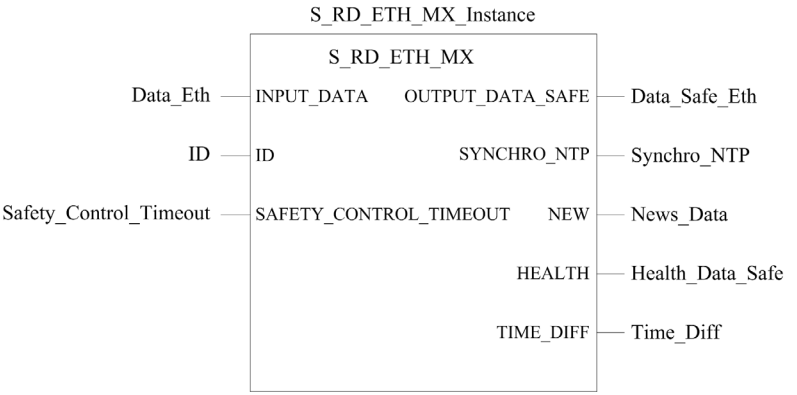
Description of the output parameter:

Parameter	Data Type	Meaning
SYNCHRO_NTP	BOOL	• 1: Indicates that NTP time synchronization is healthy. • 0: Indicates NTP time synchronization is not healthy due to internal time slippage. **NOTE:** This setting is a logical AND of bits 0, 1 and 2 of SW%39.

Configuring the S_RD_ETH_MX DFB in the Program Logic of the Receiver PAC

Representation

DFB representation:



Refer to the *Unity Pro Safety Block Library* for an extended description of this DFB.

Description

The S_RD_ETH_MX function block is used by a receiver PAC to copy the data received in the process area to the safety area and validate the accuracy of the received data.

⚠ WARNING

LOSS OF ABILITY TO PERFORM SAFETY FUNCTIONS

The S_RD_ETH_MX DFB function block must be called at each cycle in the receiver PAC program logic, and it must be executed before the data in the cycle is used.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

The S_RD_ETH_MX function block:

- Copies the data received in the INPUT_DATA register to the OUTPUT_DATA_SAFE register if it passes the following tests:
  - The function block checks the CRC of the last data packet received, via I/O scanner over Ethernet (Modbus TCP). If the CRC is not correct, the data is considered as unsafe and it is not written to the OUTPUT_DATA_SAFE register in the safety area.
  - The function block checks the last data received to determine if it is more recent than the data already written in the OUTPUT_DATA_SAFE register in the safety area (by comparing time stamps). If the last data received is not more recent, it is not copied to the OUTPUT_DATA_SAFE register in the safety area.

- Checks the age of the data in the safety area. If the age is higher than a configurable maximum value set in the SAFETY_CONTROL_TIMEOUT input register, the data is declared unsafe and the HEALTH bit is set to 0.
  NOTE: The data age is the time difference between the time when the data is computed in the sender PAC and the time when the data is checked in the receiver PAC. The time base reference is periodically updated with the time received from an NTP server.

  If the HEALTH bit is set to 0, the data available in the OUTPUT_DATA_SAFE array is considered as unsafe. In this case, take the appropriate reactive steps.

Input Parameters

Description of the input parameters:

Parameter	Data Type	Meaning
INPUT_DATA	ARRAY[0...99] of INT	Array of data variables received in the global memory area via I/O scanning over Ethernet (Modbus TCP).. Composed of “User safety data” (from index 0 to 90) and “Reserved data” (from index 91 to 99). **NOTE:** Define these data variables as shared input variables, each with an equivalent global variable, using the **Safety Data Interface** tab in Unity Pro.
ID	INT	Communication identifier. The unique ID value used to calculate the CRC. It is set to the same value as the value used by the sender. **NOTE:** You have to assign the ID parameter a unique value that identifies the safe peer-to-peer communication between a sender and a receiver.

⚠ WARNING

LOSS OF ABILITY TO PERFORM SAFETY FUNCTIONS

The ID parameter value must be unique and fixed in the network for a sender/receiver pair.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Parameter	Data Type	Meaning
SAFETY_CONTROL_TIMEOUT	INT	Time out value (in ms). Used to check the age of the data in the safety area and determine if that data is to be considered safe. Refer to the topic *Calculating a SAFETY_CONTROL_TIMEOUT Value* *(see page 175)*, below.

Output Parameters

Description of the output parameters:

Parameter	Data Type	Meaning
OUTPUT_DATA_SAFE	ARRAY[0...99] of INT	Safety data variables array structure. Composed of “User safety data” (from index 0 to 90) and “Reserved data” (from index 91 to 99).
SYNCHRO_NTP	BOOL	• 1: Indicates that NTP time synchronization is healthy. • 0: Indicates NTP time synchronization is not healthy due to internal time slippage.
NEW	BOOL	Set to 1 to indicate that new safe data has been refreshed during the current cycle. • 1: Indicates that new safe data has been refreshed during the current cycle. • 0: Indicates no new safe data has been refreshed.
HEALTH	BOOL	• 1: Indicates the data in the **User Safety Data** area is safe. • 0: Indicates the data in the **User Safety Data** area is not safe. Refer to the topic *Understanding the HEALTH Bit (see page 175)*, below.

⚠ WARNING

LOSS OF ABILITY TO PERFORM SAFETY FUNCTIONS

You must test the HEALTH bit value of the S_RD_ETH_MX DFB at each cycle before using any safe data to manage the safety function.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Parameter	Data Type	Meaning
TIME_DIFF	INT	Returns the age (in ms) of the data received and written in the OUTPUT_DATA_SAFE output parameter. Set to 0 if the internal NTP time is not initialized or if no correct data has yet been received.

Calculating a SAFETY_CONTROL_TIMEOUT Value

When calculating a SAFETY_CONTROL_TIMEOUT value, consider the following:

- Minimum value: SAFETY_CONTROL_TIMEOUT > T1
- Recommended value: SAFETY_CONTROL_TIMEOUT > 2 * T1

T1 = CPU_{sender} MAST cycle time + CPU_{sender} SAFE cycle time + Repetitive_rate + Network transmission time + $CPU_{receiver}$ MAST cycle time + $CPU_{receiver}$ SAFE cycle time

Where:

- *CPU_{sender} MAST cycle time* is the MAST cycle time of the sender PAC.
- *CPU_{sender} SAFE cycle time* is the SAFE cycle time of the sender PAC.
- *Repetitive_rate* is the time rate for the I/O scanner write query from the sender PAC to the receiver PAC.
- *Network transmission time* is the time consumed on the Ethernet network for the data transmission from the sender PAC to the receiver PAC.
- *$CPU_{receiver}$ MAST cycle time* is the MAST cycle time of the receiver PAC.
- *$CPU_{receiver}$ SAFE cycle time* is the SAFE cycle time of the receiver PAC.

Note that the value defined for the SAFETY_CONTROL_TIMEOUT parameter has a direct effect on the robustness and availability of the safe peer-to-peer communication. If the SAFETY_CONTROL_TIMEOUT parameter value greatly exceeds T1, the communication will be tolerant to various delays (for example network delays) or corrupted data transmissions.

You are responsible for configuring your Ethernet network so the load that does not cause an excessive delay on the network during data transmission, which could lead to the expiration of the timeout. To help safeguard your safe peer-to-peer communications from any excessive delays due to other non-safety data transmitted on the same network, consider using a dedicated Ethernet network for the safe peer-to-peer protocol.

When commissioning your project, you have to estimate the safe peer-to-peer communication performance by checking the values provided in the output parameter TIME_DIFF and evaluating the margin using the value defined in the SAFETY_CONTROL_TIMEOUT parameter.

Understanding the HEALTH Bit

When the HEALTH bit value equals:

- 1: The integrity of the data is correct (CRC) and the age of the data is less than the value set in the SAFTETY_CONTROL_TIMEOUT input register.
  NOTE: The age of the data considered is the time between:
- The beginning of the cycle where the data are computed in the sender PAC.
- The beginning of the cycle where the data are checked in the receiver PAC.
- 0: New valid data are not received in the required time interval (the timer expires and the HEALTH bit is set to 0).
  NOTE: If the HEALTH bit is set to 0, the data in the output array OUTPUT_DATA_SAFE is considered to be unsafe; respond accordingly.

Section 11.3
M580 CPU to Safety I/O Communication

M580 Safety PAC to I/O Communications

Communication Between the PAC and I/O

The M580 safety CPU and Copro together control all backplane exchanges, while the safety I/O respond to the commands of the CPU and Copro. Safety I/O modules can be installed in either a BMXXBP•••• X Bus rack or a BMEXBP•••• Ethernet rack.

Communications between the safety PAC and safety I/O modules in the local main rack are made via the backplane.

Communications between the safety PAC and safety I/O modules installed in an RIO drop are made through an adapter module installed on the RIO drop, either:

- a BMECRA31210 adapter, for an Ethernet rack, or
- a BMXCRA31210 adapter, for an X Bus rack.

NOTE: A BMXCRA31200 adapter cannot be used to connect safety I/O modules to the M580 safety PAC.

Communications from the safety PAC and safety /O modules, both in the local main rack and in an RIO drop, are made via the black channel *(see page 168)*.

NOTE: If you are installing safety I/O modules in an RIO drop, enable the NTP service and configure the safety CPU as either an NTP server, or as an NTP client (with another system device configured as the NTP server). If one of these two designs is not implemented, the time settings of the safety I/O modules and CPU will not be synchronized and black channel communication will not operate properly.
If you are installing safety I/O modules on local rack (or in an extension of the local rack), it is not required to enable the NTP service.

⚠ CAUTION

RISK OF UNINTENDED OPERATION

If you are placing safety I/O modules in an RIO drop, the current time needs to be configured for the PAC. Enable the NTP service for your M580 system and configure the safety CPU as an NTP server or an NTP client. Failure to do so will cause the inputs and outputs of safety I/O modules in RIO drops to enter the safe, or de-energized state.

Failure to follow these instructions can result in injury or equipment damage.

Schneider Electric recommends that you configure your safety system to include two redundant NTP servers, so that if the connection to the primary NTP server is lost, the safety modules automatically connect to the backup NTP server.

Refer to the topic *NTP Tab* in the *Modicon M580 Hardware Reference Manual* for information on how to configure the NTP service for an M580 CPU.

Optionally, you can use BMXNRP0200 or BMXNRP0201 fiber optic repeater modules to extend the physical link between the CPU and Copro in the local rack and the adapter in the RIO drop. Fiber optic repeater modules enhance RIO network noise immunity and increase cabling distance while maintaining the full dynamic range of the network and the safety integrity level.

The communication protocol between the safety I/O and PAC enables their exchanges. It permits both devices to check the accuracy of received data, detect corrupted data, and determine if the transmitting module becomes non-operational. Thus, a safety loop may include any non-interfering *(see page 23)* RIO adapters and backplane.

⚠ CAUTION

LOSS OF TIME SYNCHRONIZATION

During operation of the safety system, do not change the time setting in the NTP server or the CPU. Changing the time during operations can cause a loss of communication and a safety system shut-down.

Failure to follow these instructions can result in injury or equipment damage.

Supplying Power to the Safety I/O

The safety I/O is supplied 24 VDC and 3.3 VDC power over the backplane by the BMXCPS4002S safety power supply module *(see page 121)*. The safety power supply module monitors the power it provides so as not to exceed 36VDC.

Power for Non-Safety Functions:

5 VDC power provided by the backplane is applied by each safety I/O module to its non-safety functions.

External Power for Digital Safety I/O:

An external power supply, not greater than 60 VDC, is required for non-safety processes (sensor, actuator), and can be a protected extra-low voltage (PELV) overvoltage category II type power supply. The non-safety process power supply is supervised by the safety I/O module for overvoltage and undervoltage conditions.

Chapter 12
Diagnosing an M580 Safety System

Introduction

This chapter provides information on diagnostics that can be performed via hardware indicators (based on LED status) and system bits or words for an M580 safety system.

What Is in This Chapter?

This chapter contains the following sections:

Section	Topic	Page
12.1	M580 Safety CPU and Coprocessor Diagnostics	180
12.2	BMXCPS4002S Safety Power Supply Diagnostics	193
12.3	BMXSAI0410 Analog Input Diagnostics	194
12.4	BMXSDI1602 Digital Input Diagnostics	199
12.5	BMXSDO0802 Digital Output Diagnostics	205
12.6	BMXSRA0405 Digital Relay Output Diagnostics	212

Section 12.1
M580 Safety CPU and Coprocessor Diagnostics

Introduction

This section describes diagnostics available for both the BMEP584040S safety CPU and the BMEP58CPROS3 safety coprocessor.

What Is in This Section?

This section contains the following topics:

Topic	Page
Blocking Condition Diagnostics	181
Non-blocking Condition Diagnostics	184
M580 Safety CPU LED Diagnostics	186
M580 Safety Coprocessor LED Diagnostics	189
Memory Card Access LED	191

Blocking Condition Diagnostics

Introduction

Blocking conditions caused during the execution of the safety or the process program result from either the detection of system errors, or of the HALT state of a task in which the error was detected.

NOTE: The M580 safety PAC presents two independent HALT states:

- Process HALT applies to the non-SAFE tasks (MAST, FAST, AUX0, and AUX1). When any process task enters the HALT state, all other process tasks also enter the HALT state.
- SAFE HALT applies only to the SAFE task.

Refer to the *M580 Safety PAC Operating States* *(see page 228)* topic for a description of the HALT and STOP states.

Diagnostics

When the CPU detects a blocking condition causing a system error, a description of the detected error is provided in system word `%SW124`.

When the CPU detects a blocking condition causing a HALT state, a description of the detected error is provided in system word `%SW125`.

`%SW124` system word values and corresponding blocking condition description:

`%SW124` Value (hex)	Blocking Condition Description
5AF2	RAM detected error in memory check
5AFB	Safety firmware code error detected
5AF6	Safety watchdog overrun detected on CPU
5AFF	Safety watchdog overrun detected on Coprocessor
5B01	Coprocessor not detected at startup

`%SW125` system word values and corresponding blocking condition description:

`%SW125` Value (hex)	Blocking Condition Description
0•••	execution of an unknown function
0002	SD card signature feature (used with `SIG_CHECK` and `SIG_WRITE` functions)
2258	execution of the HALT instruction
2259	execution flow different than the reference flow
23••	execution of a CALL function towards an undefined subroutine
5AF3	comparison error detected by CPU
5AF9	instruction error detected at start-up or runtime
5AFA	comparison error detected on CRC value
5AFC	comparison error detected by coprocessor

%SW125 Value (hex)	Blocking Condition Description
5AFD	internal error detected by coprocessor; sub-code in %SW126: 1 (unknown result), 2 (CRC application), 7 (incorrect activity counter)
5AFE	Copro synchronization error detected - CPU only; sub-code in %SW126: 3 (diagnostic), 4 (end UL), 5 (comparison), 6 (BC out), 8 (HALT during UL), 9 HALT during comparison), 10 (HALT during BC out).
81F4	SFC node incorrect
82F4	SFC code inaccessible
83F4	SFC work space inaccessible
84F4	too many initial SFC steps
85F4	too many active SFC steps
86F4	SFC sequence code incorrect
87F4	SFC code description incorrect
88F4	SFC reference table incorrect
89F4	SFC internal index calculation detected error
8AF4	SFC step status not available
8BF4	SFC memory too small after a change due to a download
8CF4	transition/action section inaccessible
8DF4	SFC work space too small
8EF4	version of the SFC code older than the interpreter
8FF4	version of the SFC code more recent than the interpreter
90F4	poor description of an SFC object: NULL pointer
91F4	action identifier not authorized
92F4	poor definition of the time for an action identifier
93F4	macro step cannot be found in the list of active steps for deactivation
94F4	overflow in the action table
95F4	overflow in the step activation/deactivation table
9690	error detected in the application CRC check (checksum)
DE87	calculation detected floating point error
DEB0	task watchdog overrun (%S11 and %S19 are set)
DEF0	division by 0
DEF1	character string transfer detected error
DEF2	capacity exceeded
DEF3	index overrun
DEF4	inconsistent task periods

%SW125 Value (hex)	Blocking Condition Description
DEF7	SFC execution detected error
DEFE	SFC steps undefined

Restarting the Application

After a blocking condition has occurred, the halted tasks need to be initialized. If the HALT occurred for a:

- process task (MAST, FAST, AUX0, or AUX1), initialization is performed by either the Unity Pro **PLC → Init** command or by setting the %S0 bit to 1.
- SAFE task, initialization is performed by the Unity Pro **PLC → Init Safety** command.

When initialized, the application behaves as follows:

- the data resume their initial value
- tasks are stopped at end of cycle
- the input image is refreshed
- outputs are controlled in fallback position

The RUN command then allows the application or tasks to be restarted.

Non-blocking Condition Diagnostics

Introduction

The system encounters a non-blocking condition when it detects an input/output error on the backplane bus (X Bus or Ethernet) or through execution of an instruction, which can be processed by the user program and does not modify the CPU operating state.

This topic describes some of the system bits and words you can use to detect the state of the safety system and its component modules.

NOTE: The available system bits and words do not include all information relating to the state of safety modules. Schneider Electric recommends that you use the DDDT structure of the safety CPU and safety IO modules to determine the state of the M580 safety system.

For information about the M580 safety CPU DDDT, refer to the topic *Standalone DDT Data Structure for M580 CPUs* in the *Modicon M580 Hardware Reference Manual*.

For information about the M580 safety I/O module DDDTs, refer to the following topics:
- BMXSAI0410 Data Structure *(see page 55)* for the safety analog input module.
- BMXSDI1602 Data Structure *(see page 85)* for the safety digital input module.
- BMXSDO0802 Data Structure *(see page 100)* for the safety digital output module.
- BMXSRA0405 Data Structure *(see page 117)* for the safety digital relay output module.

NOTE: You can also perform more advanced diagnostics of Ethernet devices by means of explicit messaging. To accomplish this, use either:

- the READ_VAR function block for Modbus TCP devices.
- the DATA_EXCH function block, specifying the CIP protocol in the ADDM block, for EtherNet/IP devices.

Conditions Linked to I/O Diagnostics

A non-blocking condition linked to the I/O is diagnosed with the following indications:
- CPU **I/O** LED pattern: steady ON
- **I/O** module LED pattern: steady ON
- system bits (type of detected error):
  - `%S10` set to 0: global I/O error detected on one of the modules on the local or remote Ethernet or X Bus rack
  - `%S16` set to 0: I/O error detected in the task in progress on an X Bus rack
  - `%S40...%S47` set to 0: I/O error detected on an X Bus rack at address 0 to 7
  - `%S117` set to 0: RIO error detected on a remote X Bus rack
  - `%S119` set to 0: I/O error detected on a local X Bus rack

- system bits and words combined with the channel having an error detected (I/O channel number and type of detected error) or I/O module Device DDT information (for modules configured in Device DDT addressing mode):
  - bit %Ir.m.c.ERR set to 1: channel error detected (implicit exchanges)
  - word %MWr.m.c.2: the word value indicates the type of error detected on the specified channel and depends on the I/O module (implicit exchanges)

Conditions Linked to Execution of the Program Diagnostics

A non-blocking condition linked to execution of the program is diagnosed with the following system bits and words:

- system bits – type of error detected:
  - %S15 set to 1: character string manipulation error detected.
  - %S18 set to 1: capacity overrun, error detected on a floating point, or division by 0. (Refer to the topic *System Bits for Safe Task Execution* *(see page 292)* for more information.) When %S18 is set to 1, %SW17 contains a description of the causal event *(see page 293)*.
  - %S20 set to 1: index overrun.

  NOTE: If the configurable system bit %S78 is set in the program, the SAFE task enters the HALT state when system bit %S18 set to 1.

- system word – nature of the error detected:
  - %SW125 (always updated)

M580 Safety CPU LED Diagnostics

CPU LEDs

Use the LEDs on the front face of the CPU to diagnose the state of the PAC, as follows:

PAC State	LED Names and Colors:							
	RUN	ERR	IO[1]	ETH MS	ETH NS	DL	SRUN	SMOD
	Green	Red	Red	Green/Red	Green/Red	Green	Green	Green
Power OFF								
Power ON ● Autotest								
Not configured					 No cable plugged and connected to another powered device  Otherwise			
Configured: ● No external error detected							–	–
● External error detected				–	–		–	–
● No Ethernet link, including the Ethernet backplane							–	–
1. Not all errors detected for a safety I/O module are reported via LEDs. Check the DDDTs for safety I/O modules for additional information.								

PAC State	LED Names and Colors:							
	RUN	ERR	IO[1]	ETH MS	ETH NS	DL	SRUN	SMOD
	Green	Red	Red	Green/Red	Green/Red	Green	Green	Green
● Duplicate IP address			–				–	–
● STOP state			 Detected error on I/O module, channel or configuration  No error detected on configured input/output		 Not connected  Connected  No cable		 SAFE task running OR  SAFE task stopped	 Safety mode OR  Maintenance mode
● RUN state			–		 Not connected  Connected  No cable		 SAFE task running OR  SAFE task stopped	 Safety mode OR  Maintenance mode

1. Not all errors detected for a safety I/O module are reported via LEDs. Check the DDDTs for safety I/O modules for additional information.

PAC State	LED Names and Colors:							
	RUN	ERR	IO[1]	ETH MS	ETH NS	DL	SRUN	SMOD
	Green	Red	Red	Green/Red	Green/Red	Green	Green	Green
HALT state (recoverable error detected)			–				 SAFE task running	 Safety mode
							 SAFE task stopped	 Maintenance mode
SAFE state (non-recoverable error detected)								
OS update								
1. Not all errors detected for a safety I/O module are reported via LEDs. Check the DDDTs for safety I/O modules for additional information.								

Legend:

Symbol	Description	Symbol	Description	Symbol	Description
	Steady Green		Steady Red		OFF
	Blinking Green (500 ms ON, 500 ms OFF)		Blinking Red (500 ms ON, 500 ms OFF)	–	Not Applicable

M580 Safety Coprocessor LED Diagnostics

Coprocessor LEDs

Use the LEDs on the front face of the Coprocessor to diagnose the state of the PAC, as follows

Coprocessor State	LED Names and Colors:			
	SRUN	ERR	SMOD	DL
	Green	Red	Green	Green
Power OFF				
WAIT state (Wait for firmware download from CPU)				
Not configured (no application)				
Configured and operating in safety mode: • SAFE task stopped				
• SAFE task running				
Configured and operating in Maintenance mode: • SAFE task stopped				
• SAFE task running				
SAFE task in HALT (recoverable error detected)				
SAFE state (non-recoverable error detected)				

Legend:

Symbol	Description		Symbol	Description		Symbol	Description
	Steady Green			Steady Red			OFF
	Blinking Green (500 ms ON, 500 ms OFF)			Blinking Red (500 ms ON, 500 ms OFF)			

Memory Card Access LED

Introduction

The green memory card access LED underneath the SD memory card door indicates the CPU access to the memory card when a card is inserted. This LED can be seen when the door is open.

Dedicated LED States

By itself, the **memory card access** LEDs indicate these states:

LED Status	Description
ON	The memory card is recognized, but the CPU is not accessing it.
flashing	The CPU is accessing the memory card.
blinking	The memory card is not recognized.
OFF	The memory card can be removed from the CPU slot or the CPU does not recognize the memory card.

NOTE: Confirm that the LED is OFF before you remove the card from the slot.

Combined LED Meanings

The access card LED operates together with the **BACKUP** LED. Their combined patterns indicate the following diagnostic information:

Memory Card Status	Conditions	CPU State	Memory Card Access LED	BACKUP LED
no memory card in the slot	—	no configuration		
memory card not OK	—	no configuration		
memory card without project	—	no configuration		
memory card with a non-compatible project	—	no configuration		
– no specific circumstances or CPU state				

Memory Card Status	Conditions	CPU State	Memory Card Access LED	BACKUP LED
memory card with a compatible project	An error is detected when the project is restored from the memory card to the CPU RAM.	no configuration	during transfer:  end of transfer: 	during transfer:  end of transfer: 
	No error is detected when the project is restored from the memory card to the CPU RAM.	—	during transfer:  end of transfer: 	during transfer:  end of transfer: 
– no specific circumstances or CPU state				

This legend shows the different LED patterns:

Symbol	Meaning	Symbol	Meaning
	off		steady red
	steady green		blinking green

Section 12.2
BMXCPS4002S Safety Power Supply Diagnostics

Power Supply LED Diagnostics

Power Supply LEDs

The BMXCPS4002S safety power supply front panel includes the following diagnostic LEDs:

- **OK**: Operating Status
- **ACT**: Activity
- **RD**: Redundancy (for redundant power supply designs)

The BMXCPS4002S safety power supply LEDs can present the following diagnostic information:

LED	Description
OK	• ON (green) indicates that all of the following are true: ❍ 24 Vdc backplane voltage is OK. ❍ 3.3 Vdc backplane voltage is OK. ❍ The RESET button has not been activated. • OFF indicates that at least one of the following is true: ❍ 24 Vdc backplane voltage is not OK. ❍ 3.3 Vdc backplane voltage is not OK. ❍ The RESET button has been activated.
ACT	• ON (green) indicates the power supply is supplying power. In a redundant power supply design, the module is the primary. • OFF indicates the power supply is not supplying power. In a redundant power supply design, the module is the standby.
RD	• ON (green) indicates communication between the two power supply modules is OK. • OFF indicates at least one of the following is true: ❍ Communication between the two power supply modules is not OK. ❍ Auto-tests are being performed.

Section 12.3
BMXSAI0410 Analog Input Diagnostics

Introduction

This section describes diagnostic tools available for the BMXSAI0410 safety analog input module.

What Is in This Section?

This section contains the following topics:

Topic	Page
BMXSAI0410 DDDT Diagnostics	195
BMXSAI0410 Analog Input LED Diagnostics	196

BMXSAI0410 DDDT Diagnostics

Introduction

The BMXSAI0410 safety analog input module provides the following diagnostics using its `T_U_ANA_SIS_IN_4` *(see page 55)* device DDT elements:

- input diagnostics
- internal error detection
- channel wiring diagnostics

Input Diagnostics

The sensors connected to each channel are monitored for their ability to accurately measure 10 analog input values between 4 and 20 mA. If the input measurement tests are not successful, the `CH_HEALTH` bit in the `T_U_ANA_SIS_CH_IN` *(see page 57)* DDDT structure is set to 0, indicating it is not operational.

Internal Error Detection

The module processes the input value using two separate, parallel circuits. The two values are compared to determine if an internal error is detected in the module process. If the compared values are different, the `IC` bit in the `T_U_ANA_SIS_CH_IN` DDDT structure is set to 1, indicating it is not operational.

Refer to the architecture diagram *(see page 129)* for the BMXSAI0410 safety analog input module for a visual presentation of this process.

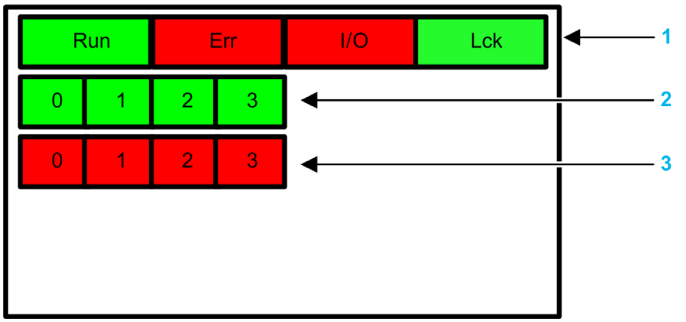
Channel Wiring Diagnostics

The wiring of the sensor to the input channel is continuously diagnosed for a cut wire condition, which is detected when the measured current is less than 3.75 mA or greater than 20.75 mA. In this case, the `OOR` bit in the `T_U_ANA_SIS_CH_IN` DDDT structure is set to 1.

BMXSAI0410 Analog Input LED Diagnostics

LED Panel

The BMXSAI0410 analog input module presents the following LED panel on its front face:



1 Module state LEDs
2 Channel state LEDs
3 Channel detected error LEDs

NOTE:

- The channel detected error LEDs are operational only after the module has been properly configured. When a channel error is detected, the corresponding LED remains ON until the underlying condition is resolved.
- Because the input module includes only four channels, LEDs in positions 4...7 are not used and are never powered on.

Module Diagnostics

Use the four LEDs at the top of the LED panel to diagnose the condition of the BMXSAI0410 analog input module:

Module LEDs				Module State	Recommended Response
Run	Err	I/O	LCK		
Blinking[1]	Blinking[1]	Blinking[1]	Blinking[1]	Auto-test at power-on.	–
Blinking[1]	ON	OFF	Blinking[1]	Auto-test at power-on has detected an internal error on input channels.	Replace the module.
OFF	ON	OFF	OFF	Internal error detected.	Replace the module if the condition persists.
OFF	Blinking[1]	OFF	X	Non-configured I/O module.	Configure the module via the CPU.

X indicates the LED state can be either ON or OFF.
1. Blinking: 500 ms ON / 500 ms OFF.
2. Flickering: 50 ms ON / 50 ms OFF.

Module LEDs				Module State	Recommended Response
Run	Err	I/O	LCK		
X	X	ON	X	External error detected on input channel.	Refer to *Channel Diagnostics* *(see page 198)* (below).
ON	Blinking[1]	X	X	No communication between CPU and I/O module.	Verify that: ● The CPU is an M580 safety CPU and that it is operational. ● The backplane is operational (if I/O module is on main rack). ● The cable between the CPU and I/O module is operational and properly connected (if the I/O module is on an extended or remote rack).
ON	Flickering[2]	X	OFF	Communication not safe and configuration unlocked.	Debug the condition using the DDDT variables *(see page 55)* for the I/O module instance.
ON	Flickering[2]	X	ON	Communication not safe and configuration locked.	Verify that: ● The locked configuration in the module is equal to the module configuration stored in application in the CPU as configured using Unity Pro. ● Debug the condition using the DDDT variables *(see page 55)* for the I/O module instance.
ON	ON	OFF	X	Input channel Internal error detected	Replace the module if the condition persists.
ON	OFF	OFF	OFF	Communication with CPU is OK and the configuration is unlocked.	–
ON	OFF	OFF	ON	Communication with CPU is OK and the configuration is locked.	–
X indicates the LED state can be either ON or OFF. 1. Blinking: 500 ms ON / 500 ms OFF. 2. Flickering: 50 ms ON / 50 ms OFF.					

Channel Diagnostics

Use all the LEDs on the BMXSAI0410 analog input module to diagnose channel status:

Module LEDs				Channel LEDs		Channel State	Recommended Response
Run	Err	I/O	LCK	Channel State (LED 0...3)	Detected Error (LED 0...3)		
ON	OFF	Off	X	ON	OFF	The input current is in the range 4...20 mA on the channel.	–
ON	OFF	ON	X	OFF	OFF	The input current is out of the range 4...20 mA on the channel.	Verify that the external power-supply, the external cabling, and the sensor are operational.
ON	ON	OFF	X	OFF	ON	The channel is not operational.	Replace the module if the condition persists.
X indicates the LED state can be either ON or OFF.							

Section 12.4
BMXSDI1602 Digital Input Diagnostics

Introduction

This section describes diagnostic tools available for the BMXSDI1602 safety digital input module.

What Is in This Section?

This section contains the following topics:

Topic	Page
BMXSDI1602 DDDT Diagnostics	200
BMXSDI1602 Digital Input LED Diagnostics	202

BMXSDI1602 DDDT Diagnostics

Introduction

The BMXSDI1602 safety digital input module provides the following diagnostics using its `T_U_DIS_SIS_IN_16` *(see page 85)* device DDT elements:

- input diagnostics
- internal error detection
- channel wiring diagnostics
- overvoltage and undervoltage diagnostics

Input Diagnostics

Each input channel is tested for operational effectiveness at the start of every cycle (or scan). Each channel is forced to the energized state and tested to check that the energized state was achieved. Then the channel is forced to the de-energized state and again is tested to check that the de-energized state was achieved.

If the channel does not successfully toggle between energized and de-energized, the `CH_HEALTH` bit in the `T_U_DIS_SIS_CH_IN` *(see page 87)* DDDT structure is set to 0 indicating it is not operational.

Internal Error Detection

Each cycle, the module performs an input diagnostic sequence. The module processes the input value using two separate, identical circuits. The two values are compared to determine if an internal error exists in the module internal process. If the compared values are different, the `IC` bit in the `T_U_DIS_SIS_CH_IN` DDDT structure is set to 1 indicating it is not operational.

Refer to the architecture diagram *(see page 130)* for the BMXSDI1602 safety digital input module for a visual presentation of this process.

Channel Wiring Diagnostics

The wiring of the sensor to the input channel can be continuously diagnosed for any of the following conditions:

- cut wire
- short circuit to 24 Vdc
- short circuit to 0 Vdc (open circuit)
- cross circuit between two parallel channels

The availability of these diagnostics depends on the power source employed by the specific wiring design *(see page 65)*, and on the diagnostic function being enabled in the configuration page of the module.

If one of these conditions is detected, the T_U_DIS_SIS_CH_IN DDDT structure sets the associated bit value to 1, as follows:

- the OC bit is set to 1 if an open (cut) wire or short circuit to 0 Vdc ground condition is detected.
- the SC bit is set to 1 if a short circuit to the 24 Vdc source or cross circuit between two channels is detected.

Overvoltage and Undervoltage Diagnostics

The module continuously tests for an overvoltage and an undervoltage condition. The following threshold values apply:

- Undervoltage threshold = 18.6 Vdc
- Overvoltage threshold = 33 Vdc

If either condition is detected, the module sets the PP_STS bit in the T_U_DIS_SIS_IN_16 device DDT to 0.

BMXSDI1602 Digital Input LED Diagnostics

LED Panel

The BMXSDI1602 digital input module presents the following LED panel on its front face:



1 Module state LEDs
2 Channel state LEDs for Rank A
3 Channel detected error LEDs for Rank A
4 Channel state LEDs for Rank B
5 Channel detected error LEDs for Rank B

NOTE: When a channel error is detected, the corresponding LED remains ON until the underlying condition is resolved.

Module Diagnostics

Use the four LEDs at the top of the LED panel to diagnose the condition of the BMXSDI1602 digital input module:

Module LEDs				Module State	Recommended Response
Run	Err	I/O	LCK		
Blinking	Blinking[1]	Blinking[1]	Blinking[1]	Auto-test at power-on.	–
Blinking	ON	OFF	Blinking[1]	Auto-test at power-on has detected an internal error on input channels.	Replace the module.
Blinking	ON	ON	Blinking[1]	• Auto-test module at power-on has detected an internal error on input channels; or • External 24VDC power supply is out of range	Verify the external pre-actuator 24 Vdc power supply is operational, and connect the 24 Vdc supply.

X indicates the LED state can be either ON or OFF.
1. Blinking: 500 ms ON / 500 ms OFF.
2. Flickering: 50 ms ON / 50 ms OFF.

Module LEDs				Module State	Recommended Response
Run	Err	I/O	LCK		
OFF	ON	OFF	OFF	Internal error detected.	Replace the module if the condition persists.
OFF	Blinking[1]	OFF	X	Non-configured I/O module.	Configure the module via the CPU.
X	XX	ON	X	• External 24 Vdc power supply is out of range; or • External error detected on input channel.	• Verify the external pre-actuator 24 Vdc power supply is operational. • Refer to *Channel Diagnostics (see page 204)*.
ON	Blinking[1]	X	X	No communication between CPU and module.	Verify that: • The CPU is an M580 safety CPU and that it is operational. • The backplane is operational (if I/O module is on main rack). • The cable between the CPU and I/O module is operational and properly connected (if the I/O module is on an extended or remote rack).
ON	Flickering[2]	X	OFF	Communication not safe and configuration unlocked.	Debug the condition using the DDDT variables *(see page 85)* for the I/O module instance.
ON	Flickering[2]	X	ON	Communication not safe and configuration locked.	• Verify the locked configuration in the module is equal to the module configuration stored in application in the CPU as configured using Unity Pro. • Debug the condition using the DDDT variables *(see page 85)* for the I/O module instance.
ON	ON	OFF	X	Input channel Internal error detected.	Replace the module if the condition persists.

X indicates the LED state can be either ON or OFF.
1. Blinking: 500 ms ON / 500 ms OFF.
2. Flickering: 50 ms ON / 50 ms OFF.

Module LEDs				Module State	Recommended Response
Run	Err	I/O	LCK		
ON	OFF	OFF	OFF	Communication with CPU is OK and the configuration is unlocked.	–
ON	OFF	OFF	ON	Communication with CPU is OK and the configuration is locked.	–
X indicates the LED state can be either ON or OFF. 1. Blinking: 500 ms ON / 500 ms OFF. 2. Flickering: 50 ms ON / 50 ms OFF.					

Channel Diagnostics

Use all the LEDs on the BMXSDI1602 digital input module to diagnose channel status:

Module LEDs				Channel LEDs		Channel State	Recommended Response
Run	Err	I/O	LCK	Channel State (LED 0...7, Rank A/B)	Detected Error (LED 0...7, Rank A/B)		
ON	OFF	OFF	X	ON	OFF	Input state on.	–
ON	OFF	OFF	X	OFF	OFF	Input state off.	–
ON	ON	OFF	X	OFF	ON	Input state off. An internal error is detected on the channel.	To change the module if condition is persistent
ON	ON	ON	X	OFF	ON	External 24 Vdc power supply is out of range.	Verify the external pre-actuator 24 Vdc power supply is operational.
ON	OFF	ON	X	X	Blinking[1]	The input is in either: ● An open circuit condition; or ● A short-circuit condition with the 0 Vdc.	Verify that the cabling is operational and properly connected.
ON	OFF	ON	X	X	Flickering[2]	The input is in either: ● A short-circuit condition with the 24 Vdc; or ● A short-circuit condition with the 0 Vdc.	Verify that the cabling is operational and properly connected.
X indicates the LED state can be either ON or OFF.							

Section 12.5
BMXSDO0802 Digital Output Diagnostics

Introduction

This section describes diagnostic tools available for the BMXSDO0802 safety digital output module.

What Is in This Section?

This section contains the following topics:

Topic	Page
BMXSDO0802 DDDT Diagnostics	206
BMXSDO0802 Digital Output LED Diagnostics	208

BMXSDO0802 DDDT Diagnostics

Introduction

The BMXSDO0802 safety digital output module provides the following diagnostics using its `T_U_DIS_SIS_OUT_8` *(see page 100)* device DDT elements:

- output diagnostics
- internal error detection
- channel wiring diagnostics
- overvoltage and undervoltage diagnostics

Output Diagnostics

Each output channel is tested for operational effectiveness at the start of each cycle (or scan). The test consists of toggling the output contact states (from ON to OFF, or OFF to ON) for a time too short to cause an actuator response (less than 1 ms). If the channel does not successfully toggle between energized and de-energized, the `CH_HEALTH` bit in the `T_U_DIS_SIS_CH_OUT` *(see page 102)* DDDT structure is set to 0, indicating it is not operational.

Internal Error Detection

The module processes the output value using two separate, identical circuits. Each circuit reads the midpoint voltage on the channel. The two values are compared, and if the values are not the expected values, an internal detected error is flagged by setting the `IC` bit in the `T_U_DIS_SIS_CH_OUT` DDDT structure to 1, indicating it is not operational.

Refer to the architecture diagram *(see page 131)* for the BMXSDO0802 safety digital output module for a visual presentation of this process.

Channel Wiring Diagnostics

The wiring of the actuator to the output channel can be continuously diagnosed for any of the following conditions:

- cut wire
- short circuit to 24 Vdc
- short circuit to 0 Vdc (open circuit)
- cross circuit between two parallel channels
- channel overload

  NOTE: Channel overload can be detected only if the output is energized.

The availability of these diagnostics depends on the diagnostic function being enabled in the configuration page of the module.

If one of these conditions is detected, the T_U_DIS_SIS_CH_OUT DDDT structure sets the associated bit value to 1, as follows:

- the OC bit is set to 1 if an open (cut) wire condition is detected.
- the SC bit is set to 1 if a short circuit to the 24 Vdc source or cross circuit between two channels is detected.
- the OL bit is set to 1 if a short circuit to the 0 Vdc ground or a channel overload condition is detected.

Overvoltage and Undervoltage Diagnostics

The module continuously tests for an overvoltage and an undervoltage condition. The following threshold values apply:

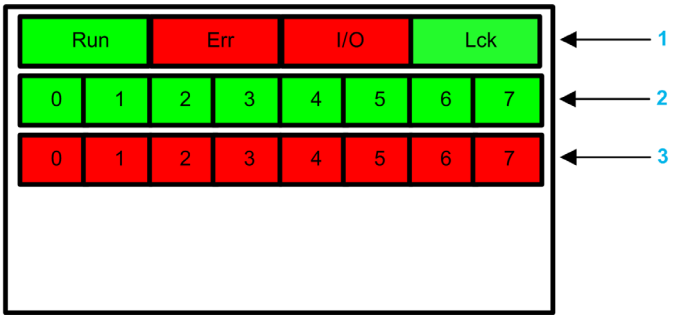
- Undervoltage threshold = 18 Vdc
- Overvoltage threshold = 31.8 Vdc

If either condition is detected, the module sets the PP_STS bit in the T_U_DIS_SIS_OUT_8 device DDT to 0.

BMXSDO0802 Digital Output LED Diagnostics

LED Panel

The BMXSDO0802 digital output module presents the following LED panel on its front face:



1 Module state LEDs
2 Channel state LEDs
3 Channel detected error LEDs

NOTE: When a channel error is detected, the corresponding LED remains ON until the underlying condition is resolved.

Module Diagnostics

Use the four LEDs at the top of the LED panel to diagnose the condition of the BMXSDO0802 digital output module:

Module LEDs				Module State	Recommended Response
Run	Err	I/O	LCK		
Blinking[1]	Blinking[1]	Blinking[1]	Blinking[1]	Auto-test at power-on.	–
Blinking[1]	ON	OFF	Blinking[1]	Auto-test at power-on has detected an internal error on output channels.	Replace the module.
Blinking[1]	ON	ON	Blinking[1]	• Auto-test module at power-on has detected an internal error on output channels; or • External 24VDC power supply is out of range	Verify the external pre-actuator 24 Vdc power supply is operational, and connect the 24 Vdc supply.
X indicates the LED state can be either ON or OFF. 1. Blinking: 500 ms ON / 500 ms OFF. 2. Flickering: 50 ms ON / 50 ms OFF.					

Module LEDs				Module State	Recommended Response
Run	Err	I/O	LCK		
OFF	ON	OFF	OFF	Internal error detected.	Replace the module if the condition persists.
OFF	Blinking[1]	OFF	X	Non-configured I/O module.	Configure the module via the CPU.
X	X	ON	X	• External 24 Vdc power supply is out of range; or • External error detected on output channel.	• Verify the external pre-actuator 24 Vdc power supply is operational. • Refer to *Channel Diagnostics* *(see page 210)* (below).
ON	Blinking[1]	X	X	No communication between CPU and module. The module is in Fallback state (or in Reset if module has never been operating normally).	Verify that: • The CPU is an M580 safety CPU and that it is operational. • The backplane is operational (if I/O module is on main rack). • The cable between the CPU and I/O module is operational and properly connected (if the I/O module is on an extended or remote rack).
ON	Flickering[2]	X	OFF	Communication not safe and configuration unlocked. The module is in Fallback state (or in Reset if module has never been operating normally).	To verify the variables available to debug safe communication in DDDT
ON	Flickering[2]	X	ON	Communication not safe and configuration locked. Module is in Fallback state.	• Verify the locked configuration in the module is equal to the module configuration stored in application in the CPU as configured using Unity Pro. • Debug the condition using the DDDT variables *(see page 100)* for the I/O module instance.
ON	ON	OFF	X	Internal error detected on an output channel.	Replace the module if the condition persists.

X indicates the LED state can be either ON or OFF.
1. Blinking: 500 ms ON / 500 ms OFF.
2. Flickering: 50 ms ON / 50 ms OFF.

Module LEDs				Module State	Recommended Response
Run	Err	I/O	LCK		
ON	OFF	OFF	OFF	Communication with CPU is safe and the configuration is unlocked	–
ON	OFF	OFF	ON	Communication with CPU is safe and the configuration is locked.	–

X indicates the LED state can be either ON or OFF.
1. Blinking: 500 ms ON / 500 ms OFF.
2. Flickering: 50 ms ON / 50 ms OFF.

Channel Diagnostics

Use all the LEDs on the BMXSDO0802 digital output module to diagnose channel status:

Module LEDs				Channel LEDs		Channel State	Recommended Response
Run	Err	I/O	LCK	Channel State (LED 0...7)	Detected Error (LED 0...7		
ON	OFF	OFF	X	ON	OFF	Output state on.	–
ON	OFF	OFF	X	OFF	OFF	Output state off.	–
ON	ON	OFF	X	OFF	ON	Output state off. Internal error detected on output channel.	Replace the module if the condition persists.
ON	ON	ON	X	OFF	ON	External pre-actuator 24VDC power supply is out of range	Verify the 24 Vdc power supply is operational.
ON	OFF	ON	X	OFF	Blinking[1]	The output is in: • An open circuit condition; or • A short-circuit condition with the 0 Vdc; or • In voltage overload.	Verify that the cabling is operational and properly connected.

X indicates the LED state can be either ON or OFF.
1. Blinking: 500 ms ON / 500 ms OFF.
2. Flickering: 50 ms ON / 50 ms OFF.

Module LEDs				Channel LEDs		Channel State	Recommended Response
Run	Err	I/O	LCK	Channel State (LED 0...7)	Detected Error (LED 0...7		
ON	OFF	ON	X	ON	Flickering[2]	The output is in: ● A short-circuit condition with the 24 Vdc; or ● A short-circuit condition with another active output channel.	Verify that the cabling is operational and properly connected.
X indicates the LED state can be either ON or OFF. 1. Blinking: 500 ms ON / 500 ms OFF. 2. Flickering: 50 ms ON / 50 ms OFF.							

Section 12.6
BMXSRA0405 Digital Relay Output Diagnostics

Introduction

This section describes diagnostic tools available for the BMXSRA0405 safety digital relay output module.

What Is in This Section?

This section contains the following topics:

Topic	Page
BMXSRA0405 DDDT Diagnostics	213
BMXSRA0405 Digital Relay Output LED Diagnostics	214

BMXSRA0405 DDDT Diagnostics

Introduction

The BMXSRA0405 safety digital output relay module provides the following diagnostics using its `T_U_DIS_SIS_OUT_4` *(see page 117)* device DDT elements:

- output contact diagnostics
- internal error detection

Output Contact Diagnostics

Depending on the application number that has been configured for the module, the module can automatically test its ability to toggle the output contact states (from ON to OFF, or OFF to ON) for a time too short to cause an actuator response. If the channels does not successfully toggle between energized and de-energized, the `CH_HEALTH` bit in the `T_U_DIS_SIS_CH_ROUT` *(see page 119)* DDDT structure is set to 0, indicating it is not operational.

NOTE: Application numbers 2, 4, 6, and 8 perform this automatic signal test. Application numbers 1, 3, 5, an 7 do not, and therefore require a daily manual transition of the output channel state to confirm its operability.

Output Command Diagnostics (Internal Error Detection)

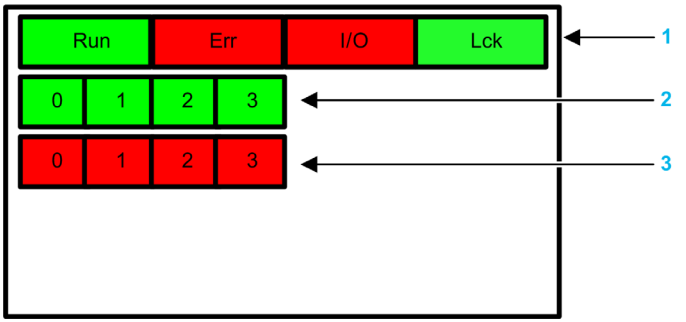
The relay command is processed using two separate, parallel circuits. The values of the circuits are compared. If the compared values are different, the channel is determined to be non-operational and the `IC` bit in the `T_U_DIS_SIS_CH_ROUT` DDDT structure is set to 1.

Refer to the architecture diagram *(see page 132)* for the BMXSRA0405 safety digital output relay module for a visual presentation of this process.

BMXSRA0405 Digital Relay Output LED Diagnostics

LED Panel

The BMXSRA0405 digital relay output module presents the following LED panel on its front face:



1 Module state LEDs
2 Channel state LEDs
3 Channel detected error LEDs

NOTE:

- When a channel error is detected, the corresponding LED remains ON until the underlying condition is resolved.
- Because the relay output module has only four channels, LEDs in positions 4...7 are not used and are never powered on.

Module Diagnostics

Use the four LEDs at the top of the LED panel to diagnose the condition of the BMXSRA0405 digital relay output module:

Module LEDs				Module State	Recommended Response
Run	Err	I/O	LCK		
Blinking[1]	Blinking[1]	Blinking[1]	Blinking[1]	Auto-test at power-on.	–
Blinking[1]	ON	Blinking[1]	Blinking[1]	Auto-test at power-on has detected an internal error on output channels.	–
OFF	ON	OFF	OFF	Internal error detected.	Replace the module if the condition persists.
OFF	Blinking[1]	OFF	X	Non-configured I/O module.	Configure the module via the CPU.

X indicates the LED state can be either ON or OFF.
1. Blinking: 500 ms ON / 500 ms OFF.
2. Flickering: 50 ms ON / 50 ms OFF.

Module LEDs				Module State	Recommended Response
Run	Err	I/O	LCK		
ON	Blinking[1]	OFF	X	No communication between CPU and module.The module is in Fallback state.	Verify that: ● The CPU is an M580 safety CPU and that it is operational. ● The backplane is operational (if I/O module is on main rack). ● The cable between the CPU and I/O module is operational and properly connected (if the I/O module is on an extended or remote rack).
ON	Flickering[2]	OFF	OFF	No communication between CPU and module. The module is in Fallback state (or in Reset if module has never been operating normally).	Debug the condition using the DDDT variables *(see page 117)* for the I/O module instance.
ON	Flickering[2]	OFF	ON	Communication not safe and configuration locked. The module is in Fallback state (or in Reset if module has never been operating normally).	● Verify the locked configuration in the module is equal to the module configuration stored in application in the CPU as configured using Unity Pro. ● Debug the condition using the DDDT variables *(see page 117)* for the I/O module instance.
ON	ON	OFF	X	Internal error detected on the output channel.	Replace the module if the condition persists.
ON	OFF	OFF	OFF	Communication with CPU is safe and the configuration is unlocked.	–
ON	OFF	OFF	ON	Communication with CPU is safe and the configuration is locked.	–

X indicates the LED state can be either ON or OFF.
1. Blinking: 500 ms ON / 500 ms OFF.
2. Flickering: 50 ms ON / 50 ms OFF.

Channel Diagnostics

Use all the LEDs on the BMXSRA0405 digital relay output module to diagnose channel status:

Module LEDs				Channel LEDs		Channel State	Recommended Response
Run	Err	I/O	LCK	Channel State (LED 0...3)	Detected Error (LED 0...3)		
ON	OFF	OFF	X	ON	OFF	Output relay is closed.	–
ON	OFF	OFF	X	OFF	OFF	Output relay is open.	–
ON	ON	OFF	X	OFF	ON	The output relay is not operational.	Replace the module if the condition persists.
X indicates the LED state can be either ON or OFF.							

Chapter 13
Operating an M580 Safety System

Introduction

This chapter provides information on how to operate an M580 safety system.

What Is in This Chapter?

This chapter contains the following sections:

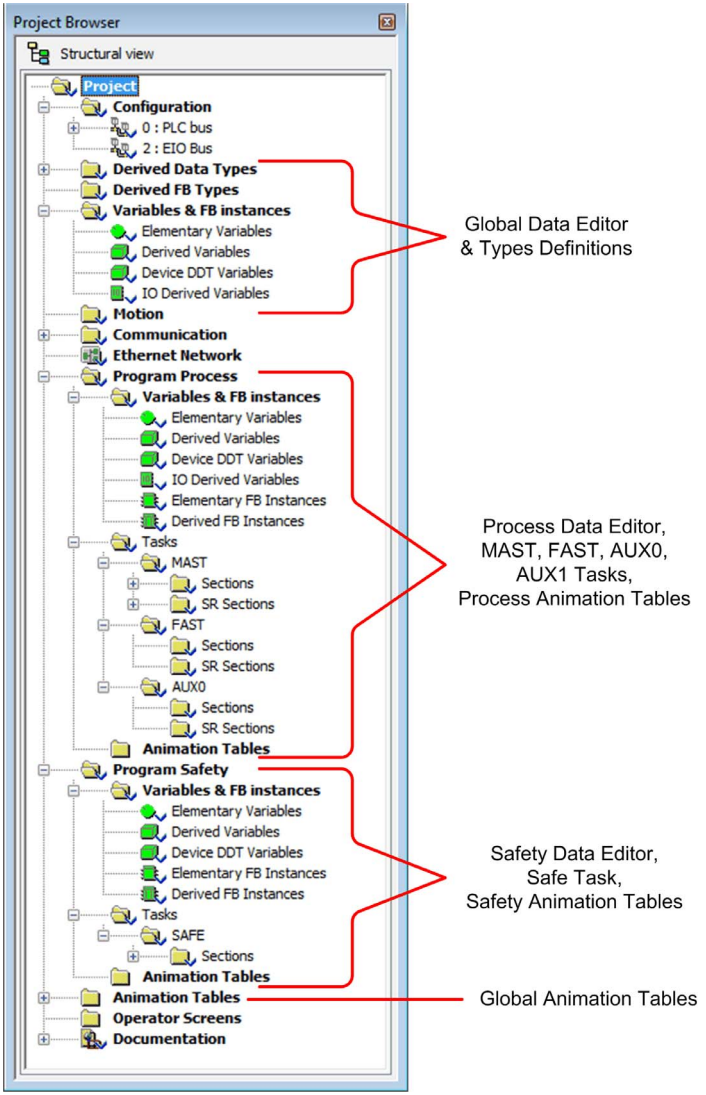
Section	Topic	Page
13.1	Process, Safety and Global Data Areas in Unity Pro	218
13.2	Operating Modes, Operating States, and Tasks	222
13.3	Building an M580 Safety Project	240
13.4	Locking M580 Safety I/O Module Configurations	241
13.5	Initializing Data in Unity Pro	244
13.6	Working with Animation Tables in Unity Pro	245
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13.10	Modifications to Unity Pro for the M580 Safety System	274

Section 13.1
Process, Safety and Global Data Areas in Unity Pro

Data Separation in Unity Pro

Data Areas in Unity Pro

The **Structural View** of the **Project Browser** displays the separation of data in Unity Pro. As shown below, each data area has its own data editor and collection of animation tables:



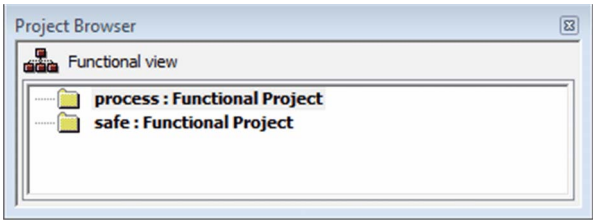
Looking at the **Project Browser** you will notice that:

- The safe area contains a Safety Data Editor, safety logic, and function block instances used by the SAFE task. However, note that:
  - I/O events, timer events, and sub-routines are not supported in a safety program.
  - IODDT variables are not supported by the SAFE task, and are not included in the safe area.
- The process area contains a Process Data Editor, process logic, and function block instances used by the non-safe tasks (i.e., MAST, FAST, AUX0 and AUX1).
- The global area contains a Global Data Editor, derived data and function block types instantiated in the process and safety programs.

NOTE: The term *Global Data* used in this topic refers to the application wide – or global – scope of data objects in a safety project. It does not refer to the Global Data service that is supported by many Schneider Electric Ethernet modules.

Project Browser in Functional View

The **Functional View** of the Unity Pro **Project Browser** for an M580 safety system presents two functional projects – one for the process namespace, one for the safe namespace:



Management of each functional project in an M580 safety system is the same as managing a project in the functional view of an M580 non-safety system, except for animation tables and code sections.

Effect on Structural View:

When you add a code section or animation table to a functional project, it becomes associated with the namespace associated with that functional project. Adding a code section or animation table to:

- the **process : Functional Project** adds it to the process namespace of the project in structural view.
- the **safe : Functional Project** adds it to the safe namespace of the project in structural view.

Availability of Language and Task Selections:

When you create a new code section for a functional project (by selecting **Create → New Section...**), the available **Language** and **Task** selections depend on the functional project:

When you create a new code section for a functional project (by selecting **Create → New Section...**), the available **Language** and **Task** selections depend on the associated functional project:

Functional Project	Available Languages and Tasks	
	Languages[1]	Tasks[2]
process : Functional Project	• IL • FBD • LD • LL984 segment • SFC • ST	• MAST • FAST • AUX0 • AUX1
safe : Functional Project	• FBD • LD	• SAFE

1. Selected in the **General** tab of the new section dialog.
2. Selected in the **Localization** tab of the new section dialog. The MAST task is available by default. Other sections are available for selection only after they have been created in the process program.

Section 13.2
Operating Modes, Operating States, and Tasks

Introduction

This section describes the operating modes, operating states, and tasks supported by the M580 safety PAC.

What Is in This Section?

This section contains the following topics:

Topic	Page
M580 Safety PAC Operating Modes	223
M580 Safety PAC Operating States	228
Start Up Sequences	233
M580 Safety PAC Tasks	237

M580 Safety PAC Operating Modes

Two Operating Modes

The M580 safety PAC presents two operating modes:

- Safety mode: the default operating mode used for safety operations.
- Maintenance mode: an optional operating mode that can be entered temporarily to debug and modify the application program, or change the configuration.

Unity Pro XLS software is the exclusive tool you can use to manage operating mode transitions.

Safety Mode and its Limitations

Safety mode is the default mode of safety PAC. When the safety PAC is powered ON with a valid application present, the PAC enters safety mode. Safety mode is used to control execution of the safety function. You can upload, download, run and stop the project in safety mode.

When the M580 safety PAC is operating in safety mode, the following functions are **not** available:

- Downloading a changed configuration from Unity Pro to the PAC.
- Editing and/or forcing safety variable values and safety I/O states.
- Debugging application logic, by means of breakpoints, watchpoints, and step-through code execution.
- Using animation tables or UMAS requests (for example, from an HMI) to write to safety variables and safety I/O.
- Changing the configuration settings of safety modules via CCOTF. (Note that the use of CCOTF for non-interfering modules is supported.)
- Performing online modification of the safety application.
- Using link animation.

NOTE: In safety mode, all safety variables and safety I/O states are read-only. You cannot directly edit the value of a safety variable.

You can create a global variable, and use it to pass a value between a linked process (non-safe) variable and a linked safety variable using the interface tabs of the Process Data Editor and the Safety Data Editor. After the link is made, the transfer is executed as follows:

- At the beginning of each SAFE task, the non-safe variable values are copied to the safe variables.
- At the end of the SAFE task, the safe output variable values are copied to the non-safe variables.

Maintenance Mode Functionality

Maintenance mode is comparable to the normal mode of a non-safety M580 CPU. It is used only to debug and tune the application SAFE task. Maintenance mode is temporary because the safety PAC automatically enters safety mode if communication between Unity Pro and the PAC is lost, or upon the execution of a disconnect command. In maintenance mode, persons with the appropriate permissions can both read and write to safety variables and safety I/O that are configured to accept edits.

In maintenance mode, dual execution of SAFE task code is performed, but the results are not compared.

When the M580 safety PAC is operating in maintenance mode, the following functions are available:

- Downloading a changed configuration from Unity Pro to the PAC.
- Editing and/or forcing safety variable values and safety I/O states.
- Debugging application logic, by means of breakpoints, watchpoints, and step-through code execution.
- Using animation tables or UMAS requests (for example, from an HMI) to write to safety variables and safety I/O.
- Changing the configuration via CCOTF.
- Performing online modification of the safety application.
- Using link animation.

In maintenance mode, the SIL level of the Safety PLC is not guaranteed.

⚠ WARNING

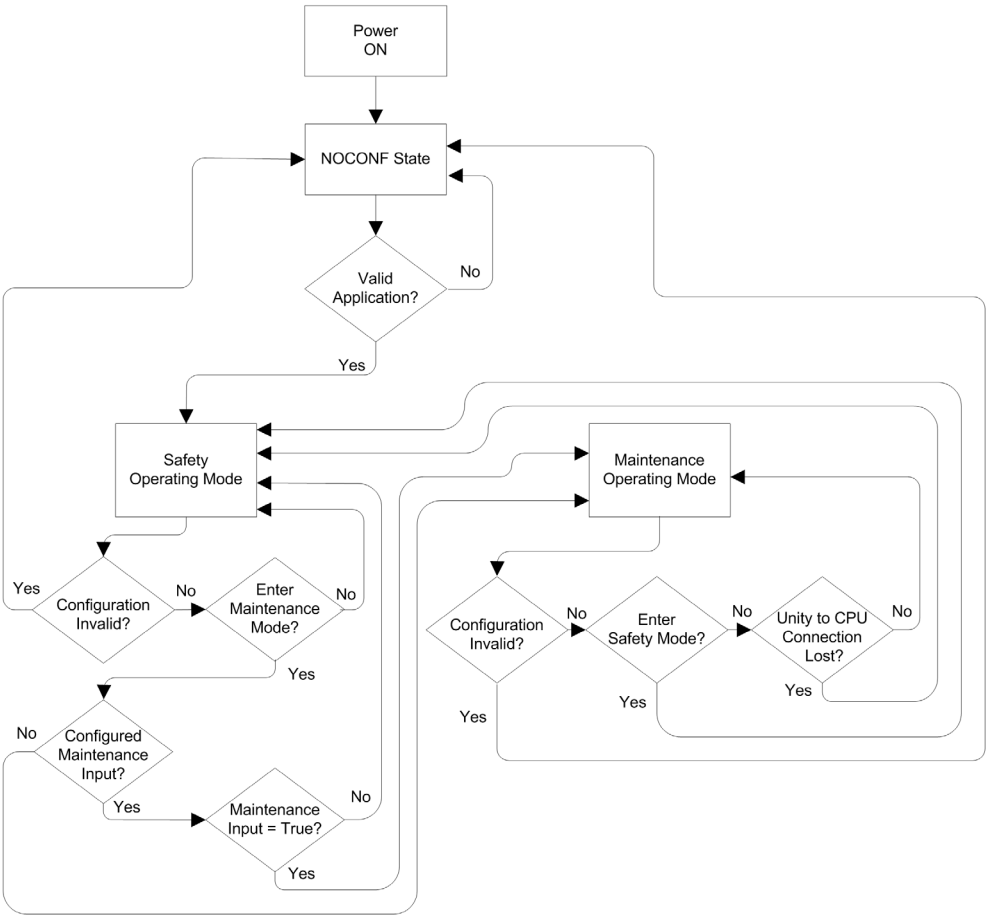
LOSS OF THE SAFETY INTEGRITY LEVEL

While the safety PAC is in maintenance mode, you need to take appropriate measures to ensure the safe state of the system.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Operating Mode Transitions

The following diagram shows how the M580 safety PAC enters, then transitions between safety mode and maintenance mode:



When switching between safety mode and maintenance mode:

- It is OK to switch from maintenance mode to safety mode with forcing ON. In this case, the forced variable value or I/O state remains forced after the transition until another transition from safety to maintenance mode occurs.
- The transition from maintenance mode to safety mode can be accomplished in the following ways:
  - Manually, by menu or toolbar command in Unity Pro.
  - Automatically, by the safety PAC, when communication between Unity Pro and the PAC is lost for about 50 seconds.

- The maintenance input function, when it is configured, operates as a check on the transition from safety mode to maintenance mode. The maintenance input function is configured in Unity Pro in the CPU **Configuration** tab by:
  - Selecting the **Maintenance Input** setting, and
  - Entering the topological address of an input bit (%I) for a non-interfering digital input module on the local rack.

  When the maintenance input is configured, the transition from safety mode to maintenance mode takes into account the state of the designated input bit (%I). If the bit is set to 0 (false), the PAC is locked in safety mode. If the bit is set to 1 (true), a transition to maintenance mode can occur.

Switching Between Safety Mode and Maintenance Mode in Unity Pro

Switching the safety PAC from maintenance mode to safety mode is not possible if:

- The PAC is in debug mode.
- A breakpoint is activated in a SAFE task section.
- A watchpoint is set in a SAFE task section.

When debug mode is not active, no SAFE task breakpoint is activated, and no SAFE task watchpoint is set, you can manually activate a transition between safety mode and maintenance mode, as follows:

- To switch from safety mode to maintenance mode, either:
  - Select **PLC → Maintenance**, or
  - Click the  toolbar button.
- To switch from maintenance mode to safety mode, either:
  - Select **PLC → Safety**, or
  - Click the  toolbar button.

NOTE: Entering and exiting safety mode events are logged in the SYSLOG server in the CPU.

Determining the Operating Mode

You can determine the current operating mode of an M580 safety PAC using either the **SMOD** LEDs of the CPU and coprocessor, or Unity Pro.

When the **SMOD** LEDs of the CPU and coprocessor are:

- *Flashing* ON, the PAC is in maintenance mode.
- *Solid* ON, the PAC is in safety mode.

When Unity Pro is connected to the PAC, it identifies the operating mode of the M580 safety PAC in several places:

- System words %SW12 (coprocessor) and %SW13 (CPU) *(see page 293)* together indicate the operating mode of the PAC, as follows:
  - if %SW12 is set to 16#A501 (hex) and %SW13 is set to 16#501A (hex), the PAC is in maintenance mode.
  - if either or both of these system words is set to 16#5AFE (hex), the PAC is in safety mode.
- Both the **Task** and **Information** sub-tabs of the CPU **Animation** tab display the operating mode of the PAC.
- The task bar, at the bottom of the Unity Pro main window, indicates the operating mode as either MAINTENANCE or SAFETY.

M580 Safety PAC Operating States

Operating States

The M580 safety PAC presents the following operating states:

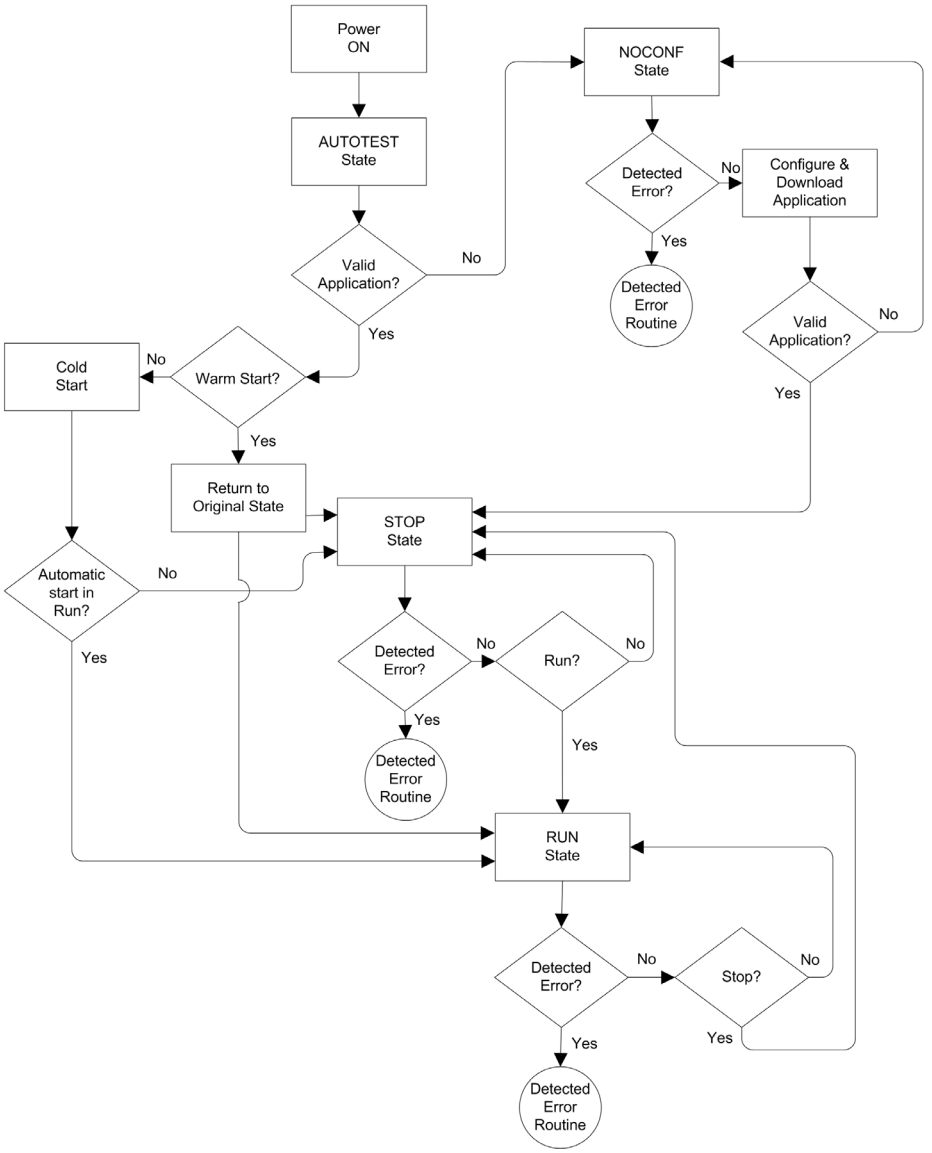
Operating State	Applies to...	Description
AUTOTEST	PAC	The CPU is executing internal self-tests. **NOTE:** If extended racks are connected to the main local rack and line terminators are not plugged into the unused connectors on the rack extender module, the CPU remains in AUTOTEST after the self-tests have completed.
NOCONF	PAC	The application program is not valid.
STOP	PAC or Task	The PAC has a valid application and no error is detected, but operation has stopped because: ● At startup **Automatic start in Run** is not set (safety mode *(see page 223)*). ● Execution stopped by execution of a STOP command (safe *(see page 223)* or maintenance *(see page 224)* mode). ● Breakpoints were set in maintenance mode, then the connection between Unity Pro and the CPU was lost for more than 50 seconds. The CPU reads the inputs associated with each task, but does not refresh outputs, which enter their fallback state. The CPU can be restarted when you are ready. **NOTE:** Issuing a STOP command in Unity Pro stops all tasks. The STOP event is recorded in the SYSLOG server of the CPU.
HALT	Task	The M580 safety PAC presents two independent HALT states: ● Process HALT applies to the non-SAFE tasks (MAST, FAST, AUX0, and AUX1). When any process task enters the HALT state, all other process tasks also enter the HALT state. The SAFE task is not affected by a process HALT condition. ● SAFE HALT applies only to the SAFE task. Process tasks are not affected by a SAFE HALT condition. In each case, task operations are halted because an unexpected blocking condition has been encountered, resulting in a recoverable *(see page 184)* condition. The CPU reads the inputs associated with each halted task, but does not refresh outputs, which are in fallback state.

Operating State	Applies to...	Description
RUN	PAC or Task	With a valid application and no error detected, the CPU reads the inputs associated with each task, executes the code associated with each task, and refreshes the associated outputs. ● in safety mode *(see page 223)*: the safety function is performed, and all limitations are applied. ● in maintenance mode *(see page 224)*: the PAC operates like any non-safety CPU. Dual execution of SAFE task code is performed, but the results are not compared. **NOTE:** Issuing a RUN command in Unity Pro starts all tasks. The RUN event is recorded in the SYSLOG server of the CPU
WAIT	PAC	The CPU is in a transitory state while it backs up data when a power down condition is detected. The CPU starts again only when power is restored and the supply reserve is replenished. Because WAIT is a transitory state, it may not be visible. The CPU performs a warm restart *(see page 235)* to exit the WAIT state.
ERROR	PAC	The CPU is stopped because an non-recoverable *(see page 181)* hardware or system error is detected. The ERROR state triggers the safety function *(see page 15)*. When the system is ready to be restarted, perform a cold start *(see page 235)* of the CPU to exit the ERROR state, either by cycling power or performing a RESET.
OS DOWNLOAD	PAC	A CPU or COPRO firmware download is in progress.

Refer to the *M580 CPU LED Diagnostics (see page 186)* and *M580 Safety Coprocessor LED Diagnostics (see page 186)* topics for information on the operating states of the PAC.

Operating State Transitions

The transitions between the several states in an M580 safety PAC are described, below:



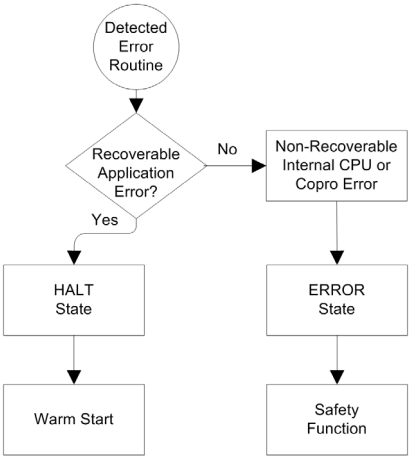
Refer to the topic *Detected Error Processing (see page 231)* for information on how the safety system handles detected errors.

Detected Error Processing

The M580 safety PAC handles the following kinds of CPU detected errors:

- Recoverable application detected errors: These events cause the related task(s) to enter the HALT state.
  NOTE: Because the MAST, FAST, and AUX tasks operate in the same memory area, an event that causes one of these tasks to enter HALT state causes the other non-safe tasks also to enter HALT state. Because the SAFE task operates in a separate memory area, the non-safe tasks are not affected if the SAFE task enters HALT state.
- Non-recoverable application detected errors: Internal CPU or coprocessor detected errors: These events cause the PAC to enter the ERROR state. The safety function is applied to the affected portion of the safety loop.

The logic of the detected error handling process is described below:



The impact of detected errors on individual tasks is described below:

Detected Error Type	Task State			
	FAST	SAFE	MAST	AUX
FAST task watchdog overrun	HALT	RUN[1]	HALT	HALT
SAFE task watchdog overrun	RUN	HALT[2]	RUN	RUN
MAST task watchdog overrun	HALT	RUN	HALT	HALT
AUX task watchdog overrun	HALT	RUN	HALT	HALT

1.Because FAST task has a higher priority than the SAFE task, delay of the FAST task may cause the SAFE task to enter HALT or ERROR state instead of RUN state.
2. The ERROR and HALT states on the SAFE task causes the safe outputs to be set to their user configurable state (fallback or maintain).
3. The safety watchdog is set equal to 1.5 times the SAFE task watchdog.

Detected Error Type	Task State			
	FAST	SAFE	MAST	AUX
CPU dual code execution detected error	RUN	HALT[2]	RUN	RUN
Safety watchdog overrun[3]	ERROR	ERROR[2]	ERROR	ERROR
CPU internal detected error	ERROR	ERROR[2]	ERROR	ERROR

1.Because FAST task has a higher priority than the SAFE task, delay of the FAST task may cause the SAFE task to enter HALT or ERROR state instead of RUN state.
2. The ERROR and HALT states on the SAFE task causes the safe outputs to be set to their user configurable state (fallback or maintain).
3. The safety watchdog is set equal to 1.5 times the SAFE task watchdog.

Task Bar Safety Status Viewer

When Unity Pro is connected to the M580 safety PAC, the task bar includes a field describing the combined operating states of the SAFE task and the process tasks (MAST, FAST, AUX0, AUX1), as follows:

Process task(s) state	SAFE task state	Message
STOP (all process tasks in STOP state)	STOP	STOP
STOP (all process tasks in STOP state)	RUN	RUN
STOP (all process tasks in STOP state)	HALT	SAFE HALT
RUN (at lease one process task in RUN state)	STOP	RUN
RUN (at lease one process task in RUN state)	RUN	RUN
RUN (at lease one process task in RUN state)	HALT	SAFE HALT
HALT	STOP	PROC HALT
HALT	RUN	PROC HALT
HALT	HALT	HALT

Start Up Sequences

Introduction

The M580 safety PAC can enter the start-up sequence in the following circumstances:

- At initial power-up.
- In response to a power interruption.

Depending on the type of task, and the context of the power interruption, the M580 safety PAC may perform either a cold start *(see page 235)* or a warm start *(see page 235)* when power is restored.

Initial Start-Up

At initial start-up, the M580 safety PAC performs a cold start. All tasks, including both the SAFE task and the non-safe (MAST, FAST, AUX0, AUX1) tasks, enter the STOP state unless **Automatic start in RUN** is enabled, in which case all tasks enter the RUN state.

Start-Up after a Power Interruption

The M580 safety power supply provides a power reserve that continues to supply all modules on the rack for up to 10 ms in case of a power interruption. When the power reserve is depleted, the M580 safety PAC performs a complete power cycle.

Before powering down the system, the safety CPU stores the following data that defines the operating context at power down:

- Date and time of the power down (stored in %SW54...%SW58).
- State of each task.
- State of event timers.
- Values of running counters.
- Signature of the application.
- Application data (current values of application variables)
- Application check sum.

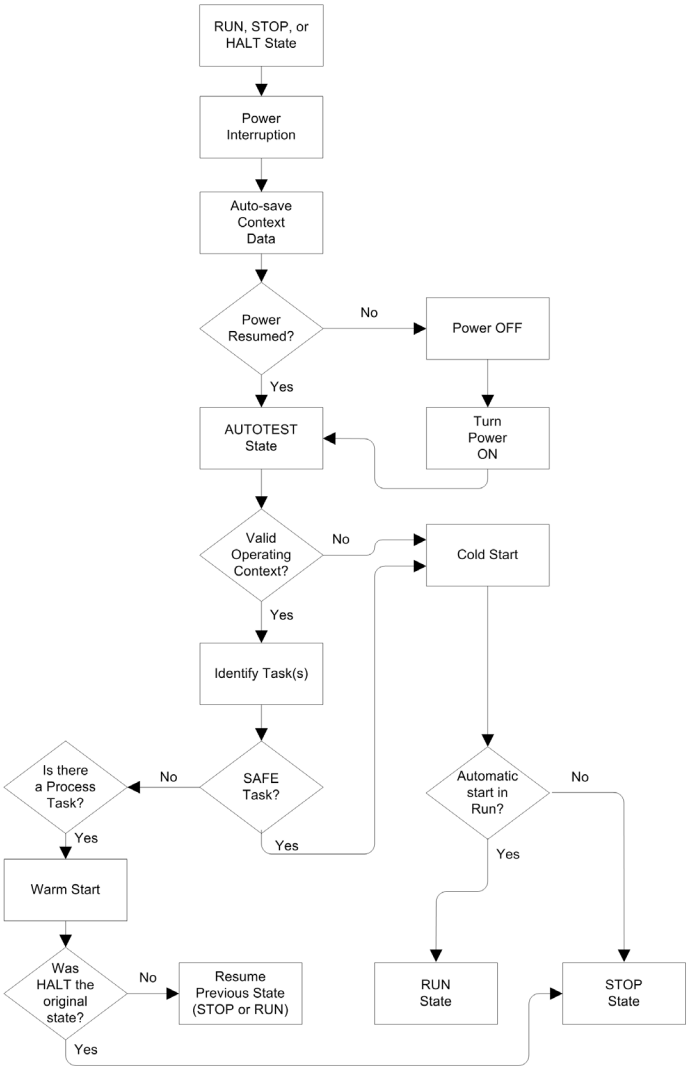
After power down, the start-up can be either automatic (if power was restored before completion of the shut-down) or manual (if not).

Next, the M580 safety PAC performs self-tests and checks the validity of the operating context data that was saved at power down, as follows:

- The application check sum is verified.
- The SD memory card is read to confirm that is contains a valid application.
- If the application in the SD memory card is valid, the signatures are checked to confirm they are identical.
- The saved application signature is verified by comparing it to the stored application signature.

If the operating context is valid, the non-safe tasks perform a warm start. If the operating context is not valid, the non-safe tasks perform a cold start. In either case, the SAFE task performs a cold start.

This start-up sequence after a power interruption is presented, below:



Cold Start

A cold start causes all tasks, including both the SAFE task and the non-safe (MAST, FAST, AUX0, AUX1) tasks, enter the STOP state, unless **Automatic start in RUN** is enabled, in which case all tasks enter the RUN state.

A cold start performs the following operations:

- Application data (including internal bits, I/O data, internal words, and so forth) are assigned the initial values defined by the application.
- Elementary functions are set to their default values.
- Elementary function blocks and their variables are set to their default values.
- System bits and words are set to their default values.
- Initializes all forced variables by applying their default (initialized) values.

A cold start can be executed for data, variables and functions in the process namespace by selecting **PLC → Init** in Unity Pro *(see page 244)*, or by setting the system bit %S0 (COLDSTART) to 1. The %S0 system bit has no effect on the data and functions belonging to the safe namespace.

Warm Start

A warm start causes each process task – including the (MAST, FAST, AUX0, AUX1) tasks – to re-enter its operating state as of the time of the power interruption. By contrast, a warm start causes the SAFE task to enter the STOP state, unless **Automatic start in RUN** is selected.

NOTE: If a task was in the HALT state or in breakpoint at the time of power interruption, that task enters the STOP state after the warm start.

A warm start performs the following operations:

- Restores the last held value to process namespace variables.
- Initializes safe namespace variables by applying their default (initialized) values.
- Initializes all forced variables by applying their default (initialized) values.
- Restores the last held value to application variables.
- Sets %S1 (WARMSTART) to 1.
- Connections between the PAC and CPU are reset.
- I/O modules are re-configured (if necessary) using their stored settings.
- Events, the FAST task, and the AUX tasks are disabled.
- The MAST task is re-started from the beginning of the cycle.
- %S1 is set to 0 at the conclusion of the first execution of the MAST task.
- Events, the FAST task, and the AUX tasks are enabled.

If a task was in the process of execution at the time of power interruption, after warm start the task resumes execution at the beginning of the task.

⚠ WARNING
UNEXPECTED EQUIPMENT OPERATION You are responsible to confirm that selecting **Automatic start in RUN** is compliant with the correct behavior of your system. If it is not, de-activate this feature. **Failure to follow these instructions can result in death, serious injury, or equipment damage.**

M580 Safety PAC Tasks

Introduction

An M580 safety PAC can execute single-task and multi-task applications. Unlike a single-task application which only executes the MAST task, a multi-task application defines the priority of each task.

The M580 safety PAC supports the following tasks:

- FAST
- SAFE
- MAST
- AUX0
- AUX1

Task Characteristics

The tasks supported by the M580 safety PAC present the following task characteristics:

Task Name	Priority	Time Model	Period Range	Period Default	Watchdog
FAST	1	Periodic	1...255 ms	100 ms	10...500 ms[2]
SAFE	2	Periodic	10...255 ms	250 ms	10...500 ms[2]
MAST[1]	3	Cyclic or Periodic	1...255 ms	250 ms	10...1500 ms[2]
AUX0	4	Periodic	10...2550 ms	2000 ms	100...5000 ms[2]
AUX1	5	Periodic	10...2550 ms	2000 ms	100...5000 ms[2]

1. MAST task is required and cannot be deactivated.
2. If CCOTF is enabled (by selecting **Online modification in RUN or STOP** in the **Configuration** tab of the CPU properties dialog), the minimum **Watchdog** setting is 64 ms.

Task Priority

M580 Safety PACs execute pending tasks according to their priority. When a task is running, it can be interrupted by another task with a higher relative priority. For example, when a periodic task is scheduled to execute its code, it would interrupt a lower priority task, but would wait until the completion of a higher priority task.

Task Configuration Considerations

All the non-safe tasks (MAST, FAST, AUX0, and AUX1) operate in the same memory area, while the SAFE task operates in its own, separate memory area. As a result:

- If one non-safe task exceeds its watchdog, all non-safe tasks enter HALT state, while the SAFE task continues to be operational.
- If the SAFE task exceeds its watchdog, only the SAFE task enters HALT state, while the non-safe tasks continue to be operational.

When creating and configuring tasks for your application, consider the following task features:

SAFE task:

Design this periodic task to execute only safety-related code sections for safety I/O modules. Because the SAFE task is assigned a lower priority than the FAST task, execution of the SAFE task may be interrupted by the FAST task.

Define the maximum execution time for the SAFE task by setting the appropriate watchdog value. Consider the time required to execute code and to read and write safe data. If the time to execute the SAFE task exceeds the watchdog setting, the SAFE task enters HALT state, and the %SW125 system word displays the detected error code 16#DEB0.

NOTE:

- Because FAST task has a higher priority than the SAFE task, you may want to include a component for FAST task delay time in the SAFE task watchdog setting.
- If the overrun of the SAFE task execution equals the “Safety watchdog” (which is a value equal to one and one-half times the SAFE task watchdog setting), the CPU and Copro will enter the ERROR state and the safety function will be applied.

MAST task:

This task can be configured as either cyclic or periodic. When operating in cyclic mode, define a maximum execution time by inputting an appropriate MAST watchdog value. Add a small time interval to this value at the end of each cycle to allow for the execution of other lower priority system tasks. Because the AUX tasks carry a lower priority than MAST, if this time slot is not provided, the AUX tasks may never be executed. Consider adding a time interval equal to 10% of cycle execution time, with a minimum of 1 ms and a maximum of 10 ms.

If the time to execute a cyclic MAST task exceeds the watchdog setting, the MAST task and all other non-SAFE tasks enter HALT state, and the %SW125 system word displays the detected error code 16#DEB0.

When operating in periodic mode, it is possible for the MAST task to exceed its period. In that case the MAST task runs in cyclic mode and the system bit %S11 is set.

FAST task:

The purpose of this periodic task is to execute a high-priority part of the application. Define a maximum execution time by setting the FAST watchdog value. Because the FAST task interrupts execution of all other tasks – including the SAFE task – it is recommended to configure the execution time of the FAST task to be as short as possible. A FAST task watchdog value not much greater than the FAST period is recommended.

If the time to execute the FAST task exceeds the watchdog setting, the FAST task and all other non-SAFE tasks enter HALT state, and the %SW125 system word displays the detected error code 16#DEB0.

AUX tasks:

AUX0 and AUX1 are optional periodic tasks. Their purpose is to execute a low priority part of the application. The AUX tasks are executed only after execution of the MAST, SAFE and FAST tasks has finished.

Define a maximum execution time for the AUX tasks by setting the appropriate watchdog value. If the time to execute an AUX task exceeds the watchdog setting, the AUX task and all other non-SAFE tasks enter HALT state, and the %SW125 system word displays the detected error code 16#DEB0.

Section 13.3
Building an M580 Safety Project

Building an M580 Safety Project

Building an M580 Safety Project

The Unity Pro XLS **Build** menu presents three different build commands, as follows:

Command	Description
Build Changes	Compiles only the changes that have been made to the application program since the previous build command, and adds them to the previously generated application program.
Rebuild All Project	Re-compiles the entire application program, replacing the previously generated build of the application program. **NOTE:** For M580 safety I/O modules, this command does not generate a new module unique identifier (MUID) value. Instead, the previously generated MUID value is retained.
Renew Ids & Rebuild All	Re-compiles the entire application program, replacing the previously generated build of the application program. **NOTE:** ● Execute this command only when the safety I/O modules are unlocked *(see page 241)*. ● For M580 safety I/O modules, this command generates a new module unique identifier (MUID) value and replaces the existing MUID value with the new value.

Section 13.4
Locking M580 Safety I/O Module Configurations

Locking M580 Safety I/O Module Configurations

Locking a Safety I/O Module Configuration

Each safety I/O module has a configuration locking button, which you can find at the top front of the module. The purpose of the locking function is to help prevent unintended changes to I/O module configuration. For example, locking the I/O module's current configuration can stop an attempt to assign the module a fake configuration, or merely help protect against configuration mistakes.

To achieve the intended safety integrity level (SIL), lock each safety I/O module after it has been configured, but before you begin or resume operations.

⚠ WARNING

RISK OF UNINTENDED OPERATION

You must lock each safety I/O module after it is configured but before beginning operations, or risk an unintended degradation in your project's safety integrity level.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

The lock and unlock mechanisms work as follows:

- To lock a safety I/O module configuration, press and hold down the lock button for more than 3 seconds, then release the button.
- To unlock a safety I/O module configuration, press and hold down the lock button for more than 3 seconds, then release the button.

Scenarios for Locking Safety I/O Module configurations

The procedure you follow to lock the configuration of SIL3 safety I/O modules will vary, depending on the scenario, which can be:

- First configuration of the I/O modules
- Fast device replacement of I/O modules
- Perform a change configuration on the fly (CCOTF) for I/O modules

The procedure for each scenario is described below.

First configuration of SIL3 I/O safety modules:

Step	Action
1	Connect Unity Pro to the M580 safety PAC.
2	Use the **Transfer project from PLC** command to load the project from the PAC into Unity Pro.
3	In the **PLC bus** window in Unity Pro, open each SIL3 safety I/O module and confirm that each module is accurately configured.
4	In an animation table in Unity Pro, view the DDDT for each SIL3 safety I/O module and confirm that the configuration of each module is the same as in step 3, above.
5	Lock the configuration of each SIL3 I/O module by pressing and holding down the configuration locking button for more than 3 seconds, then release the button.
6	Check in an animation table the validity of the lock bit status (CONF_LOCKED) for each SIL3 I/O module.

Fast device replacement of a SIL3 I/O safety module:

Step	Action
1	Replace the SIL3 safety I/O module with a new one.
2	Connect Unity Pro to the M580 safety PAC in maintenance operating mode *(see page 224)*.
3	In the **PLC bus** window in Unity Pro, open each SIL3 safety I/O module and confirm that each module is accurately configured.
4	In an animation table in Unity Pro, view the DDDT for each SIL3 safety I/O module and confirm that the configuration of each module has not changed and is the same as in step 3, above.
5	Lock the configuration of each SIL3 I/O module by pressing and holding down the configuration locking button for more than 3 seconds, then release the button.
6	Check in an animation table the validity of the lock bit status (CONF_LOCKED) for each SIL3 I/O module.

Performing CCOTF to add a new SIL3 I/O safety module:

Step	Action
1	Connect Unity Pro to the M580 safety PAC in maintenance operating mode *(see page 224)*.
2	Add a new SIL3 safety I/O module to the configuration.
3	Execute the **Build → Build Changes** command.
4	In the **PLC bus** window in Unity Pro, open each SIL3 safety I/O module and confirm that each module is accurately configured.
5	In an animation table in Unity Pro, view the DDDT for each SIL3 safety I/O module and confirm that the configuration of each module has not changed and is the same as in step 3, above.
6	Lock the configuration of each SIL3 I/O module by pressing and holding down the configuration locking button for more than 3 seconds, then release the button.

Step	Action
7	Check in an animation table the validity of the lock bit status (CONF_LOCKED) for each SIL3 I/O module.
8	In the **PLC** menu of Unity Pro, command the PAC to enter safety mode *(see page 223)*.

Section 13.5
Initializing Data in Unity Pro

Initializing Data in Unity Pro for the M580 Safety PAC

Two Init Commands

The **PLC** menu in Unity Pro provides two separate commands for the initialization of data:

- The **Init** command initializes data for the process (or non-safe) namespace, which can be used by the MAST, FAST, AUX0 and AUX1 tasks. You can execute this command if the PAC is operating in either safety or maintenance mode while the PAC is in the STOP state. This command is the equivalent of setting the system bit %S0 (COLDSTART) to 1.
  NOTE: Setting the %S0 bit to 1 initializes data in the process namespace only. It does not affect data in the safe namespace.
- The **Init Safety** command initializes data only for the safe namespace, which data can be used exclusively by the SAFE task. You can execute this command only if the SAFE task is operating in maintenance mode while the SAFE task is in the STOP or HALT state. Executing this command when the SAFE task is in the HALT state causes the SAFE task to restart in the STOP state.

Both the **Init** and the **Init Safety** commands perform a cold start. *(see page 235)*

Section 13.6
Working with Animation Tables in Unity Pro

Animation Tables and Operator Screens

Introduction

A M580 safety PAC supports three kinds of animation tables, each associated with one of the following data areas:

- Process area animation tables can include only data in the process namespace.
- Safety area animation tables can include only data in the safe namespace.
- Global animation tables can include data for the entire application, including data created for the safe and process namespaces, and global variables.
  NOTE: In a global animation table, data variable names include a prefix indicating the source namespace, as follows:
- A data variable from the Safe namespace is displayed as “SAFE.<varname>”.
- A data variable from the Process namespace is displayed as “PROCESS.<variable name>”.
- A data variable from the Global (or Application) namespace displays only its <variable name>, with no namespace prefix.

Both process and safety data from an M580 safety PAC are also accessible by external processes (for example, SCADA or HMI).

Your ability to create and modify an animation table, and the ability to execute animation table functions, depend on the namespace of the affected variables and the operating mode of the safety project.

Conditions for Creating and Editing Animation Tables

Creating and editing animation tables involves adding or deleting data variables. Your ability to add data variables to, or delete data variables from an animation table depends on:

- The namespace (safe or process) in which the data variable resides.
- The operating mode (safety or maintenance) of the M580 safety PAC.

When Unity Pro is connected to the M580 safety PAC, you can create and edit animation tables as follows:

- Adding process namespace variables to, or deleting process namespace variables from a process or global animation table is supported while the M580 safety PAC is operating in either safety mode or maintenance mode.
- Adding safe namespace variables to, or deleting safe namespace variables from a safety animation table is supported while the M580 safety PAC is operating in maintenance mode.
- Adding safe namespace variables to, or deleting safe namespace variables from a safety animation table is supported while the M580 safety PAC is operating in safety mode only if the project settings do not include animation tables in the upload information.

NOTE: Animation tables are included in, or excluded from, upload information in Unity Pro by selecting **Tools → Project Settings...** to open the **Project Settings...** window, then navigating to **Project Settings → General → PLC embedded data → Upload information → Animation tables**.

Conditions for Operating Animation Tables

You can use animation tables to force a variable value, unforce a variable value, modify a single variable value, or modify multiple variable values. Your ability to perform these functions depends on the namespace in which a variable resides and the operating mode of the M580 safety PAC, as follows:

- Process or global variable values can be read or written in both safety operating mode and maintenance operating mode.
- Safety variable values can be read or written in maintenance operating mode.
- Safety variable values can only be read in safety operating mode.

Process for Creating Animation Tables in the Safety or Process Namespace in Unity Pro

Unity Pro provides two ways to create animation tables for either the safety or process namespace:

- From a safety or process code section window, right click in the code window, then select either:
  - **Initialize Animation Table** to add the data object to an existing animation table in safety or process namespace, or
  - **Initialize New Animation Table** to add the data object to a new animation table in the safety or process namespace.

  In each case, all the variables in the code section are added to the existing or new animation table.
- From the **Project Browser**, in either the process or safety data area, right click on the **Animation Tables** folder, then select **New Animation Table**. Unity Pro creates a new, empty animation table. You can then add individual variables from the namespace (safety or process) related to the table.

Process for Creating Globally Scoped Animation Tables

Create a global animation table in the **Project Browser** by right clicking the global **Animation Tables** folder, then selecting **New Animation Table**. You can add variables to the new animation table in several ways:

- *Drag and drop*: You can drag a variable from a data editor and drop it into the global animation table. Because the scope of the animation table includes the entire application, you can drag the variable from the **Safety Data Editor**, the **Process Data Editor**, or the **Global Data Editor**.
- *Instance Selection dialog*: You can double-click in a row in the animation table, then click the ellipsis button to open the **Instance Selection** dialog. Use the filtering list in the top right part of the dialog to select a one of the following project areas:
  - SAFE: to display data objects associated with the safety area.
  - PROCESS: to display data object associated with the process area.
  - APPLICATION: to display higher-level application scope data objects.

  Select a data object, then click **OK** to add the item to the animation table.

NOTE: Data objects added to a global animation table from the:
- Process area have the prefix “PROCESS” affixed to the variable name (for example PROCESS.variable_01
- Safety area have the prefix “SAFE” affixed to the variable name (for example SAFE.variable_02
- Global area have no such prefix added to the variable name.

Displaying Data on Operator Screens

You can display data on an operator screen – such as an HMI, SCADA or FactoryCast application – in the same way that you link to data in an animation table. The data variables available for selection are those variables that are included in the Unity Pro data dictionary.

You can enable the data dictionary by opening the **Tools → Project Settings...** window, then in the **Scope → common** area of the window, selecting **General → PLC embedded data → Data dictionary**.

The data dictionary makes data variables available to operator screens as follows:
- Safe namespace variables always include the “SAFE” prefix, and can be reached only by using the format “SAFE.<variable name>”.
- Global or application namespace variables do not include a prefix, and can be reached only by using the “<variable name>” without a prefix.
- The **Usage of Process Namespace** setting determines how an operator screen can reach Process namespace variables.
  - If you select **Usage of Process Namespace**, the operator screen can read process area variables only by using the format “PROCESS.<variable name>”.
  - If you de-select **Usage of Process Namespace**, the operator screen can read process area variables only by using the format “<variable name>” without the PROCESS prefix.
    NOTE: If two variables are declared with the same name, one in the Process namespace and one in the Global namespace, only the variable from the Global Namespace is accessible by an HMI, SCADA, or Factory Cast application.

You can use the **Instance Selection** dialog to access individual data objects.

⚠ CAUTION
UNEXPECTED VARIABLE VALUE ● Be sure that your application has the correct project settings. ● Verify the syntax to access the variables in the different namespaces. **Failure to follow these instructions can result in injury or equipment damage.**

To prevent from accessing the incorrect variable:

- Use different names for the variables you declare in the Process namespace and in the Global namespace, or
- select **Usage of Process Namespace** and use the following syntax to access the variables with the same name:
  - “PROCESS.<variable name>” for variables declared in the Process namespace.
  - “<variable name>” without a prefix for variables declared in the Global namespace

Trending Tool

The Unity Pro Trending Tool is not supported for use with an M580 safety project.

Section 13.7
Adding Code Sections

Adding Code to an M580 Safety Project

Working with Tasks in Unity Pro

In the process namespace, Unity Pro includes the MAST task by default. The MAST task cannot be removed. However, you can add the tasks FAST, AUX0, and AUX1. Note that creating a task in the process part of a safety project is the same as creating a task in a non-safety project. Refer to the topic *Create and Configure a Task* in the *Unity Pro Operating Modes* manual for more information.

In the safe namespace, Unity Pro includes the SAFE task by default. The SAFE task cannot be removed, and no other tasks can be added to the **Program Safety** section of the **Project Browser** in Unity Pro. You can add multiple sections to the SAFE task.

Configuring the SAFE Task Properties

The SAFE task supports only periodic task execution (cyclic execution is not supported). Both the SAFE task **Period** and **Watch Dog** settings are input in the **Properties of SAFE** dialog and can support the following value range:

- SAFE task period: 10...255 ms with a default of 20 ms.
- SAFE task watchdog: 10...500 ms, in increments of 10 ms, with a default of 250 ms.

Set the SAFE task **Period** to a time period that is greater than the minimum transfer time of SAFE task data. The minimum transfer time of SAFE task data is a function of the size of the data to be transferred by the SAFE task.

The SAFE task **Period** can be calculated according to the following formula:

Minimum SAFE task period = greater of 10 ms or ((0.156 x Data Size) + 2 ms)

Sample results of calculating the minimum SAFE task period are set forth below:

Data Size (KB)	Minimum SAFE Task Period (ms)	Data Size (KB)	Minimum SAFE Task Period (ms)
0	10	550	88
50	10	600	96
100	18	650	103
150	25	700	111
200	33	750	119
250	41	800	127
300	49	850	135
350	57	900	142
400	64	950	150
450	72	1000	158
500	80	–	–

NOTE: Configure the SAFE task watchdog with a value that is greater than the SAFE task **Period**.

Refer to the topic *Process Safety Time* *(see page 139)*, for an information regarding how the SAFE task configuration affects process safety time.

Refer to the topic *M580 Safety PAC Tasks* *(see page 237)* for information describing the execution priority of the SAFE task.

Creating Code Sections

Right click on the **Section** folder for a task and select **New Section...** to open a configuration dialog. For the safety and process tasks, the following program languages are available:

Language	Safety tasks	Process tasks			
	SAFE	MAST	FAST	AUX0	AUX1
IL	–	✓	✓	✓	✓
FBD	✓	✓	✓	✓	✓
LD	✓	✓	✓	✓	✓
LL984 segment	–	✓	✓	✓	✓
SFC	–	✓	✓	✓	✓
ST	–	✓	✓	✓	✓
✓: available – : not available					

Except for these limitations on programming language availability for the SAFE task, the new section configuration dialog operates the same as for a non-safety M580 project. Refer to the topic *Properties Dialog Box for FBD, LD, IL, or ST Sections Unity Pro Operating Modes* manual for more information.

Adding Data to Code Sections

Because the SAFE task is separated from the process tasks, only data accessible in the **Safety Data Editor** is available to be added to a SAFE task code section. This data includes:

- Non-located safety variables (i.e. with no %M or %MW address) created in the **Safety Data Editor**.
- Safety system bits.
- Safety system words.
- Data objects that are part of M580 safety module device DDT structures.

Similarly, data available to non-safety task code sections includes all data within the scope of the process namespace. This includes all project data except:

- Data exclusively available to the SAFE namespace (see above).
- Data objects created in the **Global Data Editor**.

Code Analysis

When you analyze or build a project, Unity Pro displays a detected error message if:

- Data belonging to the process namespace is included in the SAFE task.
- Data belonging to the safe namespace is included in a process task (MAST, FAST, AUX0, AUX1).
- Located bits (%M) or words (%MW) are included in a SAFE task section.

Section 13.8
Application Security Management

Introduction

Unity Pro lets you restrict access to the M580 safety PAC to users with assigned passwords. This section references the password assignment processes available in Unity Pro.

What Is in This Section?

This section contains the following topics:

Topic	Page
Application Protection	253
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Loss of Password	260

Application Protection

Overview

Unity Pro provides a password mechanism to help guard against unauthorized access to the safety application. Unity Pro uses the password when you:

- Open the application in Unity Pro.
- Connect to the PAC in Unity Pro.

NOTE: Use of a password is optional. The application protection by password mechanism is disabled if no password is configured in the application (the default setting).

Application protection by password helps prevent unwanted application modification, download, or the opening of (.STU and .STA files). The password is stored encrypted in the application.

Password

An M580 safety project is not password protected by default. If you do not manually assign a password when you create the safety project, Unity Pro applies an empty string as the password.

In this case, when you next open your Unity Pro M580 safety project, the **Password** dialog opens. To access your project, enter **no** password text, thereby accepting the empty string, and click **OK**. Thereafter, you can follow the steps set forth below to create a new password.

It is possible to create or change a password at any time.

The password is case-sensitive and has a length from 8 to 16 characters. The password robustness is increased when it contains a mix of upper and lower case, alphabetical, numerical, and special characters.

Creating a Password

Procedure for creating the application protection password:

Step	Action
1	In the project browser right-click **Project**.
2	Select **Properties** command from the popup menu. **Result**: The **Properties of Project** window appears.
3	Select **Protection** tab.
4	In the **Application** field, click **Change password ...**. **Result**: The **Modify Password** window appears.
5	Enter the new password in the **Entry** field.
6	Enter the confirmation of the new password in the **Confirmation** field.
7	Click **OK** to confirm.
8	Click **OK** or **Apply** in the **Properties of Project** window to confirm all changes. If you click **Cancel** in the **Properties of Project** window, all changes are canceled.

Changing the Password

Procedure for changing the application protection password:

Step	Action
1	In the project browser right-click **Project**.
2	Select **Properties** command from the popup menu. **Result**: The **Properties of Project** window appears.
3	Select **Protection** tab.
4	In the **Application** field, click **Change password ...**. **Result**: The **Modify Password** window appears.
5	Enter previous password in the **Old password** field.
6	Enter the new password in the **Entry** field.
7	Enter the confirmation of the new password in the **Confirmation** field.
8	Click **OK** to confirm.
9	Click **OK** or **Apply** in the **Properties of Project** window to confirm all changes. If you click **Cancel** in the **Properties of Project** window, all changes are canceled.

Deleting the Password

Procedure for clearing the application protection password:

Step	Action
1	In the project browser right-click **Project**.
2	Select **Properties** command from the popup menu. **Result**: The **Properties of Project** window appears.
3	Select **Protection** tab.
4	In the **Application** field, click **Clear password...**. **Result**: The **Password** window appears.
5	Enter the password in the **Password** field.
6	Click **OK** to confirm.
7	Click **OK** or **Apply** in the **Properties of Project** window to confirm all changes. If you click **Cancel** in the **Properties of Project** window, all changes are canceled.

Auto-Lock Feature

There is an optional auto-lock feature that limits access to the Unity Pro software programming tool after a configured time of inactivity. You can activate the auto-lock feature with the check box **Auto-lock** and select the time-out for the time of inactivity via **Minutes before lock**.

The default values are:

- **Auto-lock** is not activated
- **Minutes before lock** is set to 10 minutes (possible values: 1...999 minutes)

If the auto-lock feature is enabled and the configured inactivity time elapses, a modal dialog box is displayed requiring the entry of the application password. Behind the modal dialog box, all opened editors remain open in the same position. As a result, anybody can read the current content of the Unity Pro windows but cannot continue to work with Unity Pro.

NOTE: If you have not assigned a password to the project, the modal dialog box is not displayed.

Password Request Condition

Open an existing application (project) in Unity Pro:

Password Management	
When an application file is opened, an **Application Password** dialog box opens.	
Enter the password.	
Click **OK**.	If the password is correct, the application opens.
	If the password is wrong, a message box indicates an incorrect password was entered, and a new **Application Password** dialog box opens.
If you click **Cancel**, the application is not opened	

Accessing the application in Unity Pro after an auto-lock, when Unity Pro is not connected to the PAC or when the project in Unity Pro is EQUAL to the project in the PAC:

Password Management	
When auto-lock time is elapsed, an **Application Password** dialog box opens:	
Enter the password.	
Click **OK**.	If the password is correct, Unity Pro becomes active again.
	If the password is wrong, a message box indicates an incorrect password was entered, and a new **Application Password** dialog box opens.
If you click **Close**, the application is closed without being saved.	

Accessing the application in the PAC after an auto-lock, when Unity Pro is connected to the PAC and the application in Unity Pro is DIFFERENT from the application in the PAC:

Password Management	
On connection, if Unity Pro software application and the CPU application are not equal, an **Application Password** dialog box opens:	
Enter the password.	
Click **OK**.	If the password is correct, the connection is established.
	If the password is wrong, a message box indicates an incorrect password was entered, and a new **Application Password** dialog box opens.
If you click **Cancel**, the connection is not established.	
NOTE: On connection, if Unity Pro software application and the CPU applications are equal, there is no password request. If no password has been initially entered (left empty on project creation), click **OK** to establish the connection on password prompt.	

NOTE: After three failed password attempts, you will have to wait an increasing amount of time between each subsequent password attempt. The wait period increases from 15 seconds to 1 hour, with the wait increment increasing by a factor of 2 after each successive failed attempt.

NOTE: In case of password loss, contact Schneider Electric support *(see page 260)*.

Section Protection

At a Glance

The section protection function is accessible from the **Properties** screen of the project in offline mode.

This function is used to help protect the program sections.

NOTE: The section protection is not active as long as the protection has not been activated in the project.

NOTE:
The project protection is effective to the marked sections only. This does not prevent:

- Connecting to the CPU
- Uploading application from the CPU
- Changing the configuration
- Adding new sections
- Changing the logic in a new section (without section protection)

Activating Protection and Creating Password

Procedure for activating the protection of sections and creating the password:

Step	Action
1	In the project browser right-click **Project**.
2	Select **Properties** command from the popup menu. **Result**: The **Properties of Project** window appears.
3	Select **Protection** tab.
4	In the **Sections** field, activate the protection by checking the **Protection active** box. **Result**: The **Modify Password** dialog box appears:
5	Enter a password in the **Entry** field.
6	Enter the confirmation of the password in the **Confirmation** field.
7	Select **Crypted** check box if an enhanced password protection is required. **NOTE:** A project with a crypted password cannot be edited with a Unity Pro version lower than 4.1.
8	Click **OK** to confirm.
9	Click **OK** or **Apply** in the **Properties of Project** window to confirm all changes. If you click **Cancel** in the **Properties of Project** window, all changes are canceled.

Notes

If a section is configured with a protection (read or read/write), when protection has been activated this will be indicated by a locked padlock at the section level.

If the section is configured with a protection but the protection is disabled, an open padlock is displayed at the section level.

Changing the Password

Procedure for changing the project sections protection password:

Step	Action
1	In the project browser right-click **Project**.
2	Select **Properties** command from the popup menu. **Result**: The **Properties of Project** window appears.
3	Select **Protection** tab.
4	In the **Sections** field, click **Change password ...**. **Result**: The **Modify Password** dialog box appears:
5	Enter previous password in the **Old password** field.
6	Enter the new password in the **Entry** field.
7	Enter the confirmation of the new password in the **Confirmation** field.
8	Select **Crypted** check box if an enhanced password protection is required. **NOTE:** A project with a crypted password cannot be edited with a Unity Pro version lower than 4.1.
9	Click **OK** to confirm.
10	Click **OK** or **Apply** in the **Properties of Project** window to confirm all changes. If you click **Cancel** in the **Properties of Project** window, all changes are canceled.

Deleting the Password

Procedure for deleting the project sections protection password:

Step	Action
1	In the project browser right-click **Project**.
2	Select **Properties** command from the popup menu. **Result**: The **Properties of Project** window appears.
3	Select **Protection** tab.
4	In the **Sections** field, click **Clear password...**. **Result**: The **Access control** dialog box appears:
5	Enter the previous password in the **Password** field.

Step	Action
6	Click **OK** to confirm.
7	Click **OK** or **Apply** in the **Properties of Project** window to confirm all changes. If you click **Cancel** in the **Properties of Project** window, all changes are canceled.

Loss of Password

Overview

If you forget your password, proceed as indicated in the following procedures and contact Schneider Electric support.

Unity Pro Application Password

Schneider Electric support needs a number displayed from the **Password** dialog box reached in following conditions:

- At open time, select the application and the **Password** dialog box is displayed.
- At auto-lock time, the **Password** dialog box is displayed. If you do not remember the password, select **Close**. Open the application again and the **Password** dialog box is displayed.
  NOTE: When the application is closed without entering a password after an auto-lock, all modifications are lost.

Procedure for resetting the application password:

Step	Action
1	**Condition:** The **Password** dialog box is displayed.
2	Press `SHIFT+F2`. **Result:** A grayed number is displayed in the right side of the **Password** dialog box.
3	Give this number to Schneider Electric support.
4	Receive the generated password from Schneider Electric support. **NOTE:** The password is a temporary password, available as long as the application is not modified.
5	Enter this password.
6	Modify the password (old password = password provided by Schneider Electric support).
7	Click **Build → Build Changes**.
8	**Save** the application.

CPU Application Password

Procedure for resetting the CPU application password if the respective **.STU* file is available:

Step	Action
1	Open the respective **.STU* file.
2	When the password dialog box is displayed press `SHIFT+F2`. **Result:** A grayed number is displayed in the right side of the **Password** dialog box.
3	Give this number to Schneider Electric support.
4	Receive the generated password from Schneider Electric support. **Note:** The password is a temporary password, available as long the application is not modified.
5	Enter this password.

Step	Action
6	Modify the password (old password = password provided by Schneider Electric support).
7	**Connect** to the PLC.
8	Click **Build → Build Changes**.
9	**Save** the application.

Procedure for resetting the CPU application password if the respective **.STU* file is not available:

Step	Action
1	**Condition:** At connection time, the **Password** dialog box is displayed.
2	Press `SHIFT+F2`. **Result:** A grayed number is displayed in the right side of the **Password** dialog box.
3	Give this number to Schneider Electric support.
4	Receive the generated password from Schneider Electric support. **Note:** The password provided by Schneider Electric support is a temporary password, available as long as the application is not modified.
5	Enter this password.
6	Upload the application from CPU.
7	**Save** the application.
8	Modify the password (old password = the one provided by Schneider Electric support).
9	Click **Build → Build Changes**.
10	**Save** the application.

Section 13.9
Workstation Security Management

Introduction

Schneider Electric provides the **Security Editor** access management tool that you can use to limit and control access to the workstation on which your Unity Pro software is installed. This section describes the features of this tool that uniquely relate to M580 safety projects.

What Is in This Section?

This section contains the following topics:

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Managing Access to Unity Pro

Introduction

Schneider Electric provides the **Security Editor** configuration tool that lets you manage access to the Unity Pro software installed on a workstation. Using he *Security Editor* configuration tool to manage access to the Unity Pro software is optional.

NOTE: Access management relates to the hardware – typically a workstation – on which Unity Pro software is installed and not to the project, which has its own protection system.

For more information about the **Security Editor**, refer to the *Access security management* section of the *Unity Pro Operating Modes* manual.

NOTE: Safety user profiles also require rights to access the process part of the safety application. When you create or modify a user profile, it is your responsibility to confirm that all necessary modifications are properly made.

Categories of Users

The **Security Editor** supports two categories of users:

- **Super User (Supervisor):**
  The super user is the only person to manage access security for the software. The super user specifies who can access the software and their access rights. During installation of Unity Pro on the workstation, only the super user can access the security configuration without any limitation of rights (without a password).
  NOTE: The user name reserved for the super user is Supervisor.
- **Users:**
  Software users are defined in the list of users by the super user, if Unity Pro access security is active. If your name is in the user list, you can access a software instance by entering your name (exactly as it appears on the list) and your password.

User Profile

The user profile comprises all of the access rights for a user. The user profile can be custom-defined by the super user, or can be created by applying a preconfigured profile that comes with the **Security Editor** tool.

Preconfigured User Profiles

The **Security Editor** offers the following preconfigured user profiles, which apply to either the safety program or the process program:

Profile	Applicable program type		Description
	Process	Safety	
ReadOnly	✓	✓	The user can only access the project in read mode, except for the PAC address, which can be modified. The user can also copy or download the project.
Operate	✓	–	The user has the same rights as with a **ReadOnly** profile, with the added possibility of modifying process program execution parameters (constants, initial values, task cycle times, etc.).
Safety_Operate	–	✓	The user has similar rights as with the **Operate** profile, but with respect to the safety program, except that: ● Transferring data values to the PAC is not permitted. ● Commanding the safety program to enter maintenance mode is permitted.
Adjust	✓	–	The user has the same rights as with an **Operate** profile, with the added possibility of uploading a project (transfer to the PAC) and modifying the PAC operating mode (**Run**, **Stop**, ...)
Safety_Adjust	–	✓	The user has similar rights as with the **Adjust** profile, but with respect to the safety program, except that: ● Transferring data values to the PAC is not permitted. ● Commanding the safety program to enter maintenance mode is permitted.
Debug	✓	–	The user has the same rights as with an **Adjust** profile, with the added possibility of using the debugging tools.
Safety_Debug	–	✓	The user has similar rights as with the **Debug** profile, but with respect to the safety program, except that: ● Stopping or starting the program is not permitted. ● Updating initialization values is not permitted. ● Transferring data values to the PAC is not permitted. ● Forcing inputs, outputs or internal bits is not permitted. ● Commanding the safety program to enter maintenance mode is permitted.
Program	✓	–	The user has the same rights as with a **Debug** profile, with the added possibility of modifying the program.

Profile	Applicable program type		Description
	Process	Safety	
Safety_Program	–	✓	The user has similar rights as with the **Program** profile, but with respect to the safety program, except that: ● Stopping or starting the program is not permitted. ● Updating initialization values is not permitted. ● Transferring data values to the PAC is not permitted. ● Restoring the project to the PAC from a saved backup is not permitted. ● Forcing inputs, outputs or internal bits is not permitted. ● Commanding the safety program to enter maintenance mode is permitted.
Disabled	✓	✓	User cannot access the project.

Assigning a Preconfigured User

The super user can assign a preconfigured user, derived from a preconfigured profile, to a specific user in the **Users** tab of the **Security Editor**. The following preconfigured user selections are available:

- safety_user_Adjust
- safety_user_Debug
- safety_user_Operate
- safety_user_Program
- user_Adjust
- user_Debug
- user_Operate
- user_Program

Refer to the topic *User Functions* in the *Unity Pro Operating Modes* manual for more information about how a super user can assign a preconfigured profile to a user.

Access rights

Introduction

Unity Pro access rights are classified in the following categories:

- project services
- adjustment/debugging
- libraries
- global modification
- elementary modification of a variable
- elementary modification of DDT compound data
- elementary modification of a DFB type
- elementary modification of a DFB instance
- bus configuration editor
- input/output configuration editor
- runtime screens
- cyber security
- safety

This topic presents the access rights available for each of the preconfigured user profiles.

Project services

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Create a new project	–	–	–	–	–	–	✓	✓
Open an existing project	✓	✓	✓	✓	✓	✓	✓	✓
Save a project	–	–	–	–	–	–	✓	✓
SaveAs a project	✓	✓	✓	✓	✓	✓	✓	✓
Import a project	–	–	–	–	–	–	✓	✓
Build off-line	–	–	–	–	–	–	✓	✓
Build on-line STOP	–	–	–	–	–	–	✓	✓
Build on-line RUN	–	–	–	–	–	–	✓	✓
Start, stop or initialize the PAC*	✓	–	✓	–	–	–	✓	✓

* Only process tasks are started or stopped. For a non-safety PAC, this means the PAC is started or stopped. For an M580 safety PAC, this means that tasks other than the SAFE task are started or stopped.
✓ : Included
– : not included

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Update init values with current values (only non-safe data)	–	–	✓	–	–	–	✓	✓
Transfer project from PAC	✓	✓	✓	✓	✓	✓	✓	✓
Transfer project to PAC	✓	✓	✓	✓	–	–	✓	✓
Transfer data values from file to PAC (only non-safe data)	✓	–	✓	–	✓	–	✓	✓
Restore project backup in PAC	–	–	–	–	–	–	✓	✓
Save to project backup in PAC	–	–	–	–	–	–	✓	✓
Set address	✓	✓	✓	✓	✓	✓	✓	✓
Modify options	✓	✓	✓	✓	✓	✓	✓	✓
* Only process tasks are started or stopped. For a non-safety PAC, this means the PAC is started or stopped. For an M580 safety PAC, this means that tasks other than the SAFE task are started or stopped. ✓ : Included – : not included								

Adjustment/Debugging

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Modify variable values	✓	–	✓		✓		✓	✓
Modify safety variable values	–	✓	–	✓	–	✓	–	✓
Force internal bits	–	–	✓	–	–	–	✓	✓
Force outputs	–	–	✓	–	–	–	✓	✓
Force inputs	–	–	✓	–	–	–	✓	✓
Task management	–	–	✓	–	–	–	✓	✓
SAFE Task management	–	–	–	✓	–	–	–	✓
✓ : Included – : not included								

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Task cycle time modification	✓	–	✓		✓	–	✓	✓
SAFE Task cycle time modification	–	✓	–	✓	–	✓	–	✓
Suppress message in viewer	✓	✓	✓	✓	✓	✓	✓	✓
Debug the executable	–	–	✓	✓	–	–	✓	✓
Replace a project variable	–	–	–	–	–	–	✓	✓
Replace a safety project variable	–	–	–	–	–	–	–	✓
✓ : Included – : not included								

Libraries

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Create libraries or families	–	–	–	–	–	–	✓	✓
Create safety libraries or families	–	–	–	–	–	–	–	✓
Delete libraries or families	–	–	–	–	–	–	✓	✓
Delete safety libraries or families	–	–	–	–	–	–	–	✓
Put an object into library	–	–	–	–	–	–	✓	✓
Put an object into safety library	–	–	–	–	–	–	–	✓
Delete an object from library	–	–	–	–	–	–	✓	✓
Delete an object from safety library	–	–	–	–	–	–	–	✓
✓ : Included – : not included								

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Get an object from a library	–	–	–	–	–	–	✓	✓
Get an object from the safety library	–	–	–	–	–	–	–	✓
✓ : Included – : not included								

Global modification

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Modify the documentation	✓	✓	✓	✓	✓	✓	✓	✓
Modify the functional view	–	–	–	–	–	–	✓	✓
Modify the animation tables	✓	✓	✓	✓	✓	✓	✓	✓
Modify constants value	✓	–	✓	–	✓	–	✓	✓
Modify safety constants value	–	✓	–	✓	–	✓	–	✓
Modify the program structure	–	–	–	–	–	–	✓	✓
Modify the safety program structure	–	–	–	–	–	–	–	✓
Modify program sections	–	–	–	–	–	–	✓	✓
Modify safety program sections	–	–	–	–	–	–	–	✓
Modify project settings	–	–	–	–	–	–	✓	✓
✓ : Included – : not included								

Elementary modification of a variable

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Variable add/remove	–	–	–	–	–	–	✓	✓
Safety Variables add/remove	–	–	–	–	–	–	–	✓
Variable main attributes modifications	–	–	–	–	–	–	✓	✓
Safety Variables main attributes modifications	–	–	–	–	–	–	–	✓
Variable minor attributes modifications	✓	–	✓	–	✓	–	✓	✓
Safety Variables minor attributes modifications	–	✓	–	✓	–	✓	–	✓
✓ : Included – : not included								

Elementary modification of DDT compound data

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
DDT add/remove	–	–	–	–	–	–	✓	✓
DDT modifications	–	–	–	–	–	–	✓	✓
✓ : Included – : not included								

Elementary modification of a DFB type

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
DFB type add/remove	–	–	–	–	–	–	✓	✓
Safety DFB type add/remove	–	–	–	–	–	–	–	✓
DFB type structure modification	–	–	–	–	–	–	✓	✓
Safety DFB type structure modification	–	–	–	–	–	–	–	✓
DFB type sections modification	–	–	–	–	–	–	✓	✓
Safety DFB type sections modification	–	–	–	–	–	–	–	✓
✓ : Included – : not included								

Elementary modification of a DFB instance

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
DFB instance modification	–	–	–	–	–	–	✓	✓
Safety DFB instance modification	–	–	–	–	–	–	–	✓
DFB instance minor attributes modification	✓	–	✓	–	✓	–	✓	✓
Safety DFB instance minor attributes modification	–	✓	–	✓	–	✓	–	✓
✓ : Included – : not included								

Bus configuration editor

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Modify the configuration	–	–	–	–	–	–	✓	✓
Modify the safety configuration	–	–	–	–	–	–	–	✓
I/O sniffing	–	–	–	–	–	–	✓	✓
✓ : Included – : not included								

Input/output configuration editor

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Modify the I/O configuration	–	–	–	–	–	–	✓	✓
Modify the safety I/O configuration	–	–	–	–	–	–	–	✓
Adjust the I/O	✓	–	✓	–	✓	–	✓	✓
Adjust the safety I/O	–	✓	–	✓	–	✓	–	✓
Save_param	–	–	✓	–	–	–	✓	✓
Restore_param	–	–	✓	–	–	–	✓	✓
✓ : Included – : not included								

Runtime screens

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Modify screens	–	–	–	–	–	–	✓	✓
Modify messages	–	–	–	–	–	–	✓	✓
Add/remove screens or families	–	–	–	–	–	–	✓	✓
✓ : Included – : not included								

Cyber Security

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Create or modify application password	–	–	–	–	–	–	✓	✓
Enter Maintenance mode	–	✓	–	✓	–	✓	–	✓
Adapt Auto-Lock timeout	✓	✓	✓	✓	✓	✓	✓	✓
✓ : Included – : not included								

Safety

The access rights for this category are as follows:

Access right	Preconfigured User Profile							
	Adjust	Safety_ Adjust	Debug	Safety_ Debug	Operate	Safety_ Operate	Program	Safety_ Program
Enter Maintenance mode	–	✓	–	✓	–	✓	–	✓
✓ : Included – : not included								

Section 13.10
Modifications to Unity Pro for the M580 Safety System

Introduction

This section describes Unity Pro functionality that has been modified or limited for the M580 safety system.

What Is in This Section?

This section contains the following topics:

Topic	Page
Transferring and Importing M580 Safety Projects and Code in Unity Pro	275
Saving & Restoring Data Between a File and the PAC	276
CCOTF for an M580 Safety PAC	277
Changes to M580 Safety PAC Tools	278

Transferring and Importing M580 Safety Projects and Code in Unity Pro

Transferring a Safety Project from Unity Pro to the Safety PAC

You can use the **PLC → Transfer Project to PLC** command to transfer the project from Unity Pro to the PAC when:

- Unity Pro is connected in programming mode to the M580 safety PAC, and
- A project is open in Unity Pro, and
- All PAC tasks are in STOP state.

NOTE: You can transfer a safety application only to a safety PAC. A safety application cannot be transferred to a non-safety PAC.

Transferring a Safety Project from the Safety PAC to Unity Pro

Similarly, you can use the **PLC → Transfer Project from PLC** command to transfer the project from the PAC to Unity Pro when:

- Unity Pro is connected in programming mode to the M580 safety PAC, and
- No project is open in Unity Pro.

You can transfer content relating to any task (SAFE, MAST, FAST, AUX0, or AUX1) in either safety or maintenance operating mode.

Importing Projects and Code Sections in Unity Pro

Unity Pro XLS supports the importing of both entire projects (via **File → Open**) and code sections (via **Tasks → Import...** or **Sections → Import...**), subject to the following conditions:

- Only functions or function block types, which exist in either the safety library (**Data Scope Editor → <Libset> → Safety**) or the custom library (**Data Scope Editor → <Libset> → Custom Lib**, can be included in a code section handled by the SAFE task.
- Only functions or function block types, which exist in libraries other than the safety library, can be included in a non-SAFE code section handled by a process task (MAST, FAST, AUX0, or AUX1) .

Saving & Restoring Data Between a File and the PAC

Save and Restore Functions for Non-Safety Data

Unity Pro supports the commands **PLC → Save Data from PLC to File** and **PLC → Restore Data from File to PLC** for process and global area data. However, the data saved and restored does not include variables and function block instances created in the safe namespace.

For information on how to use these commands for non-safe data, refer to the topic *Save/Restore Data Between a File and the PLC* in the *Unity Pro Operating Modes* document.

CCOTF for an M580 Safety PAC

Changing a Configuration on the Fly

The change configuration on the fly (CCOTF) feature makes it possible to change a Unity Pro configuration while the PAC is running. Supported functions can include:

- Adding a drop.
- Adding an I/O module.
- Deleting an I/O module.
- Editing the configuration of an I/O module, including:
  - Change a parameter setting.
  - Add a channel function.
  - Delete a channel function.
  - Change a channel function.

The CCOTF feature is enabled by selecting **Online modification in RUN or STOP** in the **Configuration** tab of the CPU module.

The basic functionality of CCOTF has been implemented in the M580 safety PAC, with the limitations described below.

For a full description of CCOTF, refer to the *Modicon M580 Change Configuration on the Fly User Guide*.

Limitations of CCOTF for an M580 Safety PAC

The CCOTF feature is implemented in the M580 safety PAC, with limitations that are based on the specific function and type of I/O module, as follows:

CCOTF Function	I/O Module Type & Operating Mode			
	Non-Interfering I/O		SIL3 Safety I/O	
	Maintenance mode	Safety mode	Maintenance mode	Safety mode
Add drop	✓	✓	✓[1]	✓
Add module	✓	✓	✓[1]	X
Delete module	✓	✓	✓	X
Edit I/O module configuration	✓	✓	X	X

✓: Allowed
X: Not allowed
1.Adding both a drop and a safety module requires two CCOTF sessions: one CCOTF session to add the drop, the second CCOTF session to add the safety module. These actions cannot be performed in a single CCOTF session.

NOTE: Edits made in a single CCOTF session can relate only to a single task (SAFE, MAST, FAST, AUX0, or AUX1).

Changes to M580 Safety PAC Tools

Introduction

The M580 safety PAC supports the use of several related tools. Some of these tools have been modified for use together with the M580 safety PAC. This topic addresses some of these tools.

Memory Usage

The **Memory Usage** screen presents the following information:

- the physical distribution of the PAC (internal memory and memory card)
- the space taken up in the memory by a project (data, program, configuration, system)

For the M580 safety PAC, this screen specifically provides two new parameters – **Safety Declared Data** and **Safety Executable code** – which are described below.

NOTE: You can also use the **Pack** command in this screen to reorganize the memory where possible.

For more information, refer to the topic *Memory Usage* in the *Unity Pro Operating Modes* user manual.

For the M580 safety PAC, the following parameters are displayed:

Parameter	Description
User Data	This field indicates the memory space (in words) taken up by user data (objects relating to configuration): ● **Data**: located data associated with the processor (%M, %MW, %S, %SW, etc.) or the input/output modules. ● **Declared Data**: unlocated data (declared in the process data editor) saved after power cut. ● **Unsaved Declared Data**: unlocated data (declared in the process data editor) not saved after power cut. ● **Safety Declared Data**: unlocated data (declared in the safety data editor) not saved after power cut.
User Program	This field indicates the memory space (in words) taken up by the project program: ● **Constants**: static constants associated with the processor (%KW) and the input/output modules; initial data values. ● **Executable code**: executable code of the process area part of the project program, EFs, EFBs and DFB types. ● **Upload information**: information for uploading a project (graphic code of languages, symbols, etc.). ● **Safety Executable code**: executable code of the safety area part of the project program, EFs, EFBs and DFB types.

Parameter	Description
Other	This field indicates the memory space (in words) taken up by other data relating to the configuration and the project structure: ● **Configuration**: other data relating to configuration (hardware configuration, software configuration). ● **System**: data used by the operating system (task stack, catalogs, etc.), ● **Diagnostic**: information relating to process or system diagnostics, diagnostics buffer. ● **Data Dictionary**: Dictionary of symbolized variables with their characteristic (address, type, and so forth).
Internal Memory	This field shows the organization of the PAC internal memory. It also indicates the memory space available (**Total**), the largest possible contiguous memory space (**Greatest**) and the level of fragmentation (due to online modifications).

Event Viewer

Event Viewer is an MS-Windows utility that captures events logged by Unity Pro. You can use *Event Viewer* to display a history of logged events.

Access *Event Viewer* in MS-Windows in the *Administrative Tools* folder of the *Control Panel*. When you open the utility, select **Show Action Pane**, then click **Create Custom View** to open that dialog. There, you can create a custom view for Unity Pro events.

NOTE: In the **Create Custom View** dialog, first select **By source**, then select **TraceServer** as the source to display Unity Pro events.

Appendices

Introduction

The appendices contain information on the IEC 61508 and its SIL policy. Further, technical data of the Safety and non-interfering modules are provided and example calculations are carried out.

What Is in This Appendix?

The appendix contains the following chapters:

Chapter	Chapter Name	Page
A	IEC 61508	283
B	System Objects	291

Appendix A
IEC 61508

Introduction

This chapter provides information on the Safety concepts of the IEC 61508 in general and its SIL policy in particular.

What Is in This Chapter?

This chapter contains the following topics:

Topic	Page
General Information on the IEC 61508	284
SIL Policy	286

General Information on the IEC 61508

Introduction

Safety-Related Systems are developed for use in processes in which risks to humans, environment, equipment and production are to be kept at an acceptable level. The risk depends on the severity and likelihood, thereby defining the necessary measures of protection.

Concerning the Safety of processes, there are 2 sides to be considered:

- the regulations and requirements defined by official authorities in order to help protect humans, environment, equipment, and production
- the measures by which these regulations and requirements are fulfilled

IEC 61508 Description

The technical standard defining the requirements for Safety-Related Systems is

- the IEC 61508.

It deals with the Functional Safety of electrical, electronic or programmable electronic Safety-Related Systems. A Safety-Related System is a system that is required to perform 1 or more specific functions to ensure risks are kept at an acceptable level. Such functions are defined as Safety Functions. A system is defined functionally Safe if random, systematic, and common cause failures do not lead to malfunctioning of the system and do not result in injury or death of humans, spills to the environment and loss of equipment and production.

The standard defines a generic approach to all lifecycle activities for systems that are used to perform Safety Functions. It constitutes procedures to be used for the design, the development, and the validation of both hardware and software applied in Safety-Related Systems. Further, it determines rules concerning both the management of Functional Safety and documentation.

IEC 61511 Description

The Functional Safety requirements defined in the IEC 61508 are refined specifically for the process industry sector in the following technical standard:

- the IEC 61511: Functional safety - safety instrumented systems for the process industry sector

This standard guides the user in the application of a Safety-Related System, starting from the earliest phase of a project, continuing through the start up, covering modifications and eventual decommissioning activities. In summary, it deals with the Safety Lifecycle of all components of a Safety-Related System used in the process industry.

Risk Description

The IEC 61508 is based on the concepts of risk analysis and Safety Function. The risk depends on severity and probability. It can be reduced to a tolerable level by applying a Safety Function that consists of an electrical, electronic or programmable electronic system. Further, it should be reduced to a level that is as low as reasonably practicable.

In summary, the IEC 61508 views risks as follows:

- Zero risk can never be reached.
- Safety is to be considered from the beginning.
- Intolerable risks are to be reduced.

SIL Policy

Introduction

The SIL value evaluates the robustness of an application against failures, thus indicating the ability of a system to perform a Safety Function within a defined probability. The IEC 61508 specifies 4 levels of Safety performance depending on the risk or impacts caused by the process for which the Safety-Related System is used. The more dangerous the possible impacts are on community and environment, the higher the Safety requirements are to lower the risk.

SIL Value Description

Discrete level (1 out of a possible 4) for specifying the Safety Integrity requirements of the Safety Functions to be allocated to the Safety-Related Systems, where Safety Integrity Level 4 has the highest level of Safety Integrity and Safety Integrity Level 1 has the lowest, see *SILs for Low Demand, page 287*.

SIL Requirements Description

To achieve Functional Safety, 2 types of requirements are necessary:

- Safety Function requirements, defining what Safety Functions have to be performed
- Safety Integrity requirements, defining what degree of certainty is necessary that the Safety Functions are performed

The Safety Function requirements are derived from hazard analysis and the Safety Integrity ones from risk assessment.

They consist of the following quantities:

- Mean time between failures
- Probabilities of failure
- Failure rates
- Diagnostic coverage
- Safe failure fraction
- Hardware fault tolerance

Depending on the level of Safety Integrity, these quantities must range between defined limits.

SIL Rating Description

As defined in the IEC 61508, the SIL value is limited by both the Safe Failure Fraction (SFF) and the hardware fault tolerance (HFT) of the subsystem that performs the Safety Function. A HFT of n means that n+1 faults could cause a loss of the Safety Function, the Safe state cannot be entered. The SFF depends on failure rates and diagnostic coverage.

The following table shows the relation between SFF, HFT, and SIL for complex Safety-Related subsystems according to IEC 61508-2, in which the failure modes of all components cannot be completely defined:

SFF	HFT=0	HFT=1	HFT=2
SFF ≤ 60%	-	SIL1	SIL2
60% < SFF ≤ 90%	SIL1	SIL2	SIL3
90% < SFF ≤ 99%	SIL2	SIL3	SIL4
SFF > 99%	SIL3	SIL4	SIL4

There are 2 ways to reach a certain Safety Integrity Level:

- via increasing the HFT by providing additional independent shutdown paths
- via increasing the SFF by additional diagnostics

SIL-Demand Relation Description

The IEC 61508 distinguishes between low demand mode and high demand (or continuous) mode of operation.

In low demand mode, the frequency of demand for operation made on a Safety-Related System is not greater than 1 per year and not greater than twice the proof test frequency. The SIL value for a low demand Safety-Related System is related directly to its average probability of failure to perform its Safety Function on demand or, simply, probability of failure on demand (PFD).

In high demand or continuous mode, the frequency of demand for operation made on a Safety-Related System is greater than 1 per year and greater than twice the proof test frequency. The SIL value for a high demand Safety-Related System is related directly to its probability of a dangerous failure occurring per hour or, simply, probability of failure per hour (PFH).

SILs for Low Demand

The following table lists the requirements for a system in low demand mode of operation:

Safety Integrity Level	Probability of Failure on Demand
4	$\geq 10^{-5}$ to $< 10^{-4}$
3	$\geq 10^{-4}$ to $< 10^{-3}$
2	$\geq 10^{-3}$ to $< 10^{-2}$
1	$\geq 10^{-2}$ to $< 10^{-1}$

SILs for High Demand

The following table lists the requirements for a system in high demand mode of operation:

Safety Integrity Level	Probability of Failure per Hour
4	≥ 10^{-9} to < 10^{-8}
3	≥ 10^{-8} to < 10^{-7}
2	≥ 10^{-7} to < 10^{-6}
1	≥ 10^{-6} to < 10^{-5}

For SIL3, the required probabilities of failure for the complete Safety integrated system are:

- PFD ≥ 10^{-4} to < 10^{-3} for low demand
- PFH ≥ 10^{-8} to < 10^{-7} for high demand

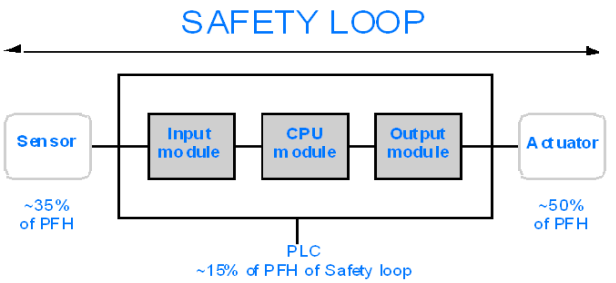
Safety Loop Description

The Safety loop to which the M580 Safety PAC consists of the following 3 parts:

- Sensors
- M580 Safety PAC with safety power supply, safety CPU, safety Coprocessor, and safety I/O modules
- Actuators

A backplane or a remote connection that includes a switch or a CRA does not destroy a Safety Loop. Backplanes, switches, and CRA modules are part of a the black channel. This means that the data exchanged by I/O and PAC cannot be corrupted without detection by the receiver.

The following figure shows a typical Safety loop:



As shown in the figure above, the contribution of the PAC is only 10-20% because the probability of failure of sensors and actuators is usually quite high.

A conservative assumption of 10% for the Safety PAC's contribution to the overall probability leaves more margin for the user and results in the following required probabilities of failure for the Safety PAC:

- PFD ≥ 10^{-5} to < 10^{-4} for low demand
- PFH ≥ 10^{-9} to < 10^{-8} for high demand

PFD Equation Description

The IEC 61508 assumes that half of the failures end in a Safe state. Therefore, the failure rate λ is divided into

- λ_S - the safe failure and
- λ_D - the dangerous failure, itself composed of
  - λ_{DD} - dangerous failure detected by the internal diagnostic
  - λ_{DU} - dangerous failure undetected.

The failure rate can be calculated by using the mean time between failures (MTBF), a module specific value, as follows:

$\lambda = 1/MTBF$

The equation for calculating the probability of failure on demand is:

$PFD(t) = \lambda_{DU} \times t$

t represents the time between 2 proof tests.

The probability of failure per hour implies a time interval of 1 hour. Therefore, the PFD equation is reduced to the following one:

$PFH = \lambda_{DU}$

Appendix B
System Objects

Introduction

This chapter describes the system bits and words of the M580 Safety PAC.

NOTE: The symbols associated with each bit object or system word mentioned in the descriptive tables of these objects are not implemented as standard in the software, but can be entered using the data editor.

What Is in This Chapter?

This chapter contains the following topics:

Topic	Page
M580 Safety System Bits	292
M580 Safety System Words	293

M580 Safety System Bits

System Bits for SAFE Task Execution

The following system bits apply to the M580 safety PAC. For a description of system bits that apply to both the M580 safety PAC and non-safety M580 PACs, refer to the presentation of *System Bits* in the *Unity Pro System Bits and Words Reference Manual*.

These system bits are related to the execution SAFE task, but are not directly accessible in process program code. They can be accessed only via the `S_SYST_READ_TASK_BIT_MX` and `S_SYST_RESET_TASK_BIT_MX` blocks.

Bit Symbol	Function	Description	Initial State	Type
%S17 `CARRY`	Rotate shift output	During a rotate shift operation in the SAFE task, this bit takes the state of the outgoing bit.	0	R/W
%S18 `OVERFLOW`	Overflow or arithmetic error detected	Normally set to 0, this bit is set to 1 in the event of a capacity overflow if there is: ● A result greater than + 32 767 or less than - 32 768, in single length. ● A result greater than + 65 535, in unsigned integer. ● A result greater than + 2 147 483 647 or less than - 2 147 483 648, in double length ● A result greater than +4 294 967 296, in double length or unsigned integer. ● Division by 0. ● The root of a negative number. ● Forcing to a non-existent step on a drum. ● Stacking up of an already full register, emptying of an already empty register.	0	R/W
%S21 `1RSTTASKRUN`	First SAFE task scan in RUN	Tested in the SAFE task, this bit indicates the first cycle of this task. It is set to 1 at the start of the cycle and reset to 0 at the end of the cycle. **NOTE:** The first cycle of the task status can be read using the `SCOLD` output of the `S_SYST_STAT_MX` system function block.	0	R/W

M580 Safety System Words

System Words for M580 Safety PACs

The following system words apply to the M580 safety PAC. For a description of system words that apply to both the M580 safety PAC and non-safety M580 PACs, refer to the presentation of *System Words* in the *Unity Pro System Bits and Words Reference Manual*.

These system words and values are related to the SAFE task. They can be accessed from application program code in the non-safety sections (MAST, FAST, AUX0 or AUX1), but not from code in the SAFE task section.

Word	Function	Type
%SW4	Period of the SAFE task defined in the configuration. The period is not modifiable by the operator.	R
%SW12	Indicates the operating mode of the Copro module: ● 16#A501 = maintenance mode ● 16#5AFE = safety mode Any other value is interpreted as a detected error.	R
%SW13	Indicates the operating mode of the CPU: ● 16#501A = maintenance mode ● 16#5AFE = safety mode Any other value is interpreted as a detected error.	R
%SW42	SAFE task current time. Indicates the execution time of the last cycle of the SAFE task (in ms).	R
%SW43	SAFE task max time. Indicate the longest task execution time of the SAFE task since the last cold start (in ms).	R
%SW44	SAFE task min time. Indicate the shortest task execution time of the SAFE task since the last cold start (in ms).	R
%SW124	Contains the cause of the detected error in the M580 Safety PAC: ● 0x5AF2: RAM detected error in memory check. ● 0x5AFB: Safety firmware code error detected. ● 0x5AF6: Safety watchdog overrun error detected on CPU. ● 0x5AFF: Safety watchdog overrun error detected on coprocessor. ● 0x5B01: Coprocessor not detected at start-up. **NOTE:** The above does not constitute a complete list. Refer to the *Unity Pro System Bits and Words Reference Manual* for more information.	R

Word	Function	Type
%SW125	Contains the cause of the detected error in the M580 Safety PAC: ● 0x5AF3: Comparison error detected by main CPU. ● 0x5AFC: Comparison error detected by coprocessor. ● 0x5AFD: Internal error detected by coprocessor. ● 0x5AFE: Synchronization error detected between CPU and coprocessor. ● 0x9690: Application program checksum error detected. **NOTE:** The above does not constitute a complete list. Refer to the *Unity Pro System Bits and Words Reference Manual* for more information.	R
%SW126 %SW127	These two system words contain information that is for Schneider Electric internal use to help analyze a detected error in more detail.	R
%SW128	Force time synchronization between internal CPU time and Safe time into the safe IO modules and Safe CPU task: ● Value change from 16#1AE5 to 16#E51A forces synchronization. ● Other sequences and values do not force synchronization.	R/W
%SW142	Contains the safety COPRO firmware version in 4 digits BCD: for example firmware version 21.42 corresponds to %SW142 = 16#2142.	R
%SW148	Count of error correcting code (ECC) errors detected by the CPU.	R
%SW169	AppId. Contains an ID of the safety code part of the application. The ID is automatically modified when the safe application code is modified. **NOTE:** The SAFE program unique identifier can be read using the `SAID` output of the `S_SYST_STAT_MX` system function block.	R

Glossary

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NOTE: For terms taken from the IEC 61508 standard, refer to the standard for complete definitions.

%I

According to the CEI standard, `%I` indicates a language object of type discrete IN.

%IW

According to the CEI standard, `%IW` indicates a language object of type analog IN.

%M

According to the CEI standard, `%M` indicates a language object of type memory bit.

%MW

According to the CEI standard, `%MW` indicates a language object of type memory word.

%S

ccording to the CEI standard, `%SW` indicates a language object of type system bit.

%SW

According to the CEI standard, `%SW` indicates a language object of type system word.

1oo2D diagnostic configuration

X out of Y. For example 1 out of 2. Voting and redundancy capacity of a Safety-Related System.

D in 1oo2D refers to diagnostics. Hence, D in 1oo2D means 1 out of 2 with diagnostics.

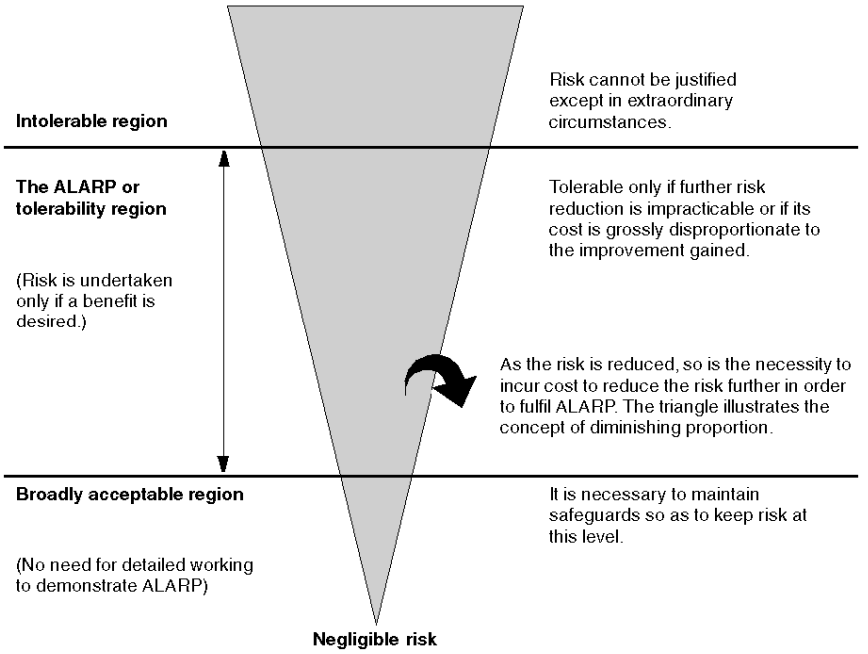
A

adapter

An adapter is the target of real-time I/O data connection requests from scanners. It cannot send or receive real-time I/O data unless it is configured to do so by a scanner, and it does not store or originate the data communications parameters necessary to establish the connection. An adapter accepts explicit message requests (connected and unconnected) from other devices.

ALARP

(*as low as reasonably practicable*) (Definition IEC 61508)



ARRAY

An `ARRAY` is a table containing elements of a single type. This is the syntax: `ARRAY [<limits>] OF <Type>`

Example: `ARRAY [1..2] OF BOOL` is a one-dimensional table with two elements of type `BOOL`.

`ARRAY [1..10, 1..20] OF INT` is a two-dimensional table with 10x20 elements of type `INT`.

ART

(*application response time*) The time a CPU application takes to react to a given input. ART is measured from the time a physical signal in the CPU turns on and triggers a write command until the remote output turns on to signify that the data has been received.

AUX

An (AUX) task is an optional, periodic processor task that is run through its programming software. The AUX task is used to execute a part of the application requiring a low priority. This task is executed only if the MAST and FAST tasks have nothing to execute. The AUX task has two sections:

- IN: Inputs are copied to the IN section before execution of the AUX task.
- OUT: Outputs are copied to the OUT section after execution of the AUX task.

B

BCD

(*binary-coded decimal*) Binary encoding of decimal numbers.

BOOL

(*boolean type*) This is the basic data type in computing. A `BOOL` variable can have either of these values: 0 (`FALSE`) or 1 (`TRUE`).

A bit extracted from a word is of type `BOOL`, for example: `%MW10.4`.

BOOTP

(*bootstrap protocol*) A UDP network protocol that can be used by a network client to automatically obtain an IP address from a server. The client identifies itself to the server using its MAC address. The server, which maintains a pre-configured table of client device MAC addresses and associated IP addresses, sends the client its defined IP address. The BOOTP service utilizes UDP ports 67 and 68.

broadcast

A message sent to all devices in a broadcast domain.

C

CCF

(*common cause failure*) Failure, which is the result of 1 or more events, causing coincident failures of 2 or more separate channels in a multiple channel system, leading to system failure. (Definition IEC 61508) The common cause factor in a dual channel system is the crucial factor for the probability of failure on demand (PFD) for the whole system.

CCOTF

(*change configuration on the fly*) A feature of Unity Pro that allows a module hardware change in the system configuration while the system is operating. This change does not impact active operations.

CIP™

(*common industrial protocol*) A comprehensive suite of messages and services for the collection of manufacturing automation applications (control, safety, synchronization, motion, configuration and information). CIP allows users to integrate these manufacturing applications with enterprise-level Ethernet networks and the internet. CIP is the core protocol of EtherNet/IP.

cold start

Cold start refers to starting the computer from power off.

CRC

(*cyclic redundancy check*)

D

DDDT

(*device derived data type*) A DDT predefined by the manufacturer and not modifiable by user. It contains the I/O language elements of an I/O module.

DDT

(*derived data type*) A derived data type is a set of one or more types of basic data types, for example an array or structure.

determinism

For a defined application and architecture, you can predict that the delay between an event (change of value of an input) and the corresponding change of a controller output is a finite time *t*, smaller than the deadline required by your process.

device network

An Ethernet-based network within an RIO network that contains both RIO and distributed equipment. Devices connected on this network follow specific rules to allow RIO determinism.

DFB

(*derived function block*) DFBs are function blocks that can be defined by the user in ST, IL, LD or FBD language. Using these DFB types in an application makes it possible to:

- simplify the design and entry of the program
- make the program easier to read
- make it easier to debug
- reduce the amount of code generated

diagnostic coverage

Fractional decrease in the probability of dangerous hardware failures resulting from the operation of the automatic diagnostic tests. (Definition IEC 61508) The fraction of the possible dangerous failures λ_D is divided into failures which are detected by diagnostics and failures which remain undetected.

$\lambda_D = \lambda_{DD} + \lambda_{DU}$

The diagnostic coverage (DC) defines the fraction of the dangerous failures which are detected.

$\lambda_{DD} = \lambda_D$. DC

$\lambda DU = \lambda_D (1 - DC)$

The definition may also be represented in terms of the following equation, where DC is the diagnostic coverage, λ_{DD} is the probability of detected dangerous failures and λ_D total is the probability of total dangerous failures:

$$DC = \frac{\sum \lambda_{DD}}{\sum \lambda_{Dtotal}}$$

DIO

(*distributed I/O*) Also known as distributed equipment. DRSs use DIO ports to connect distributed equipment.

DIO cloud

A group of distributed equipment that is not required to support RSTP. DIO clouds require only a single (non-ring) copper wire connection. They can be connected to some of the copper ports on DRSs, or they can be connected directly to the CPU or Ethernet communications modules in the *local rack*. DIO clouds **cannot** be connected to *sub-rings*.

DIO network

A network containing distributed equipment, in which I/O scanning is performed by a CPU with DIO scanner service on the local rack. DIO network traffic is delivered after RIO traffic, which takes priority in a device network.

distributed equipment

Any Ethernet device (Schneider Electric device, PC, servers, or third-party devices) that supports exchange with a CPU or other Ethernet I/O scanner service.

DLL

(*dynamic link library*)

DNS

(*domain name server/service*) A service that translates an alpha-numeric domain name into an IP address, the unique identifier of a device on the network.

DTM

(*device type manager*) A DTM is a device driver running on the host PC. It provides a unified structure for accessing device parameters, configuring and operating the devices, and troubleshooting devices. DTMs can range from a simple graphical user interface (GUI) for setting device parameters to a highly sophisticated application capable of performing complex real-time calculations for diagnosis and maintenance purposes. In the context of a DTM, a device can be a communications module or a remote device on the network.

See FDT.

E

E/E/PES

(*electrical/electronic/programmable electronic system*) (Definition IEC 61508) System for control, protection or monitoring based on 1 or more electrical/electronic programmable electronic (E/E/PE) devices. This includes elements of the system such as power supplies, sensors and other input devices, data highways and other communication paths, and actuators and other output devices.

EDT

(*elementary data type*) An elementary data type is predefined.

EF

(*elementary function*) This is a block used in a program which performs a predefined logical function.

A function does not have any information on the internal state. Several calls to the same function using the same input parameters will return the same output values. You will find information on the graphic form of the function call in the [*functional block (instance)*]. Unlike a call to a function block, function calls include only an output which is not named and whose name is identical to that of the function. In FBD, each call is indicated by a unique [number] via the graphic block. This number is managed automatically and cannot be modified.

Position and configure these functions in your program in order to execute your application.

You can also develop other functions using the SDKC development kit.

EFB

(*elementary function block*) This is a block used in a program which performs a predefined logical function.

EFBs have states and internal parameters. Even if the inputs are identical, the output values may differ. For example, a counter has an output indicating that the preselection value has been reached. This output is set to 1 when the current value is equal to the preselection value.

EMC

(*electromagnetic compatibility*) The term refers to the origin, control, and measurement of electromagnetic effects on electronic systems.

EN

`EN` stands for **EN**able; it is an optional block input. When the `EN` input is enabled, an `ENO` output is set automatically.

If `EN` = 0, the block is not enabled; its internal program is not executed, and `ENO` is set to 0.

If `EN` = 1, the block's internal program is run and `ENO` is set to 1. If a runtime error is detected, `ENO` is set to 0.

If the `EN` input is not connected, it is set automatically to 1.

ENO

ENO stands for **E**rror **NO**tification; this is the output associated with the optional input EN.

If ENO is set to 0 (either because EN = 0 or if a runtime error is detected):

- The status of the function block outputs remains the same as it was during the previous scanning cycle that executed correctly.
- The output(s) of the function, as well as the procedures, are set to 0.

error

Discrepancy between a computed, observed or measured value or condition and the true, specified or theoretically correct value or condition. (Definition IEC 61508)

ESD

(*emergency shutdown*)

Ethernet

A 10 Mb/s, 100 Mb/s, or 1 Gb/s, CSMA/CD, frame-based LAN that can run over copper twisted pair or fiber optic cable, or wireless. The IEEE standard 802.3 defines the rules for configuring a wired Ethernet network; the IEEE standard 802.11 defines the rules for configuring a wireless Ethernet network. Common forms include 10BASE-T, 100BASE-TX, and 1000BASE-T, which can utilize category 5e copper twisted pair cables and RJ45 modular connectors.

EtherNet/IP™

A network communication protocol for industrial automation applications that combines the standard internet transmission protocols of TCP/IP and UDP with the application layer common industrial protocol (CIP) to support both high speed data exchange and industrial control.

EUC

(*equipment under control*) (Definition IEC 61508) This term designates equipment, machinery, apparatuses or plants used for manufacturing, process, transportation, medical or other activities.

explicit messaging

TCP/IP-based messaging for Modbus TCP and EtherNet/IP. It is used for point-to-point, client/server messages that include both data, typically unscheduled information between a client and a server, and routing information. In EtherNet/IP, explicit messaging is considered class 3 type messaging, and can be connection-based or connectionless.

F

failure

Termination of the ability of a functional unit to perform a required function. (Definition IEC 61508)

FAST

A FAST task is an optional, periodic processor task that identifies high priority, multiple scan requests, which is run through its programming software. A FAST task can schedule selected I/O modules to have their logic solved more than once per scan. The FAST task has two sections:

- IN: Inputs are copied to the IN section before execution of the FAST task.
- OUT: Outputs are copied to the OUT section after execution of the FAST task.

Execution of the FAST task is given priority over all other tasks.

fault

Abnormal condition that may cause a reduction in, or loss of, the capability of a functional unit to perform a required function. (Definition IEC 61508)

FBD

(*function block diagram*) An IEC 61131-3 graphical programming language that works like a flowchart. By adding simple logical blocks (`AND`, `OR`, etc.), each function or function block in the program is represented in this graphical format. For each block, the inputs are on the left and the outputs on the right. Block outputs can be linked to inputs of other blocks in order to create complex expressions.

FDR

(*fast device replacement*) A service that uses configuration software to replace an inoperable product.

FFB

(*function/function block*)

FMEA

(*failure modes and effects analysis*)

FMECA

(*failure modes and effects criticality analysis*)

FTP

(*file transfer protocol*) A protocol that copies a file from one host to another over a TCP/IP-based network, such as the internet. FTP uses a client-server architecture as well as separate control and data connections between the client and server.

full duplex

The ability of two networked devices to independently and simultaneously communicate with each other in both directions.

Functional Safety

Part of the overall safety relating to the EUC and the EUC control system which depends on the correct functioning of the E/E/PE safety-related systems, other technology safety-related systems and external risk reduction facilities. (Definition IEC 61508)

A system is defined functionally Safe if random, systematic and common cause failures do not lead to malfunctioning of the system and do not result in injury or death of humans, spills to the environment and loss of equipment or production:

- Functional Safety deals with the part of the overall Safety that depends on the correct functioning of the Safety-Related System.
- Functional Safety applies to products as well as organizations.

H

HFT

(*hardware fault tolerance*) (Definition IEC 61508)

A hardware fault tolerance of N means that N + 1 faults could cause a loss of the Safety Function, for instance:

- HFT = 0: The 1st failure could cause a loss of the Safety Function
- HFT = 1: 2 faults in combination could cause a loss of the Safety Function. (There are 2 different paths to go to a Safe state. Loss of the Safety Function means that a Safe state cannot be entered.

I

I/O scanner

An Ethernet service that continuously polls I/O modules to collect data, status, event, and diagnostics information. This process monitors inputs and controls outputs. This service supports both RIO and DIO logic scanning.

IEC

(*International Electrotechnical Commission*)

IEC 61131-3

International standard: programmable logic controllers; Part 3: programming languages

IEC 61508

The IEC 61508 standard is an international standard that addresses Functional Safety of electrical / electronic / programmable electronic Safety-Related Systems. It applies to any kind of Safety-Related System in any industry wherever there are no product standards.

IL

(*instruction list*) An IEC 61131-3 programming language that contains a series of basic instructions. It is very close to assembly language used to program processors. Each instruction is made up of an instruction code and an operand.

implicit messaging

UDP/IP-based class 1 connected messaging for EtherNet/IP. Implicit messaging maintains an open connection for the scheduled transfer of control data between a producer and consumer. Because an open connection is maintained, each message contains primarily data, without the overhead of object information, plus a connection identifier.

INT

(*INTeger*) (encoded in 16 bits) The upper/lower limits are as follows: -(2 to the power of 15) to (2 to the power of 15) - 1.

Example: `-32768`, `32767`, `2#1111110001001001`, `16#9FA4`.

IODDT

(*input/output derived data type*) A structured data type representing a module, or a channel of a CPU. Each application expert module possesses its own IODDTs.

IP address

The 32-bit identifier, consisting of both a network address and a host address assigned to a device connected to a TCP/IP network.

L

LD

(*ladder diagram*) An IEC 61131-3 programming language that represents instructions to be executed as graphical diagrams very similar to electrical diagrams (contacts, coils, etc.).

local rack

An M580 rack containing a power supply, the CPU and – in an M580 safety system – the coprocessor. A local rack consists of one or two racks: the main rack and the extended rack, which belongs to the same family as the main rack. The extended rack is optional.

M

main ring

The main ring of an Ethernet RIO network. The ring contains RIO modules and a local rack (containing a CPU with Ethernet I/O scanner service) and a power supply module.

MAST

A master (MAST) task is a deterministic processor task that is run through its programming software. The MAST task schedules the RIO module logic to be solved in every I/O scan. The MAST task has two sections:

- IN: Inputs are copied to the IN section before execution of the MAST task.
- OUT: Outputs are copied to the OUT section after execution of the MAST task.

MB/TCP

(*Modbus over TCP protocol*) This is a Modbus variant used for communications over TCP/IP networks.

Modbus

Modbus is an application layer messaging protocol. Modbus provides client and server communications between devices connected on different types of buses or networks. Modbus offers many services specified by function codes.

MTBF

(*mean time between failures*)

MTTF

(*mean time to failure*)

MTTR

(*mean time to repair*)

N

NFPA

(*National Fire Protection Association*): This is a body for establishing codes and standards for fire protection, electrical and machine Safety in the U.S.

non-interfering module

Non-interfering modules are modules that are not directly used to control the Safety Function. They do not interfere with the Safety modules (either during normal operation or if there is a fault).

NTP

(*network time protocol*) Protocol for synchronizing computer system clocks. The protocol uses a jitter buffer to resist the effects of variable latency.

P

PAC

(*programmable automation controller*) The PAC is the brain of an industrial manufacturing process. It automates a process as opposed to relay control systems. PACs are computers suited to survive the harsh conditions of an industrial environment.

PELV

(*protected extra low voltage*)

PES

(*programmable electronic system*) (Definition IEC 61508)

System for control, protection or monitoring based on 1 or more programmable electronic devices, including elements of the system such as power supplies, sensors and other input devices, data highways and other communication paths, and actuators and other output devices. PES is another term for a computer control system or PAC.

PFD

(*probability of failure on demand*) (Definition IEC 61508)

For a single channel system the average probability of a failure on demand is calculated as follows:

$$PFD(t)_{Av} = \frac{1}{2} \lambda_{DU} \bullet t$$

For a dual channel system the average probability of a failure on demand is calculated as follows:

$$PFD(t)_{Av} = \lambda_{DUCH1} \bullet \lambda_{DUCH2} \bullet t^2 + CC$$

For a dual channel system, also the Common Cause effect (CC) must be considered. The common cause effect ranges from 1% to 10% of PFD_{CH1} and PFD_{CH2}. (=1/RRF).

PFH

(*probability of failure per hour*) (Definition IEC 61508)

port mirroring

In this mode, data traffic that is related to the source port on a network switch is copied to another destination port. This allows a connected management tool to monitor and analyze the traffic.

project

A project is a user application in Unity Pro XLS.

proof test

Periodic test performed to detect failures in a safety-related system so that, if necessary, the system can be restored to an "as new" condition or as close as practical to this condition. (Definition IEC 61508)

proof test interval

The proof test interval is the time period between proof tests.

PS

(*power supply*)

PST

(*process safety time*) The process safety time is defined as the period of time between a failure occurring in EUC or the EUC control system (with the potential to give rise to a hazardous event) and the occurrence of the hazardous event if the safety function is not performed. (Definition IEC 61508)

Q

QoS

(*quality of service*) The practice of assigning different priorities to traffic types for the purpose of regulating data flow on the network. In an industrial network, QoS is used to provide a predictable level of network performance.

R

RAM

(*random access memory*)

random hardware failure

Failure, occurring at a random time, which results from 1 or more of the possible degradation mechanisms in the hardware. (Definition IEC 61508)

RIO

(*remote input/output*)

RIO drop

A rack of Ethernet I/O modules, managed by an RIO adapter, with inputs and outputs included in the RIO scan of the CPU. A drop can be a single rack or a main rack with an extended rack.

RIO network

A deterministic Ethernet-based network that includes a main ring with a local rack and CPU that performs an RIO scan of I/O modules, which may be located either in the local rack or in RIO drops. The RIO drops may be part of the main ring, one or more sub-rings, or both.

risk

Combination of the probability of occurrence of harm and the severity of that harm. (Definition IEC 61508)

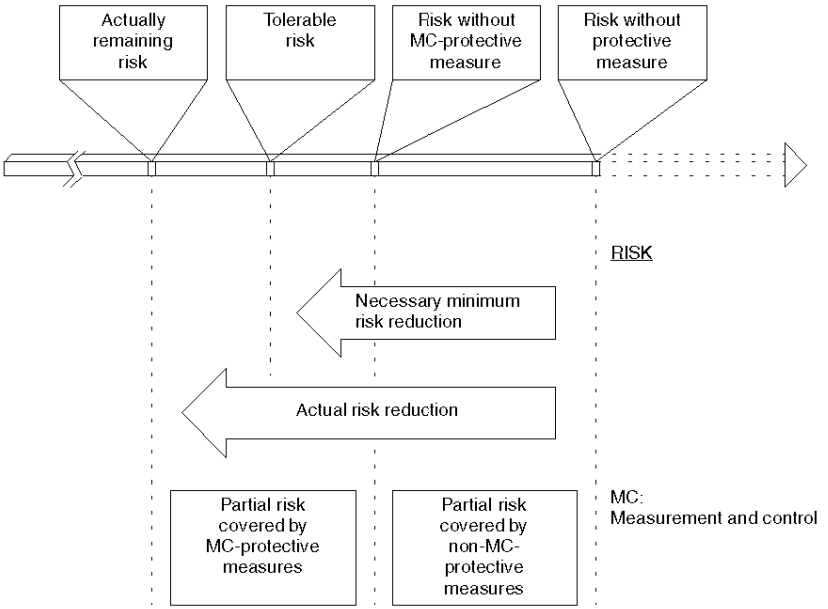
Risk is calculated using the equation R=S*H, where the letters stand for:

Letter	Meaning
R	risk
S	extent of the damage
H	frequency of occurrence of the damage

RRF

(*risk reduction factor*) (Definition IEC 61508)

The risk reduction factor equals 1/PFD.



RSTP

(*rapid spanning tree protocol*) Allows a network design to include spare (redundant) links to provide automatic backup paths if an active link stops working, without the need for loops or manual enabling/disabling of backup links.

S

Safety Function

Function to be implemented by an E/E/PE safety-related system, other technology safety-related system or external risk reduction facilities, which is intended to achieve or maintain a safe state for the EUC, in respect of a specific hazardous event. (Definition IEC 61508)

Safety Integrity

Probability of a safety-related system satisfactorily performing the required safety functions under all the stated conditions within a stated period of time. (Definition IEC 61508)

Safety PAC

M580 safety PAC (BMEP584040S CPU with BMEP58CPROS3 coprocessor)

Safety variable

Variable used to implement a Safety Function in a Safety-Related System.

Safety-Related System

This term designates a system that both

- implements the required Safety Functions necessary to achieve or maintain a Safe state for the EUC and
- is intended to achieve, on its own or using other E/E/PE Safety-Related Systems, other technology Safety-Related Systems, or external risk reduction facilities, the necessary Safety Integrity for the required Safety Functions.

SCADA

(*supervisory control and data acquisition*) SCADA systems are computer systems that control and monitor industrial, infrastructure, or facility-based processes (examples: transmitting electricity, transporting gas and oil in pipelines, and distributing water).

scanner

A scanner acts as the originator of I/O connection requests for implicit messaging in EtherNet/IP, and message requests for Modbus TCP.

service port

A dedicated Ethernet port on the M580 RIO modules. The port may support these major functions (depending on the module type):

- port mirroring: for diagnostic use
- access: for connecting HMI/Unity Pro/ConneXview to the CPU
- extended: to extend the device network to another subnet
- disabled: disables the port, no traffic is forwarded in this mode

SFC

(*sequential function chart*) An IEC 61131-3 programming language that is used to graphically represent in a structured manner the operation of a sequential CPU. This graphical description of the CPU's sequential behavior and of the various resulting situations is created using simple graphic symbols.

SFF

(*safe failure fraction*)

SFR

(*safety functional requirement*) Safety functional requirements are derived from the hazard analysis and define what the function does, for instance the safety function to be performed.

SIL

(*safety integrity level*) Discrete level (1 out of a possible 4) for specifying the safety integrity requirements of the safety functions to be allocated to the E/E/PE safety-related systems, where safety integrity level 4 has the highest level of safety integrity and safety integrity level 1 has the lowest.(Definition IEC 61508)

NOTE: For complete definitions and parameters related to SIL ratings refer to IEC 61508, "Functional safety of electrical/electronic/programmable electronic safety related systems". Provided here is a partial definition.

SIL3 project (application)

A project (application) that uses an M580 safety PAC to implement safety functions in a safety-related system.

simple daisy chain loop

Often referred to as SDCL, a simple daisy chain loop contains RIO modules only (no distributed equipment). This topology consists of a local rack (containing a CPU with Ethernet I/O scanner service), and one or more RIO drops (each drop containing an RIO adapter module).

SIR

(*safety integrity requirement*) Safety integrity requirements are derived from a risk assessment and describe the likelihood of a Safety Function to be performed satisfactorily, for instance the degree of certainty necessary for the Safety Function to be carried out.

SRS

(*safety requirements specification*) Specification containing all the requirements of the safety functions that have to be performed by the safety-related systems. (Definition IEC 61508)

SRT

(*system reaction time*) The system reaction time is the period of time between detection of a signal at the input module terminal and the reaction of setting an output at the output module terminal.

SSC

(*system safety concept*) This is a detailed description of the system architecture, configuration and diagnostics required to achieve Functional Safety.

ST

(*structured text*) An IEC 61131-3 programming language that presents structured literal language and is a developed language similar to computer programming languages. It can be used to organize a series of instructions.

Statement of Consequence

This is the last line within all special messages. It begins with "**Failure to follow these instructions**..."

sub-ring

An Ethernet-based network with a loop attached to the main ring, via a dual-ring switch (DRS) or BMENOS0300 network option switch module on the main ring. This network contains RIO or distributed equipment.

switch

A multi-port device used to segment the network and limit the likelihood of collisions. Packets are filtered or forwarded based upon their source and destination addresses. Switches are capable of full-duplex operation and provide full network bandwidth to each port. A switch can have different input/output speeds (for example, 10, 100 or 1000Mbps). Switches are considered OSI layer 2 (data link layer) devices.

systematic failure

Failure related in a deterministic way to a certain cause, which can only be eliminated by a modification of the design or of the manufacturing process, operational procedures, documentation or other relevant factors. (Definition IEC 61508)

T

TCP

(*transmission control protocol*) A key protocol of the internet protocol suite that supports connection-oriented communications, by establishing the connection necessary to transmit an ordered sequence of data over the same communication path.

TCP/IP

Also known as *internet protocol suite*, TCP/IP is a collection of protocols used to conduct transactions on a network. The suite takes its name from two commonly used protocols: transmission control protocol and internet protocol. TCP/IP is a connection-oriented protocol that is used by Modbus TCP and EtherNet/IP for explicit messaging.

TÜV

(*Technischer Überwachungsverein*) (German for Association for Technical Inspection)

U

UDP

(*user datagram protocol*) A transport layer protocol that supports connectionless communications. Applications running on networked nodes can use UDP to send datagrams to one another. Unlike TCP, UDP does not include preliminary communication to establish data paths or provide data ordering and checking. However, by avoiding the overhead required to provide these features, UDP is faster than TCP. UDP may be the preferred protocol for time-sensitive applications, where dropped datagrams are preferable to delayed datagrams. UDP is the primary transport for implicit messaging in EtherNet/IP.

UMAS

(*Unified Messaging Application Services*) UMAS is a proprietary system protocol that manages communications between Unity Pro and a controller.

UTC

(*coordinated universal time*) Primary time standard used to regulate clocks and time worldwide (close to former GMT time standard).

V

VDE

(*Verband Deutscher Elektroingenieure*) This is the German equivalent of the IEEE.

W

warm start

Warm start refers to restarting the computer without turning the power off.

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